Information, Geometry and Cryptography : The Faithfulness of Constructive Dimension, and the Non-robustness of Polynomial-time Dimension.

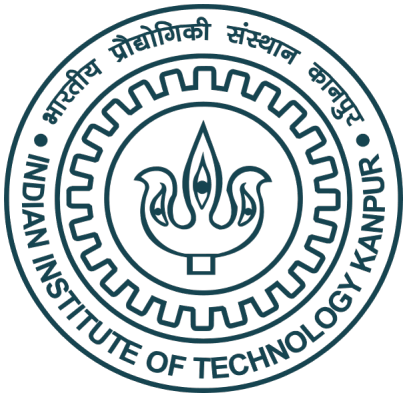
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by

Akhil S

20211262



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January 2026

Certificate

It is certified that the work contained in this thesis entitled **"Information, Geometry and Cryptography : The Faithfulness of Constructive Dimension, and the Non-robustness of Polynomial-time Dimension."** has been carried out under my supervision and that it has not been submitted elsewhere for a degree.

Month and Year of Thesis Submission: October 2025

Prof. Satyadev Nandakumar
Research Supervisor
CSE Department
Indian Institute of Technology Kanpur

Prof. Sunil Simon
Research Co-supervisor
CSE Department
Indian Institute of Technology Kanpur

Declaration

This is to certify that the thesis titled **"Information, Geometry and Cryptography : The Faithfulness of Constructive Dimension, and the Non-robustness of Polynomial-time Dimension."** has been authored by me. It presents the research conducted by me under the supervision of **Prof. Satyadev Nandakumar** and **Prof. Sunil Simon**.

To the best of my knowledge, it is an original work, both in terms of research content and narrative, and has not been submitted elsewhere, in part or in full, for a degree. Further, due credit has been attributed to the relevant state-of-the-art and collaborations with appropriate citations and acknowledgement, in line with established norms and practices.



Akhil S

Roll No. 20211262

CSE Department

Indian Institute of Technology Kanpur

Synopsis

Name of the student: **Akhil S** Roll No: **20211262**

Degree for which submitted: **PhD** Department: **CSE**

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Randomness is a fundamental concept in computation. Martin-Löf (1966) gave a robust definition of randomness, classifying infinite binary sequences as random or non-random. Lutz introduced *constructive dimension*, an effective analogue of Hausdorff dimension. Mayordomo (2002) established its equivalence with Kolmogorov complexity rates. Thus, constructive dimension quantifies the density of information in infinite binary sequences.

In the first part of this thesis, we introduce *constructive* Φ*-dimension*, an analogue of constructive dimension sensitive to the method of representation. We use this to study a constructive analogue of the notion of *faithfulness* from Hausdorff dimension. A mode of representation Φ is faithful to constructive dimension if constructive Φ-dimension and constructive dimension are equal for any point.

- **Constructive Φ-dimension.** We define the notion of constructive Φ-dimension by effectivizing Hausdorff Φ-dimension. We show equivalent characterizations of constructive Φ-dimension via Kolmogorov complexity and s-gales. We further establish a point-to-set principle for Φ-dimension.

- **Continued fractions.** We show that continued fraction representation is non-faithful to constructive dimension. In particular, we show the existence of reals that have constructive dimension arbitrarily close to 0 but have continued fraction dimension at least 0.5. We also show that continued fraction representation is non-faithful to Hausdorff dimension.

- **Cantor series.** We provide a characterisation of Cantor series expansions that preserve constructive dimension, using a log-limit condition. We further prove equivalence between Hausdorff and constructive faithfulness.

In the second part of this thesis, we study constructive dimension under polynomial-time resource bounds. We examine whether the robustness properties of constructive dimension extend to polynomial-time dimension. We study polynomial-time dimension, a quantification of polynomial-time density of information defined using polynomial-time s-gales ($\mathrm{cdim_P}$). We compare it to the another notion of polynomial-time density of information defined using polynomial-time Kolmogorov complexity ($\mathcal{K}_{\mathrm{poly}}$).

- **Polynomial-time dimension.** We show that, under the existence of one-way functions, there exist sequences with $\mathcal{K}_{\mathrm{poly}}(X) < \mathrm{cdim_P}(X)$, thereby establishing non-robustness of polynomial-time dimension.

In the third part of this thesis, we study constructive dimension under the most restrictive resource-bound of finite-state computation. Finite-state dimension is the finite-state analogue of constructive dimension and measures the density of information perceived by finite-state automata. We initiate the study of a *relative* analogue of finite-state dimension.

- **Finite-state dimension.** We initiate the study of finite-state relative dimension, defining it via finite-state s-gales with finite-window oracle access to another sequence. We provide an equivalent characterization of finite-state relative dimension using conditional block entropy rates, extending the result by Bourke, Hitchcock and Vinodchandran (2005).

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Symbols

$\mathbb{N}$	Set of natural numbers
$\mathbb{Q}$	Set of rational numbers
$\mathbb{R}$	Set of real numbers
∞	Infinity
$\emptyset$	The empty set
$\in$	Element of
$\sqsubseteq$	Prefix
$\sqsubset$	Proper prefix
$\subseteq$	Subset
$\subset$	Proper subset
γ	Gauss measure
μ	Lebesgue measure
Σ	A finite alphabet, typically $\{0, 1\}$
Σ^*	Set of all finite strings from Σ
Σ^∞	Set of all infinite strings from Σ
$\mathbb{N}^*$	Set of all finite continued fraction strings
$\mathbb{N}^\infty$	Set of all infinite continued fraction strings
$X \upharpoonright n$	First n-digits of X
$X[i]$	The character at position i in string X
$X[i:j]$	The substring $X[i]X[i+1]\ldots X[j-1]$ of string X

List of Publications

Conference Publications

1. S. Nandakumar, S. Pulari and A. S, Point-to-set principle and constructive dimension faithfulness, in *49th International Symposium on Mathematical Foundations of Computer Science*, Art. No. 76, 15 pp., LIPIcs. Leibniz Int. Proc. Inform., 306, Schloss Dagstuhl. Leibniz-Zent. Inform., Wadern, ; MR4802192

2. S. Nandakumar, A. S and P. Vishnoi, Effective continued fraction dimension versus effective Hausdorff dimension of reals, in *48th International Symposium on Mathematical Foundations of Computer Science*, Art. No. 70, 15 pp., LIPIcs. Leibniz Int. Proc. Inform., 272, Schloss Dagstuhl. Leibniz-Zent. Inform., Wadern, ; MR4640580

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Journal Publications

1. S. Nandakumar, A. S and P. Vishnoi, Effective continued fraction dimension versus effective Hausdorff dimension of reals, ACM Trans. Comput. Theory 17 (2025), no. 2, Art. 15, 18 pp.; MR4927308

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Chapter 1

Introduction

Randomness is a basic tool in various aspects of computation. A mathematically robust definition of a random sequence was given by Martin-Löf in 1966 [38]. An infinite binary sequences is Martin-Löf random if and only if it is outside the maximal constructive null set. There are several equivalent characterizations of Martin-Löf randomness. For instance, a sequence is Martin-Löf random if and only if the universal computably enumerable martingale does not succeed on it. Equivalently, Martin-Löf randoms are exactly those whose prefixes have maximal values of self-delimiting Kolmogorov complexity. In other words, all but finitely many of the prefixes of a Martin-Löf random are incompressible. Martin-Löf randoms were shown to obey the statistical laws of randomness including Strong Law of Large numbers [55], Law of Iterated Logarithms [57] and Ergodic Theorems [58] [42].

While the theory of Martin-Löf randomness classifies sequences into random and non-random, it does not quantify the *density of randomness* in a non-random sequence. Lutz effectivised the notion of Hausdorff dimension in geometric measure theory using effective s-gales. This notion was termed as effective Hausdorff dimension, or constructive dimension. Mayordomo [39] showed that constructive dimension is equivalent to Kolmogorov complexity rates. Hence this effective analogue of Hausdorff dimension quantifies density of information of infinite binary sequences.

Lutz and Stull [33] established the point-to-set principle which gives a characterization of the Hausdorff dimension of a set $E \subseteq \mathbb{R}^n$ in terms of the constructive dimensions (relative to a minimizing oracle) of the points x present in E. This line of work has been fruitful, the point-to-set principle gives us information theoretic proofs of theorems in classical fractal geometry using Kolmogorov complexity [33] [37] [36] .

1.1 Φ-dimension

Martin-Löf randomness is invariant across multiple (computable) representations of the same real. Martin-Löf randomness is preserved among different base-b representations of the same real [51] and when we convert from the base-b expansion to the continued fraction expansion [45]. A quantification of this notion is whether the *rate* of information, or constructive dimension is preserved across multiple representations. Hitchcock and Mayordomo [22] show that constructive dimension of a real is preserved across base-b representations.

In the first part of the thesis (Chapter 3, 4 and 5), we extend this body of work. We define classes of representations and formulate the notion of constructive dimension that is tied to the method of representation. We also study *faithfulness* of some representations. A class of representations is said to be faithful for constructive dimension if, for every real, the corresponding effective dimension coincides with the effective Hausdorff dimension. Faithful covering sets help simplify the calculation of Hausdorff and effective dimensions of sets.

1.1.1 Constructive Φ-dimension

In Chapter 3, we formalise the notion of constructive Φ-dimension. By this, we get a general notion of constructive dimension which is sensitive to the mode of representation Φ. We do this by effectivising the notion of Hausdorff Φ-dimension from geometric measure theory. This definition is measure theoretic and uses the notion of s-dimensional null covers, where the covers come from a restricted class of covering sets Φ. This notion covers the the continued fraction, base-b and Cantor series representations. We show that constructive Φ-dimension is robust by providing equivalent characterisations using Kolmogorov complexity and s-gales. We also prove a point-to-set principle for Φ-dimension.

1.1.2 Continued fraction dimension

Continued fractions are a way of representing reals using a series of integer parts and reciprocals. In Chapter 4, We study the notion of effective continued fraction dimension introduced by Nandakumar and Vishnoi [44]. We show that this notion is equivalent to effective Φ_{CF}-dimension, when Φ_{CF} denotes the class of continued fraction cylinders over $[0, 1]$.

We then study the faithfulness of continued fraction representation. We show that the continued fraction representation does not preserve constructive dimension, and therefore is not faithful to constructive dimension. In particular, we prove that there are reals for which the effective continued fraction dimension is greater than or equal to 0.5, but the constructive dimension of the same real is arbitrarily close to 0. We then use the point-to-set principle for continued fraction dimension to show that the continued fraction representation is not faithful to Hausdorff dimension, giving an information theoretic proof of the result by Peres and Torbin [48].

1.1.3 Cantor Covering dimension

The Cantor series representation, introduced by Georg Cantor [6], generalises the notion of base-b representation. It uses a sequence of natural numbers $Q = \{n_k\}_{k\in\mathbb{N}}$ as the terms of representation.

In Chapter 5, we study the faithfulness of Cantor series representation. We show that this notion is equivalent to effective Φ_Q-dimension, where Φ_Q denotes the class of covers generated by the Cantor series representation Q.

We derive a log-limit condition on the terms of the Cantor series expansion that characterizes the faithfulness of the expansion. We then show that the notions of "faithfulness" of Cantor coverings at the Hausdorff and constructive levels are equivalent. Hence we show the necessary and sufficient conditions for Hausdorff dimension faithfulness of Cantor coverings, thereby giving an information theoretic proof of the result by Albeverio, Ivanenko, Lebid, and Torbin [1].

1.2 Robustness of Polynomial-time dimension

In the first part of the thesis, we studied the notion of constructive dimension, restricted to the method of representation. In this setting, there are no bounds placed on the computational resources (space and time) involed in the computations of constructive dimension.

In the second part of the thesis, we study the notion of constructive dimension (of infinite binary sequences, or equivalently the base-2 representation of reals) with feasible resource bounds placed on the computation.

Lutz introduced the notion of resource-bounded measure [29] to study randomness in the resource-bounded setting. This lead to the study of effective dimension in the resource bounded setting. This line of research investigates if the properties of constructive dimension hold in the resource bounded setting as well. For resource bounds at polynomial space and above, the corresponding resource bounded notions of constructive dimensions have been shown to have equivalent characterizations in terms of resource bounded notions of Kolmogorov complexity, entropy rates and compressibility [23]. These proof techniques rely on exponential search, which is possible at polynomial space and above.

Polynomial-time dimension is the notion of constructive dimension with polynomial-time resource bounds put on the model of computation. The polynomial-time bounded Kolmogorov complexity of a string x is the length of the shortest program that outputs x within a number of steps polynomial in $|x|$. Polynomial-time density of information can be quantified in terms of the polynomial-time bounded Kolmogorov complexity of the prefixes of the string, denoted $\mathcal{K}_{\text{poly}}$.

In contrast, we can also use the gambling based approach to quantify polynomial-time density of information. Polynomial-time dimension, denoted cdim_{P} is defined using polynomial-time betting algorithms called s-gales.

For the setting of polynomial-time dimension, proof techniques that rely on exponential search do not work. Hitchcock and Vindochandran showed that $\mathcal{K}_{\text{poly}}(X) \leq \text{cdim}_{\text{P}}(X)$ for any sequence $X \in \Sigma^{\infty}$. But it remained open whether these two notions are equal or different [52].

In the second part of the thesis, (Chapter 6), we show that the existence of one-way functions implies existence of sequences for which $\mathcal{K}_{\text{poly}}(X) < \text{cdim}_{\text{P}}(X)$. Thus the notion of polynomial-time dimension is non-robust, under the cryptographic assumption that One-way functions exist.

1.3 Finite-state dimension

In the thirs part of the thesis, we look at constructive dimension from the most restrictive resource bound of finite-state computation.

Finite-state dimension quantifies the density of information of a sequence measured using finite-state automata [27]. Dai, Lathrop, Lutz and Mayordomo [10] introduced this notion using finite-state betting algorithms called finite-state s-gales. They show that finite-state

dimension can be characterised using rates of compression by finite-state automata. Along with an earlier result due to Schnorr and Stimm [49], this implies that a sequence is Borel normal if and only if it has finite-state dimension 1 [10], establishing connections to metric number theory and probability.

The study of relative dimension measures the density of information in a sequence relative to another sequence. In the third part of the thesis, (Chapter 7), we initiate the study of relative finite-state dimension using finite-state betting algorithms (s-gales) equipped with a with a finite-window oracle access to another sequence. Bourke Hitchcock and Vinodchandran[4] gave a characterisation of finite-state dimension using block entropy rate. We provide an equivalent characterization of relative finite-state dimension using conditional block entropy rates, generalizing the insight in the work of Bourke, Hitchcock and Vindochandran [4]. This demonstrates the mathematical robustness of our definition.

1.4 Organization of the thesis

The thesis is organized as follows. In Chapter 2, we give the basic definitions and notation used in the rest of the thesis. In Chapter 3, we introduce constructive Φ-dimension using s-dimensional Φ-null covers and prove alternate characterisations in terms of s-gales and Kolmogorov complexity. We also provide a point-to-set principle for Φ-dimension. In Chapter 4, we study the non-faithfulness of continued fraction dimension. We show that there are reals for which constructive continued fraction dimension is strictly greater than its constructive dimension. We also prove non-faithfulness of continued fraction dimension at the level of Hausdorff dimension. In Chapter 5, we initiate the study of constructive Cantor covering dimension. We provide a condition which characterises constructive dimension faithfulness of Cantor coverings. We also show that for computable Cantor coverings, the notions of faithfulness at constructive and Hausdorff dimension levels are equivalent. In Chapter 6, we show that polynomial-time dimension is non-robust if One-way functions exist. We study the two notions of measuring polynomial-time density of information - $\mathrm{cdim_P}$ and $\mathcal{K}_{\mathrm{poly}}$, based on dimension ($s$-gales) and Kolmogorov complexity. We show that if One-way functions exist there are sequences for which $\mathrm{cdim_P} > \mathcal{K}_{\mathrm{poly}}$. In Chapter 7, we initiate the study of a relative analogue of finite-state dimension. We provide a formulation of finite-state relative dimension based on finite-state s-gales with a finite-window oracle access to another sequence. We also show an equivalent characterisation of finite-state relative dimension using relative block entropy rates.

Chapter 2

Preliminaries

2.1 Notations

$\mathbb{N}$ denotes the set of natural numbers, $\mathbb{Q}$ denotes the set of rational numbers and $\mathbb{R}$ denotes the set of real numbers.

Σ denotes a finite alphabet, with $|\Sigma| > 2$. Unless specified otherwise, we consider the binary alphabet, $\Sigma = \{0, 1\}$. We use λ to denote the empty string. Σ^∞ denotes the set of infinite strings and Σ^* denotes the set of finite strings formed using Σ. For any $\ell \in \mathbb{N}$, Σ^ℓ denotes the set of strings having length ℓ formed using Σ. $\Sigma^{<\ell}$ and $\Sigma^{\leq\ell}$ denotes the set of strings of length less than ℓ and the set of strings of length less than or equal to ℓ respectively, formed using Σ, including λ.

For a finite string $w \in \Sigma^*$, $|w|$ denotes the length of w. Given two finite strings $u, v \in \Sigma^*$, $u.v$ represents the concatenation of strings u and v, in that order.

For an infinite string $X \in \Sigma^\infty$, $X \upharpoonright n$ denotes the string formed by the first n symbols of X. For a finite string $x \in \Sigma^*$, $x[a : b]$ denotes the string x from index a to $b - 1$. $x[n]$ denotes the nth bit of x. We use the same notion for infinite strings as well.

Given $w \in \Sigma^*$, the cylinder set of w, denoted C_w is the set of strings in Σ^∞ for which w appears as a prefix. For a Borel set $S \subseteq \mathbb{R}^n$, $\mu(S)$ denotes the Lebesgue measure of S. In particular for a real interval we have $\mu([a, b]) = b - a$. In the Cantor space, for a $v \in \Sigma^*$, $\mu(v)$ refers to the Lebesgue measure of the continued fraction cylinder C_v. In particular, for $\Sigma = \{0, 1\}$, $\mu(v) = 2^{-|v|}$.

If a finite binary string x is a prefix of a finite string z or an infinite binary sequence Z, then we denote this by $x \sqsubseteq z$ or $x \sqsubseteq Z$ respectively. If x is a proper prefix of a finite string z, we denote it by $x \sqsubset z$.

log denotes the logarithm function, typically to the base $|\Sigma|$, typically taken to 2.

We call a set of strings $\mathcal{P} \subset \Sigma^*$ prefix free if there are no two distinct strings $u, v \in \mathcal{P}$ such that $u \sqsubseteq v$. Given $n \in \mathbb{N}$ we use $[n]$ to denote $\{0, 1, \ldots n-1\}$.

The Kolmogorov complexity of a finite binary string is defined as the length of the shortest program (from a fixed prefix-free set) which produces the string.

Definition 2.1. The Kolmogorov complexity of $x \in \Sigma^*$ is defined as

$$K(x) = \min_{\pi \in \Sigma^*} \{|\pi| \mid U(\pi) = x\},$$

where U is a fixed universal prefix free Turing machine.

Chapter 3

Constructive Φ-dimension

3.1 Introduction

Lutz [30] effectivised the notion of Hausdorff dimension and the corresponding notion was termed as constructive dimension. Mayordomo [40] showed that constructive dimension quantifies the asymptotic notion of information density in sequences.

Hausdorff Φ-dimension is a notion from geometric measure theory where the class of covers used is from a restricted class of covering sets Φ. In this chapter, we introduce a constructive analogue of Hausdorff Φ-dimension which we call constructive Φ-dimension. We show that the notion of constructive Φ-dimension is robust, by giving equivalent characterisations using Kolmogorov complexity (Theorem 3.31) and Φ-s-supergales (Theorem 3.39).

The point-to-set principle introduced by J. Lutz and N. Lutz [33] relates the Hausdorff dimension of a set with the constructive dimensions of points in the set, relative to a minimizing oracle. This theorem has been instrumental in answering open questions in classical fractal geometry using the theory of computing.

We prove the point-to-set principle for Φ-dimension (Theorem 3.22) for any metric space $\mathbb{X}$, and show that this generalizes the original point-to-set principle.

3.2 Notation

We use $\mathbb{X}$ to denote the metric space under consideration with metric $d(x, y)$. For a set $U \subseteq \mathbb{X}$, we use $|U|$ to denote the diameter of U, that is $|U| = \sup_{x,y \in U} d(x, y)$.

We use $\emptyset$ to denote the empty set, and we assume $|\emptyset| = 0$. We call two sets U and V incomparable if $U \not\subseteq V$ and $V \not\subseteq U$.

Given infinite sequences $X_1, \ldots X_n$, we define the *interleaved sequence* $X_1 \oplus X_2 \oplus \ldots X_n$ to be the interleaved sequence $X = X_1[0]X_2[0] \ldots X_n[0]X_1[1] \ldots X_n[1] \ldots$.

3.3 Background

3.3.1 Hausdorff Dimension

The following definitions are originally given by Hausdorff [19]. We take the definitions from Falconer [14].

Definition 3.1 (Hausdorff [19])**.** Given a set $\mathcal{F} \subseteq \mathbb{X}$, where $\mathbb{X}$ is a metric space. A collection of sets $\{U_i\}_{i \in \mathbb{N}}$ where for each $i \in \mathbb{N}$, $U_i \subseteq \mathbb{X}$ is called a *ε-cover of $\mathcal{F}$* if for all $i \in \mathbb{N}$, $|U_i| \leq \varepsilon$ and $\mathcal{F} \subseteq \bigcup_{i \in \mathbb{N}} U_i$.

Definition 3.2 (Hausdorff [19])**.** Given an $\mathcal{F} \subseteq \mathbb{X}$, for any $s > 0$, define

$$\mathcal{H}^s_\varepsilon(\mathcal{F}) = \inf \left\{ \sum_i |U_i|^s : \{U_i\}_{i \in \mathbb{N}} \text{ is a } \varepsilon\text{-cover of } \mathcal{F} \right\}.$$

As ε decreases, the set of admissible ε covers decreases. Hence $\mathcal{H}^s_\varepsilon(\mathcal{F})$ increases.

Definition 3.3 (Hausdorff [19])**.** For $s \in (0, \infty)$, the *s-dimensional Hausdorff outer measure* of $\mathcal{F}$ is defined as:

$$\mathcal{H}^s(\mathcal{F}) = \lim_{\varepsilon \to 0^+} \mathcal{H}^s_\varepsilon(\mathcal{F}).$$

Observe that for any $t > s$, if $\mathcal{H}^s(\mathcal{F}) < \infty$, then $\mathcal{H}^t(\mathcal{F}) = 0$ (see Section 2.2 in [14]).

Finally, we have the following definition of Hausdorff dimension.

Definition 3.4 (Hausdorff [19])**.** For any $\mathcal{F} \subset \mathbb{X}$, the *Hausdorff dimension* of $\mathcal{F}$ is defined as:

$$\dim(\mathcal{F}) = \inf\{s \geq 0 : \mathcal{H}^s(\mathcal{F}) = 0\}.$$

3.3.2 Constructive dimension of sequences

Lutz [31] defines the notion of effective (equivalently, constructive) dimension of an infinite binary sequence using the notion of lower semicomputable s-gales.

Definition 3.5 (Lutz [31])**.** For $s \in [0, \infty)$, a binary s-gale is a function $d : \Sigma^* \to [0, \infty)$ such that $d(\lambda) < \infty$ and for all $w \in \Sigma^*$,

$$2^s \cdot d(w) = \sum_{i \in \{0,1\}} d(wi).$$

Definition 3.6 (Lutz [31])**.** The *success set* of a binary s-gale d is

$$S^\infty(d) = \left\{ X \in \Sigma^\infty \mid \limsup_{n \to \infty} d(X \upharpoonright n) = \infty \right\}.$$

For $\mathcal{F} \subseteq \Sigma^\infty$, $\mathcal{G}(\mathcal{F})$ denotes the set of all $s \in [0, \infty)$ such that there exists a lower semicomputable (Definition 3.38) binary s-gale d with $\mathcal{F} \subseteq S^\infty(d)$.

Definition 3.7 (Lutz [31], Hitchcock [21])**.** The *constructive dimension* or *effective Hausdorff dimension* of $\mathcal{F} \subseteq \Sigma^\infty$ is

$$\mathrm{cdim}(\mathcal{F}) = \inf \mathcal{G}(\mathcal{F}).$$

The constructive dimension of a sequence $X \in \Sigma^\infty$ is $\mathrm{cdim}(X) = \mathrm{cdim}(\{X\})$.

Mayordomo [40] extends the result by Lutz [30] to give the following Kolmogorov complexity characterization of constructive dimension of sets of infinite binary sequences.

Theorem 3.8 (Lutz [30], Mayordomo [40])**.** *For any* $\mathcal{F} \subseteq \Sigma^\infty$*,*

$$\mathrm{cdim}(\mathcal{F}) = \sup_{X \in \mathcal{F}} \liminf_{n \to \infty} \frac{K(X \upharpoonright n)}{n}.$$

3.3.2.1 Constructive Dimension in Euclidean space

Lutz and Mayordomo [35] define constructive dimension of a point $x \in \mathbb{R}^n$ in Euclidean space by considering the binary expansions of these points, and porting it over to the Cantor space Σ^∞. They note that if one or more of the coordinates of x have two binary expansions, $\mathrm{cdim}(x)$ is unaffected by how we choose between these binary expansions.

Definition 3.9 (Lutz and Mayordomo [35])**.** For a real $x = (x_1, ...x_n) \in \mathbb{R}^n$, let $S_1 \in \Sigma^\infty, \ldots, S_n \in \Sigma^\infty$ respectively be one of the binary expansions of the fractional parts of each of the coordinates of x. Define $\text{bin}(x) = S_1 \oplus S_2 \cdots \oplus S_n$.

Definition 3.10 (Lutz and Mayordomo [35])**.** For an $x \in \mathbb{R}^n$, define

$$\text{cdim}(x) = n \cdot \text{cdim}(\text{bin}(x)).$$

Lutz and Mayordomo [35] gave the following Kolmogorov complexity characterization of constructive dimension of points in $\mathbb{R}^n$.

Theorem 3.11 (Lutz and Mayordomo [35])**.** *For any $x \in \mathbb{R}^n$,*

$$\text{cdim}(x) = \liminf_{r\to\infty} \frac{K_r(x)}{r}.$$

where $K_r(x) = \min_{q\in\mathbb{Q}^n}\{K(q) : d(x,q) < 2^{-r}\}$. Here d is the Euclidean metric in R^n.

3.4 Hausdorff Φ-dimension and Effective Φ-dimension

Hausdorff dimension is defined using the notion of s-dimensional outer measures. In this setting, we fix a countable family of covering sets Φ, over a metric space $\mathbb{X}$. Here a covering set $U \in \Phi$ can be any arbitrary subset of the space $\mathbb{X}$. The notion of Hausdorff Φ-dimension is defined by restricting the class of admissible covers to Φ-covers, which are the union of sets from Φ.

3.4.1 Family of covering sets

In this work, we consider a *family of covering sets* which satisfy the properties given below.

Definition 3.12 (Family of covering sets Φ)**.** Consider a metric space $\mathbb{X}$. Let Φ be a countable family of subsets of X, that is for each $U \in \Phi$, we have $U \subseteq \mathbb{X}$. Φ is called a family of covering sets over $\mathbb{X}$ if the following property holds

- Fineness: For all $x \in \mathbb{X}$ and $\varepsilon > 0$, there exists a $U \in \Phi$ such that $x \in U$ and $|U| < \varepsilon$.

We now define the notion of a Φ-cover of a set.

Definition 3.13 (Φ-cover)**.** Let Φ be a family of covering sets over $\mathbb{X}$. A Φ - cover of a set $\mathcal{F} \subseteq \mathbb{X}$ is a countable collection of sets $\{U_i\}_{i\in\mathbb{N}} \subseteq \Phi$ such that $\{U_i\}_{i\in\mathbb{N}}$ covers $\mathcal{F}$, that is $\mathcal{F} \subseteq \bigcup_{i\in\mathbb{N}} U_i$.

3.4.2 Hausdorff Φ-dimension

We call a Φ-cover of $\mathcal{F}$ a ε-cover if the diameter of elements in the cover are less than ε.

Definition 3.14. Let Φ be a family of covering sets defined over $\mathbb{X}$. Given a set $\mathcal{F} \subseteq \mathbb{X}$, a Φ-cover $\{U_i\}_{i\in\mathbb{N}}$ of $\mathcal{F}$ is called a *ε-cover of $\mathcal{F}$ using* Φ if for all $i \in \mathbb{N}$, $|U_i| \leq \varepsilon$.

From the fineness property in Definition 3.12, we have that for any set $\mathcal{F} \subseteq \mathbb{X}$ and $\varepsilon > 0$, a ε-cover of $\mathcal{F}$ using Φ always exists.

Definition 3.15. Given an $\mathcal{F} \subseteq \mathbb{X}$, for any $s > 0$, we define

$$\mathcal{H}^s_\varepsilon(\mathcal{F}, \Phi) = \inf\left\{\sum_i |U_i|^s : \{U_i\}_{i\in\mathbb{N}} \text{ is a } \varepsilon\text{-cover of } \mathcal{F} \text{ using } \Phi\right\}.$$

As ε decreases, the set of admissible ε-covers using Φ decreases. Hence $\mathcal{H}^s_\varepsilon(\mathcal{F}, \Phi)$ increases.

Definition 3.16. For $s \in (0, \infty)$, define the *s-dimensional Φ outer measure* of $\mathcal{F}$ as:

$$\mathcal{H}^s(\mathcal{F}, \Phi) = \lim_{\varepsilon\to 0^+} \mathcal{H}^s_\varepsilon(\mathcal{F}, \Phi).$$

Observe that as with the case of classical Hausdorff dimension, for any $t > s$, if $\mathcal{H}^s(\mathcal{F}, \Phi) < \infty$, then $\mathcal{H}^t(\mathcal{F}, \Phi) = 0$ (see Section 2.2 in [14]).

Finally, we have the following definition of Hausdorff Φ-dimension.

Definition 3.17. For any $\mathcal{F} \subseteq \mathbb{X}$, the *Hausdorff Φ-dimension* of $\mathcal{F}$ is defined as:

$$\dim_\Phi(\mathcal{F}) = \inf\{s \geq 0 : \mathcal{H}^s(\mathcal{F}, \Phi) = 0\}.$$

It follows from the definition that Hausdorff Φ-dimension of a set is always greater than or equal to its Hausdorff-dimension (see Definition 3.4).

Lemma 3.18. *Let Φ be a family of covering sets over $\mathbb{X}$. For all $\mathcal{F} \subseteq \mathbb{X}$,*

$$\dim_\Phi(\mathcal{F}) \geq \dim(\mathcal{F}).$$

3.4.3 Effective Φ-dimension

We effectivise the notion of Φ-dimension using effective Φ-covers of a set. Let $\delta : \mathbb{N} \to \Phi$ be a fixed enumeration of the elements in Φ.

Definition 3.19 (Effective s-dimensional Φ-null cover)**.** For an $s \in [0, \infty)$, we say that a set $\mathcal{F} \subseteq \mathbb{X}$ has an *effective s-dimensional Φ-null cover* if and only if there exists a total Turing machine $M : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$ such that:

- For all $r \in \mathbb{N}$, $\sum_i |\delta(M(i,r))|^s \leq 2^{-r}$ and
- For all $r \in \mathbb{N}$, $\bigcup_i \delta(M(i,r)) \supseteq \mathcal{F}$.

We have the following definition of effective Φ-dimension.

Definition 3.20. For any $\mathcal{F} \subset \mathbb{X}$, the *effective (constructive) Φ-dimension* of $\mathcal{F}$,

$$\mathrm{cdim}_\Phi(\mathcal{F}) = \inf\{s \geq 0 : \text{ there exists an effective } s\text{-dimensional } \Phi\text{-null cover of } \mathcal{F}\}.$$

Note that Definition 3.19 can be relativised with respect to an oracle $A \in \Sigma^\infty$. This is done by replacing the Turing machine in Definition 3.19 with a Turing machine with oracle access to A. We then say that $\mathcal{F}$ has an effective s-dimensional Φ-null cover $\mathrm{cdim}_\Phi^A(\mathcal{F})$ relative to oracle A.

Definition 3.21. For $\mathcal{F} \subseteq \mathbb{X}$ and $A \in \Sigma^\infty$, the *effective Φ-dimension* of $\mathcal{F}$ relative to A is defined as:

$$\mathrm{cdim}_\Phi^A(\mathcal{F}) = \inf\{s \geq 0 : \text{ exists an effective } s\text{-dimensional } \Phi\text{-null cover of } \mathcal{F} \text{ relative to } A\}.$$

3.5 Point-to-set principle for Φ-dimension

J. Lutz and N. Lutz [34] introduce the point-to-set principle for Hausdorff dimension. This provides an information-theoretic characterization of Hausdorff dimension, and has been fruitful in solving problems in classical dimension using tools from information theory.

We show a Φ-dimensional analogue of the point-to-set principle. This states that the Hausdorff Φ-dimension of a set is the supremum of the constructive relative Φ-dimensions of its points, over an optimal oracle A.

The proof is an adaptation of the proof from [34]. The idea is to encode indices corresponding to the s-dimensional Φ-null covers of a set into the minimising oracle A.

Theorem 3.22. *Let* Φ *be a family of covering sets over* $\mathbb{X}$*. For all* $\mathcal{F} \subseteq \mathbb{X}$*,*

$$\dim_\Phi(\mathcal{F}) = \min_{A \in \Sigma^\infty} \sup_{x \in \mathcal{F}} \operatorname{cdim}_\Phi^A(x).$$

Proof. By the definition of Φ-dimension, we have that for any $s > \dim_\Phi(\mathcal{F})$ and $r \in \mathbb{N}$, there exists a sequence of Φ-covers $\{U_i^{r,s}\}_{i\in\mathbb{N}}$ of $\mathcal{F}$ such that $\sum_{i\in\mathbb{N}} |U_i^{r,s}|^s \leq 2^{-r}$.

Consider a sequence $A \in \Sigma^\infty$ that encodes the function $f(i,r,s) = \delta^{-1}(U_i^{r,s})$ for each $i \in \mathbb{N}$ and $r \in \mathbb{N}$ and $s \in \mathbb{Q}$ such that $s > \dim_\Phi(\mathcal{F})$. For any rational $s > \dim_\Phi(\mathcal{F})$, given oracle access to A and thereby f, a machine on input $i, r, s \in \mathbb{N}$ can output $f(i,r,s)$. Since $\sum_{i\in\mathbb{N}} |U_i^{r,s}|^s \leq 2^{-r}$ and $\bigcup_i U_i^{r,s} \supseteq \mathcal{F}$, we have that $\operatorname{cdim}_\Phi^A(x) \leq s$.

Now for some $A \in \Sigma^\infty$, take a rational $s > \sup_{x\in\mathcal{F}} \operatorname{cdim}_\Phi^A(x)$. Let $\{M_j(i,r)\}_{j\in\mathbb{N}}$ be the machines with oracle access to A that enumerates the effective s-dimensional Φ-null covers of elements $x \in \mathcal{F}$. Take $\mathcal{W} = \bigcup_{j\in\mathbb{N}}\{\delta(M_j(i,r+j))\}_{i\in\mathbb{N}}$. We have $\sum_{U\in\mathcal{W}} |U|^s \leq 2^{-r}$ and $\bigcup_{U\in\mathcal{W}} U \supseteq \mathcal{F}$. Since such a sequence of Φ-covers exist, we have that $\dim_\Phi(\mathcal{F}) \leq s$. □

3.5.1 Point to Set Principle for constructive dimension

Definition 3.23 (Dyadic Family of covers)**.** Consider the Euclidean space $\mathbb{X} = \mathbb{R}^n$. The dyadic family of covers is the set of coverings

$$\Phi_B = \bigcup_{r\in\mathbb{N}} \{[\frac{m_1}{2^r}, \frac{m_1+1}{2^r}] \times \cdots \times [\frac{m_n}{2^r}, \frac{m_n+1}{2^r}]\}_{m_1,m_2\ldots,m_n\in[2^r]}.$$

Besicovitch [3] gave the following Φ-dimension characterization of Hausdorff dimension.

Lemma 3.24 (Besicovitch [3])**.** *For all* $\mathcal{F} \subseteq \mathbb{R}^n$*, we have* $\dim(\mathcal{F}) = \dim_{\Phi_B}(\mathcal{F})$.

Similarly, we have the following Φ-dimension characterization of constructive dimension.

Lemma 3.25 (Lutz and Mayordomo [35])**.** *For all* $\mathcal{F} \subseteq \mathbb{R}^n$*, we have* $\operatorname{cdim}(\mathcal{F}) = \operatorname{cdim}_{\Phi_B}(\mathcal{F})$.

From Theorem 3.22 for Φ_B and using Lemma 3.24 and 3.25, we have the following point-to-set principle from [34] relating Hausdorff and constructive dimensions.

Corollary 3.26 (J. Lutz and N. Lutz [34])**.** *For all* $\mathcal{F} \subseteq \mathbb{R}^n$,

$$\dim(\mathcal{F}) = \min_{A \subseteq \mathbb{N}} \sup_{x \in \mathcal{F}} \operatorname{cdim}^A(x).$$

From Theorem 3.22, we also get new point-to-set principles for continued fraction dimension (See Corollary 4.4) and coverings generated by Cantor series expansions (See Corollary 5.4).

3.6 Robustness of Effective Φ-dimension

In this section we give equivalent characterisations of effective Φ-dimension using Kolmogorov complexity and Φ-s-supergales. This shows that the notion of effective Φ-dimension is robust.

3.6.1 Computable covering sets

Definition 3.27 (Family of computable covering sets)**.** Let Φ be a family of covering sets defined over $\mathbb{X}$. We say Φ is a computable family of covering sets if for all $w \in \mathbb{N}$, $|\delta(w)|$ is computable.

When we say $|\delta(w)|$ is computable, we mean there exists a total Turing machine M that on input $w \in \Sigma^*$ and $r \in \mathbb{N}$, outputs an $\ell \in \mathbb{Q}$ such that $||\delta(w)| - \ell| \leq 2^{-r}$.

We show that there is a universal s-dimensional Φ-null cover.

Lemma 3.28. *Let* Φ *be a family of computable covering sets over* $\mathbb{X}$*. For any* $s \in \mathbb{Q}$*, there exists a set* $\mathcal{F}_s$ *such that* $\operatorname{cdim}_\Phi(\mathcal{F}_s) \leq s$ *and for all* $\mathcal{F} \subseteq \Phi$ *such that* $\operatorname{cdim}_\Phi(\mathcal{F}) < s$*, we have* $\mathcal{F} \subseteq \mathcal{F}_s$*.*

Proof. As Φ is computable, and since s is rational, we can consider a universal effective s-dimensional Φ-null cover in the same manner as in Theorem 6.2.5 in [11]. Let $M_0, M_1, \ldots$ be an effective listing of all two argument Turing machines. Given an $r, j \in \mathbb{N}$, enumerate (over $i \in \mathbb{N}$) the output of $M_j(i, r+j)$, till the largest $N_j \in \mathbb{N} \cup \{\infty\}$ such that $\sum_{i=0}^{N_j} |\delta(w)|^s \leq 2^{-(r+j)}$. For an $r \in \mathbb{N}$, consider $U_r = \bigcup_{i=1}^{N_j} \delta(M_j(i, r+j))$. We have that $\sum_{U \in U_r} |U|^s \leq 2^{-r}$.

We now show that $\mathcal{F}_s = \cap_{r\in\mathbb{N}} \{U_r\}_{r\in\mathbb{N}}$ is the universal effective s-dimensional Φ-null cover. It is easy to see that $\mathrm{cdim}_\Phi(\mathcal{F}_s) \leq s$. Let $\mathcal{F} \subseteq \mathbb{X}$, such that $\mathrm{cdim}_\Phi(\mathcal{F}) < s$. Let $M_j(i,r)$ be the machine that enumerates the effective s-dimensional Φ-null covers of $\mathcal{F}$. As for all $r \in \mathbb{N}$, $\mathcal{F} \subseteq \bigcup_i \delta(M_j(i, r+j))$, we have that $\mathcal{F} \subseteq \mathcal{F}_s$. □

We show that the constructive Φ-dimension of a set is the supremum of constructive Φ-dimensions of points in the set.

Theorem 3.29. *For any family of computable covering sets* Φ *over* $\mathbb{X}$*, for any* $\mathcal{F} \subseteq \mathbb{X}$*, we have*

$$\mathrm{cdim}_\Phi(\mathcal{F}) = \sup_{x\in\mathcal{F}} \mathrm{cdim}_\Phi(x).$$

Proof. Let s be a rational such that $s > \mathrm{cdim}_\Phi(\mathcal{F})$. There exists an effective s-dimensional Φ-null cover for $\mathcal{F}$. Therefore, for all points $x \in \mathcal{F}$, there exists an effective s-dimensional Φ-null cover for x. Hence $\sup_{x\in\mathcal{F}} \mathrm{cdim}_\Phi(x) \leq s$. Therefore, $\mathrm{cdim}_\Phi(\mathcal{F}) \geq \sup_{x\in\mathcal{F}} \mathrm{cdim}_\Phi(x)$.

In the other direction, consider any rational $s > \sup_{x\in\mathcal{F}} \mathrm{cdim}_\Phi(x)$. From Lemma 3.28, we have that there exists an $\mathcal{F}_s \supseteq \mathcal{F}$ such that $\mathrm{cdim}_\Phi(\mathcal{F}_s) \leq s$. Any s-dimensional Φ-null cover of $\mathcal{F}_s$ is also an s-dimensional Φ-null cover of $\mathcal{F}$. So, we have $\mathrm{cdim}_\Phi(\mathcal{F}) \leq s$. □

3.6.2 Effective Φ-dimension using Kolmogorov Complexity

We give an equivalent formulation of constructive Φ-dimension of a point using Kolmogorov complexity.

We define the notion of Kolmogorov complexity of a point x at precision r with respect to Φ. We denote this using $K_r(x, \Phi)$.

Definition 3.30. Given an $r \in \mathbb{N}$, and $x \in \mathbb{X}$, define

$$K_r(x, \Phi) = \min_{U\in\Phi}\{K(\delta^{-1}(U)) \mid x \in U \text{ and } |U| < 2^{-r}\}.$$

We now show a Kolmogorov complexity characterization of constructive Φ-dimension. The proof uses techniques from Lutz and Lutz [34] along with standard techniques [12], adapted to the Φ-dimension setting.

Theorem 3.31. *Given a family of computable covering sets* Φ*, over a space* $\mathbb{X}$*. For any* $x \in \mathbb{X}$*,*

$$\mathrm{cdim}_\Phi(x) = \liminf_{r\to\infty} \frac{K_r(x, \Phi)}{r}.$$

Proof. Take any $s \in \mathbb{Q}$ such that for infinitely many $r \in \mathbb{N}$, $K_r(x, \Phi) < rs + o(r)$. Consider a machine M that on input $k \in \mathbb{N}$, enumerates all strings w such that there exists an $r \in \mathbb{N}$ such that $K(w) \leq rs - k$ and $|\delta(w)| < 2^{-r}$. We have

$$\sum_{w \in M(k)} |\delta(w)|^s \leq \sum_{w \in M(k)} 2^{-(K(w)+k)} \leq 2^{-k}.$$

The last inequality follows from the Kraft inequality. Since for infinitely many $r \in \mathbb{N}$, $K_r(x, \Phi) < rs + o(r)$, we have that for all $k \in \mathbb{N}$, $x \in \bigcup_{w \in M(k,r)} \delta(w)$. Therefore $\mathrm{cdim}_\Phi(x) \leq s$.

For the other direction, take any $s \in \mathbb{Q}$ such that $s > \mathrm{cdim}_\Phi(x)$. From the definition, it follows that there exists a machine M that enumerates labels of covers $\{U_i^k\}$ of x such that $\sum_{i \in \mathbb{N}} |U_i^k|^s \leq 2^{-k}$. It follows that there are at most 2^{rs} sets U in $\{U_i^r\}$ such that $2^{-r} < |U| < 2^{-r+1}$. Consider the following description of the element U that specifies r along with the index of appearance of the set U within sets V such that $2^{-r} < |V| < 2^{-r+1}$. Such a description takes size atmost $rs + o(r)$. From this, it follows that for infinitely many r, $K_r(x, \Phi) \leq rs + o(r)$. □

3.6.3 Effective Φ-dimension using gales

In this section, we give a characterisation of constructive Φ-dimension using Φ-s-supergales. A Φ-s-supergale can be seen as a gambling strategy where the bets are placed on the covering sets from Φ. The definitions in this subsection are adaptations from the setting of *Nice covers* by Mayordomo [41].

Note crucially that the our setting does not use the c-cover property in [41], and therefore covers more classes Φ. One example being the set of continued fraction covers. Mayordomo's goal was to generalise constructive dimension to general metric spaces and therefore Nice-covers are designed to preserve effective dimension. This is important as the aim of this work is to study faithfulness, that is if coverings preserve effective dimension or not.

Definition 3.32 (Family of layerwise covering sets Φ)**.** We consider the metric space $\mathbb{X}$. A countable family of sets $\Phi = \bigcup_{n \in \mathbb{N}} \{U_i^n\}_{i \in \mathbb{N}}$, where for each $i \in \mathbb{N}, n \in \mathbb{N}$, $U_i^n \subseteq \mathbb{X}$, is called a *layerwise family of covering sets* if it satisfies the following properties:

- Increasing Monotonicity: For every $n \in \mathbb{N}$, $U \in \{U_i^n\}_{i \in \mathbb{N}}$ and $m \leq n$, there is a unique $V \in \{U_i^m\}_{i \in \mathbb{N}}$ such that $U \subseteq V$.

- Computable subsets: For every $n \in \mathbb{N}$, and $i \in \mathbb{N}$, the set $\{j \in \mathbb{N} : U_j^{n+1} \subseteq U_i^n\}$ is uniformly computable.
- Fineness : Given any $\epsilon > 0$, and $x \in \mathbb{X}$, there exists a $U \in \Phi$ such that $|U| < \epsilon$ and $x \in U$.
- Computable diameter: For all $i, n \in \mathbb{N}$, $|U_i^n|$ is computable.

When we say $|U_i^n|$ is computable, we mean there exists a total Turing machine M that on input $i, n, r \in \mathbb{N}$, outputs an $\ell \in \mathbb{Q}$ such that $||U_i^n| - \ell| \leq 2^{-r}$. The set $\{j \in \mathbb{N} : U_j^{n+1} \subseteq U_i^n\}$ is uniformly computable if there is a Turing machine which on input i, j, n decides if $U_j^{n+1} \subseteq U_i^n$.

Note that the number of sets $\{U_i^n\}$ in some level $n \in \mathbb{N}$ can also be finite and bounded by m. The definition still holds because in this case we take $U_j^n = \emptyset$ for $j > m$. Note that from Increasing monotonicity property, it follows that all elements $\{U_i^n\}$ in a particular level $n \in \mathbb{N}$ are incomparable.

Note that the notion of layerwise family of covering sets (Definition 3.32) is a refinement of family of computable covering sets (Definition 3.27). We take $\delta(< i, n >) = U_i^n$, where $<,>$ is the pairing function from $\mathbb{N} \times \mathbb{N} \to \mathbb{N}$.

Definition 3.33 (Mayordomo [41])**.** Let $\Phi = \bigcup_{n\in\mathbb{N}}\{U_i^n\}_{i\in\mathbb{N}}$ be a family of layerwise covering sets from Definition 3.32. For $s \in [0, \infty)$, a Φ-s-supergale is a function $d : \Phi \to [0, \infty)$ such that:

- $\sum_{U\in\{U_i^1\}_{i\in\mathbb{N}}} d(U)|U|^s < \infty$ and
- For all $n \in \mathbb{N}$ and all $U \in \{U_i^n\}_{i\in\mathbb{N}}$, the following condition holds:

$$d(U) \cdot |U|^s \geq \sum_{V\in\{U_i^{n+1}\}_{i\in\mathbb{N}}, V\subseteq U} d(V)|V|^s \, .$$

The following is the generalization of Kraft inequality for s-supergales from Mayordomo [41].

Lemma 3.34 (Generalisation of Kraft inequality [41])**.** *Let d be a Φ-s-supergale. Then for every $\mathcal{E} \subseteq \Phi$ such that the sets in $\mathcal{E}$ are incomparable, we have that*

$$\sum_{V\in\mathcal{E}} d(V)|V|^s \leq \sum_{U\in\{U_i^1\}_{i\in\mathbb{N}}} d(U)|U|^s.$$

Proof. Let $\sum_{U\in\{U_i^1\}_{i\in\mathbb{N}}} d(U)|U|^s = c$ for some $c \in \mathbb{R}$. It suffices to show that the Lemma holds when sets in $\mathcal{E}$ are subsets of level sets till a finite level N. That is, there exists an $N \in \mathbb{N}$ such that for all $U \in \mathcal{E}$, $U \in \{U_i^m\}$ for some $m \leq N$. In this case, we show that the lemma follows from induction. The base case when $N = 0$ is immediate. Now assume the lemma holds for $N-1$, consider $\mathcal{E}_N = \{V \in \mathcal{E} \mid V \in \{U_i^N\}\}$ and $\mathcal{E}_{<N} = \mathcal{E}\backslash\mathcal{E}_N$. From the increasing monotonicity property in Definition 3.32, we have that for all $V \in \mathcal{E}_N$, $V \subseteq W$ for a unique $W \in \{U_i^{N-1}\}$. Let $\mathcal{E}'_{N-1}$ be the collection of such W's. We have that elements in $\mathcal{E}_{<N} \cup \mathcal{E}'_{N-1}$ are incomparable and so $\sum_{V\in\mathcal{E}_{<N}\cup\mathcal{E}'_{N-1}} d(V)|V|^s \leq c$. From the gale condition in Definition 3.33, we have that for all $W \in \mathcal{E}'_{N-1}$, $d(W).|W|^s \geq \sum_{V\in\{U_i^N\},V\subseteq U} d(V)|V|^s$. □

Definition 3.35 (Mayordomo [41])**.** Let $\Phi = \bigcup_{n\in\mathbb{N}}\{U_i^n\}_{i\in\mathbb{N}}$ be a family of layerwise covering sets. Given $x \in \mathbb{X}$, a Φ-representation of x is a sequence $(U_n)_{n\in\mathbb{N}}$ such that for each $n \in \mathbb{N}$, $U_n \in \{U_i^n\}_{i\in\mathbb{N}}$ and $x \in \cap_n U_n$.

Note that the same x can have multiple Φ-representations. Given $x \in \mathbb{X}$, let $\mathcal{R}(x)$ be the set of Φ-representations of x.

Definition 3.36 (Mayordomo [41])**.** A Φ-s-supergale d succeeds on $x \in \mathbb{X}$ if there is a $(U_n)_{n\in\mathbb{N}} \in \mathcal{R}(x)$ such that $\limsup_{n\to\infty} d(U_n) = \infty$.

Equivalently, a Φ-s-supergale d succeeds on a point $x \in \mathbb{X}$ iff for every $k \in \mathbb{N}$, there exists a $U \in \Phi$ such that $x \in U$ and $d(U) > 2^k$.

Definition 3.37. The *success set* of d is $S^\infty(d) = \{x \in \mathbb{X} \mid d \text{ succeeds on x}\}$.

We use constructive Φ-s-gales for constructive Φ-dimension. For a Φ-s-gale d to be constructive, we require the gale function d to be lower semicomputable. Note that a lower semicomputable supergale actually takes as input (i, n) where $i \in \mathbb{N}, n \in \mathbb{N}$ to place bets on U_i^n. We omit this technicality in this paper and keep the domain of the gale as Φ for the sake of simplicity.

Definition 3.38. A function $d : \Phi \longrightarrow [0, \infty)$ is called *lower semicomputable* if there exists a total computable function $\hat{d} : \Phi \times \mathbb{N} \longrightarrow \mathbb{Q} \cap [0, \infty)$ such that the following two conditions hold.

- **Monotonicity** : For all $U \in \Phi$ and for all $n \in \mathbb{N}$, we have $\hat{d}(U, n) \leq \hat{d}(U, n + 1) \leq d(U)$.
- **Convergence** : For all $U \in \Phi$, $\lim_{n\to\infty} \hat{d}(U, n) = d(U)$.

For $\mathcal{F} \subseteq \mathbb{X}$, let $\hat{\mathcal{G}}_\Phi(\mathcal{F})$ denote the set of all $s \in [0, \infty)$ such that there exists a lower semicomputable Φ-s-supergale d with $\mathcal{F} \subseteq S^\infty(d)$.

We now give a Φ-s-supergale characterisation of constructive Φ-dimension.

Theorem 3.39. *Let* Φ *be a family of layerwise covering sets. For any* $\mathcal{F} \subseteq \mathbb{X}$*,*

$$\operatorname{cdim}_\Phi(\mathcal{F}) = \inf \hat{\mathcal{G}}_\Phi(\mathcal{F}).$$

Proof. By definition of Φ-dimension, we have that for any $s > \operatorname{cdim}_\Phi(\mathcal{F})$ and $r \in \mathbb{N}$, there exists an effective sequence of covers $\{U_i^r\}_{i\in\mathbb{N}}$ where each $U_i^r \in \Phi$ such that $\sum_{i\in\mathbb{N}} |U_i|^s \leq 2^{-r}$.

Given $r \in \mathbb{N}$, consider the following s-supergale $d_r : \Phi \to [0, \infty)$

$$d_r(U) = \frac{1}{|U|^s} \left(\sum_{\mathcal{V}\in\{U_i^r\}\,;\,\mathcal{V}\subset U} |\mathcal{V}|^s \right).$$

Note that $\sum_{U\in\{U_i^1\}_{i\in\mathbb{N}}} d(U)|U|^s \leq \sum_{\mathcal{V}\in\{U_i^r\}} |\mathcal{V}|^s \leq 2^{-r}$. Since $|\mathcal{V}|$ is computable, d_r is lower semicomputable when given an oracle access to A. It follows that for all $U \in \{U_i^r\}$, $d_r(U) \geq 1$. Finally define

$$d(U) = \sum_{r=0}^{\infty} 2^r d_{2r}(U).$$

Now, for all $x \in \mathcal{F}$ and $r \in \mathbb{N}$, there exists a $U \in \{U_i^{2r}\}$ such that $x \in U$. For that U, $d(U) \geq 2^r d_{2r}(U) = 2^r$. Therefore $\mathcal{F} \subseteq S^\infty(d)$ and so $\inf \mathcal{G}_\Phi(\mathcal{F}) \leq s$.

Now let $s > \inf \mathcal{G}_\Phi(\mathcal{F})$. There exists a lower semicomputable Φ-s-gale, say d that succeeds on all $X \in \mathcal{F}$.

Given an $r \in \mathbb{N}$, define

$$U_r = \{U \in \Phi \mid d(U) \geq 2^r \text{ and } \forall\, \mathcal{V} \in \Phi \text{ such that } U \subset \mathcal{V},\ d(\mathcal{V}) < 2^r\}.$$

Since $\mathcal{F} \subseteq S^\infty[d]$, it follows for all $r \in \mathbb{N}$, $\mathcal{F} \subseteq \bigcup_{U\in U_r} U$.

Note that by definition, U_r^δ is a disjoint set. Since d is an s-gale, using Lemma 3.34, it follows that

$$\sum_{U\in U_r} d(U) \cdot |U|^s \leq \infty.$$

We have that for all $U \in \mathcal{W}_r^\delta$, $d(U) \geq 2^r$. It follows that for some constant c,

$$\sum_{U \in \mathcal{W}_r^\delta} \mu^s(U) \leq c.2^{-r}.$$

It follows that $\mathrm{cdim}_\Phi(\mathcal{F}) \leq s$. □

We now use the first part of the proof to the universal c.e Φ-s-cover to get a universal c.e Φ-s-gale.

Lemma 3.40. *Let* Φ *be a family of layerwise covering sets. For any* $s \in \mathbb{Q}$*, there exists a universal* Φ*-*s*-supergale* $\tilde{d} : \Phi \to [0, \infty)$ *such that for any* $\mathcal{F} \subseteq \mathbb{X}$*, with* $\mathrm{cdim}_\Phi(\mathcal{F}) < s$ *we have* $\mathcal{F} \subseteq S^\infty(\tilde{d})$*.*

Proof. From Lemma 3.28, we have that there exists a set $\mathcal{F}_s \subseteq \mathbb{X}$ such that $\mathrm{cdim}_\Phi(\mathcal{F}_s) \leq s$ and for all $\mathcal{F} \subseteq \mathbb{X}$ such that $\mathrm{cdim}_\Phi(\mathcal{F}) < s$, we have $\mathcal{F} \subseteq \mathcal{F}_s$.

Using the proof of the first part of Theorem 3.39, it follows that there exists an s-supergale $\tilde{d} : \Phi \to [0, \infty)$ such that $\mathcal{F}_s \subseteq S^\infty[d]$. □

Note that the proof can be easily relativised with respect to an oracle $A \subseteq \Sigma^\infty$.

Lemma 3.41. *Let* Φ *be a family of layerwise covering sets. For any* $s \in \mathbb{Q}$ *and* $A \in \Sigma^\infty$*, there exists a universal* Φ*-*s*-supergale* $\tilde{d}^A : \Phi \to [0, \infty)$ *with oracle access to* A *such that for any* $\mathcal{F} \subseteq \mathbb{X}$*, with* $\mathrm{cdim}_\Phi^A(\mathcal{F}) < s$ *we have* $\mathcal{F} \subseteq S^\infty(\tilde{d}^A)$*.*

3.7 Effective Φ-dimension in Euclidean Space

We now look at Φ-covers in the Euclidean Space $\mathbb{R}^n$. We extend Definition 3.27 to have an additional property that for all U, there is a computable function which outputs a rational point arbitrarily close to a point $x \in U$.

Definition 3.42 (Family of computable covering sets over $\mathbb{R}^n$)**.** Let Φ be a family of covering sets defined over $\mathbb{X}$. We say Φ is a family of computable covering sets over if

- For all $w \in \mathbb{N}$, $|\delta(w)|$ is computable.
- There is a computable function $f_\delta : \mathbb{N} \times \mathbb{N} \to \mathbb{Q}$ such that
  for all $w \in \mathbb{N}$, $n \in N$, there exists an $x \in \delta(w)$ such that $|x, f_\delta(w, n)| \leq 2^{-n}$.

When we say $|\delta(w)|$ is computable, we mean there exists a total Turing machine M that on input $w \in \Sigma^*$ and $r \in \mathbb{N}$, outputs an $\ell \in \mathbb{Q}$ such that $||\delta(w)| - \ell| \leq 2^{-r}$.

Let Φ be a family of computable covering sets over $\mathbb{R}^n$. We now show that the constructive Φ-dimension of a set is always greater than or equal to its constructive dimension.

Theorem 3.43. *Let Φ be a family of computable covering sets over $\mathbb{R}^n$. For all $\mathcal{F} \subseteq \mathbb{R}$,*

$$\mathrm{cdim}_\Phi(\mathcal{F}) \geq \mathrm{cdim}(\mathcal{F}).$$

Proof. Take any $s \in \mathbb{Q}$ such that $s > \mathrm{cdim}_\Phi(\mathcal{F})$. From the Theorem 3.31 and Theorem 3.29, for all $x \in \mathcal{F}$ there exists infinitely many $r \in \mathbb{N}$ such that, $K_r(x, \Phi) < rs + o(r)$. Therefore, for each such r, there exists an $w \in \mathbb{N}$ and $U = \delta(w)$ such that $K(w) < rs + o(r)$ and $x \in U$ and $|U| < 2^{-r}$.

Let $q = f_\delta(w, t)$ where $t \in \mathbb{N}$ is the least number such that $|U| < 2^{-t}$. Since $|\delta(w)|$ and f_δ are computable, we have $K(q) < rs + o(r)$. There exists a $y \in U$ such that $d(y, q) < 2^{-t}$. Since $x \in U$, we have $d(x, y) < 2^{-t}$. From the triangle inequality, we have $d(x, q) < 2^{-(t+1)} \leq 2^{-(r+1)}$.

Therefore we have that for all $x \in \mathcal{F}$, for infinitely many $r \in \mathbb{N}$, $K_r(x) < rs + o(r)$. From Theorem 3.11, it follows that $\mathrm{cdim}(x) \leq s$. □

When the reverse inequality also holds, we have that cdim_Φ and cdim are the same notions. We call these family of coverings to be *faithful* to constructive dimension.

3.8 Faithfulness of Family of Coverings

We will first see the definition of Hausdorff dimension faithfulness. We then introduce the corresponding notion at the effective level, which we call constructive dimension faithfulness.

A family of covering sets Φ is said to be *faithful* with respect to Hausdorff dimension if the Φ dimension of every set in the space is the same as its Hausdorff dimension.

Definition 3.44. A family of covering sets Φ over the space $\mathbb{X}$ is said to be *faithful with respect to Hausdorff dimension* if for all $\mathcal{F} \subseteq \mathbb{X}$, $\dim_\Phi(\mathcal{F}) = \dim(\mathcal{F})$.

We extend the definition to the constructive level as well. A family of computable covering sets Φ over $\mathbb{R}^n$ is defined to be *faithful* with respect to constructive dimension if the constructive Φ dimension of every set is the same as its constructive dimension.

Definition 3.45. A family of computable covering sets Φ over $\mathbb{R}^n$ is said to be *faithful with respect to constructive dimension* if for all $\mathcal{F} \subseteq \mathbb{X}$, $\mathrm{cdim}_\Phi(\mathcal{F}) = \mathrm{cdim}(\mathcal{F})$.

The following lemma follows from Theorem 3.29. It states that constructive dimension faithfulness can be equivalently stated in terms of preservation of constructive dimensions of points in the set.

Lemma 3.46. *A family of computable covering sets* Φ *over* $\mathbb{R}^n$ *is faithful with respect to constructive dimension if and only if for all* $x \in \mathbb{R}^n$, $\mathrm{cdim}_\Phi(x) = \mathrm{cdim}(x)$.

Let Φ be a a family of computable covering sets Φ over $\mathbb{R}^n$. From Theorem 3.43, we have that Φ is not faithful for constructive dimension if and only if there exists an $x \in \mathbb{R}^n$ such that $\mathrm{cdim}_\Phi(x) > \mathrm{cdim}(x)$.

In Chapter 4, we study constructive dimension faithfulness of continued fraction covers. In Chapter 5, we study constructive dimension faithfulness of covers generated by Cantor series expansions.

Chapter 4

Non-faithfulness of Continued Fraction Dimension

4.1 Introduction

A family of covering sets Φ from a metric space $\mathbb{X}$ is *faithful* to constructive dimension if the constructive Φ-dimension of every set $\mathcal{F} \subseteq \mathbb{X}$ coincides with the constructive dimension of $\mathcal{F}$. Faithful covering sets Φ give us representations (see Definition 3.35) with respect to which constructive dimension of a real is preserved. This simplifies the calculation of effective dimensions of sets.

Nandakumar and Vishnoi [44] define the notion of effective dimension of continued fractions. They use the notion of an s-gale on the space of continued fraction sequences to define effective dimension. The setting of continued fraction dimension is interesting topologically since the space of continued fractions has a countably infinite alphabet and is non-compact. It is also interesting measure-theoretically since it forms a non-product measure space.

In this chapter, we show that continued fraction covers are not faithful for constructive dimension. We prove the following result: for every $0 < \varepsilon < 0.5$, there is a real whose effective (binary) Hausdorff dimension is less than ε while its effective continued fraction dimension is at least 0.5. Thus, surprisingly, there is a sharp gap between the effective dimension of a real and its effective continued fraction dimension.

We also use the point-to-set principle for continued fraction and show that the set of continued fraction covers are not faithful for Hausdorff dimension, giving an alternate proof of result by Peres and Torbin [48].

4.2 Preliminaries

We use $\mathbb{N}$ to denote the set of natural numbers.

We adopt the notation $[a_1, a_2, \dots]$, where $a_i \in \mathbb{N}$ for all $i \in \mathbb{N}$, for the continued fraction

$$\cfrac{1}{a_1 + \cfrac{1}{a_2 + \cdots}}$$

and similarly, we use the notation $[a_1, a_2, \dots, a_n]$ for finite continued fractions.

The set of finite continued fractions is denoted $\mathbb{N}^*$ and the set of all infinite continued fractions, as $\mathbb{N}^\infty$.

We adopt the notations given in Chapter 2 for continued fraction strings $u \in \mathbb{N}^*$ and $Y \in \mathbb{N}^\infty$. For a $v \in \mathbb{N}^*$, the *cylinder set* of v, denoted C_v, is defined as set of reals whose continued fraction expansions start with v. For a $w \in \Sigma^*$, C_w is defined similarly. λ denotes the empty string.

For a finite continued fraction string $v = [a_1, \dots a_k]$, we call k as the length of v. $\mathbb{N}^{\leq k}$ refers to the set of continued fraction strings having rank less than or equal to k. $\mathbb{N}^k$ refers to the set of continued fraction strings having length equal to k.

For an infinite continued fraction string $Y = [a_1, a_2, \dots]$, $Y \upharpoonright n$ denotes the continued fraction string $[a_1, a_2 \dots a_n]$.

For $v \in \mathbb{N}^*$, $\mu(v)$ refers to the Lebesgue measure of the continued fraction cylinder C_v. $\gamma(v)$ refers to the Gauss measure of the continued fraction cylinder C_v, defined by $\gamma(v) = \int_{C_v} \frac{1}{1+x}\, dx$. We use the same notation for a binary cylinder $w \in \Sigma^*$.

Wherever there is no scope for confusion, for a $v \in \mathbb{N}^*$, we use $\mu(v)$ and $\gamma(v)$ to represent $\mu(C_v)$ and $\gamma(C_v)$ respectively. The same holds for a $v \in \Sigma^*$. We also use the notation $\mu^s(v)$ and $\gamma^s(v)$ to denote $(\mu(v))^s$ and $(\gamma(v))^s$ respectively. For a continued fraction string $v = [a_1, \dots, a_n]$, $[v, i]$ denotes the continued fraction $[a_1, \dots a_n, i]$.

4.3 Effective Continued Fraction Dimension

Nandakumar and Vishnoi [46] formulate the notion of effective dimension of continued fractions using the notion of lower semicomputable continued fraction s-gales. Whereas a binary s-gale bets on the digits of the binary expansion of a number, a continued fraction s-gales places bets on the digits of its continued fraction expansion.

Definition 4.1 (Nandakumar, Vishnoi [46])**.** For $s \in [0, \infty)$, a continued fraction s-gale is a function $d : \mathbb{N}^* \to [0, \infty)$ such that $d(\lambda) < \infty$ and for all $w \in \mathbb{N}^*$, the following holds.

$$d(w)[\gamma(C_w)]^s = \sum_{i \in \mathbb{N}} d(wi)[\gamma(C_{wi})]^s.$$

The *success set* of d is $S^\infty(d) = \left\{ Y \in \mathbb{N}^\infty \mid \limsup_{n \to \infty} d(Y \upharpoonright n) = \infty \right\}$.

For $\mathcal{F} \subseteq \mathbb{N}^\infty$, $\mathcal{G}_{CF}(\mathcal{F})$ denotes the set of all $s \in [0, \infty)$ such that there exists a lower semicomputable continued fraction s-gale d with $\mathcal{F} \subseteq S^\infty(d)$.

Definition 4.2 (Nandakumar, Vishnoi [46])**.** The *effective continued fraction dimension* of $\mathcal{F} \subseteq \mathbb{N}^\infty$ is

$$\mathrm{cdim}_{\mathrm{CF}}(\mathcal{F}) = \inf \mathcal{G}_{CF}(\mathcal{F}).$$

The *effective continued fraction dimension* of a sequence $Y \in \mathbb{N}^\infty$ is defined by $\mathrm{cdim}_{\mathrm{CF}}(\{Y\})$, the effective continued fraction dimension of the singleton set containing Y.

4.3.1 Continued fraction dimension using Φ-covers

We now define continued fraction dimension for a real $y \in [0, 1]$ using the machinery from Chapter 3. Fix the metric space $\mathbb{X} = [0, 1]$ and let Φ_{CF} be the set of all continued fraction covers in $[0, 1]$. That is $\Phi_{\mathrm{CF}} = \bigcup_{n \in \mathbb{N}} \{C_{[a_1, a_2, \ldots a_n]}\}_{a_1 \ldots a_n \in \mathbb{N}}$.

We can see that Φ_{CF} forms a *family of layerwise covering sets* from Definition 3.32. The following is a Corollary of Theorem 3.39.

Corollary 4.3. *For any $Y \in \Sigma^\infty$, we have*

$$\mathrm{cdim}_{\mathrm{CF}}(Y) = \mathrm{cdim}_{\Phi_{\mathrm{CF}}}(y)$$

where $y \in [0, 1]$ is the real whose continued fraction expansion is Y.

In the remaining part of this chapter, we therefore abuse the notation and for a $y \in [0,1]$, use $\mathrm{cdim}_{\mathrm{CF}}(y)$ to denote $\mathrm{cdim}_{\Phi_{\mathrm{CF}}}(y)$.

From Theorem 3.22, we have the following point-to-set principle for continued fraction dimension.

Corollary 4.4. *For all* $\mathcal{F} \subseteq \mathbb{X}$, $\dim_{\mathrm{CF}}(\mathcal{F}) = \min_{A \in \Sigma^\infty} \sup_{x \in \mathcal{F}} \mathrm{cdim}^A_{\mathrm{CF}}(x)$.

We can see that Φ_{CF} forms a family of computable covering sets over $\mathbb{R}$ from Definition 3.42. The following is a Corollary of Theorem 3.43. Thus the effective continued fraction dimension of any real is always greater than or equal to its constructive dimension.

Corollary 4.5. *For any* $y \in [0,1]$, *we have*

$$\mathrm{cdim}_{\Phi_{\mathrm{CF}}}(y) \geq \mathrm{cdim}(y).$$

In the rest of this chapter, we show that there are reals for which this inequality is strict.

4.4 Lebesgue Measure of Continued fraction Cylinders

We first require some technical lemmas about the estimation of measures of a continued fraction cylinder in terms of digits of the continued fraction. Some of the bounds derived in this section may be of independent interest.

The following lemma from Falconer [14] gives the upper bound on the measure of two consecutive binary cylinders (dyadic intervals) that covers any given interval $[a,b) \subseteq [0,1]$.

Lemma 4.6 (Falconer [14])**.** *For any* $0 \leq a < b \leq 1$, *let* $\left[\frac{m}{2^k}, \frac{m+1}{2^k}\right)$, $\left[\frac{m+1}{2^k}, \frac{m+2}{2^k}\right)$, *where* $0 \leq m \leq 2^k - 2$, *be the smallest consecutive dyadic intervals whose union covers* $[a,b)$. *Then* $\frac{1}{2^k} \leq 2(b-a)$.

In combinatorial arguments, the Gauss measure is often difficult to deal with directly, and it is convenient to use the Lebesgue measure, and derive inequalities.

The following lemma is an easy consequence of the fact that the Gauss measure is absolutely continuous with respect to the Lebesgue measure (see for example, Nandakumar and Vishnoi [46]).

Lemma 4.7. *For any interval $B \subseteq (0,1)$, we have*

$$\frac{1}{2\ln 2}\mu(B) \leq \gamma(B) \leq \frac{1}{\ln 2}\mu(B).$$

The following equation, Proposition 1.2.7 in Kraaikamp and Iosifescu [24], is extremely useful in deriving an estimate for the Lebesgue measure of continued fraction cylinders. Note that the bounds for Gauss measure are not simple to derive directly.

Lemma 4.8 (Kraaikamp, Iosifescu [24])**.** *For any $v = [a_1, \ldots a_n]$ and $i \in \mathbb{N}$,*

$$\frac{\mu([v,i])}{\mu([v])} = \frac{s_n + 1}{(s_n + i)(s_n + i + 1)}$$

where $s_n = [a_n, \ldots a_1]$ is the rational corresponding to the reverse of string v.

The lemma given above gives the following bounds on the Lebesgue measure of a continued fraction cylinder in terms of the digits of the continued fraction.

Lemma 4.9. *For any $v = [a_1, \ldots a_k] \in \mathbb{N}^k$ we have*

$$\prod_{i=1}^{k} \frac{1}{(a_i+1)(a_i+2)} \;\leq\; \mu(v) \;\leq\; \prod_{i=1}^{k} \frac{2}{a_i\,(a_i+1)}$$

Proof. If $k = 1$, then $\mu(v) = \frac{1}{a_1(a_1+1)}$, therefore the lemma holds. For $k > 1$, let $v = [a_1, \ldots a_{k-1}]$ and consider $[v, a_k]$. From Lemma 4.8, it follows that

$$\mu([v,a_k]) = \frac{s_{k-1}+1}{(s_{k-1}+a_k)(s_{k-1}+a_k+1)}\;\; \mu(v),$$

where $s_{k-1} = [a_{k-1}, \ldots a_1]$. Since $s_{k-1} \in [0,1]$, it follows that

$$\frac{1}{(a_k+1)(a_k+2)}\;\mu(v) \leq \mu([v,a_k]) \leq \frac{2}{(a_k)(a_k+1)}\;\mu(v)$$

Therefore, by induction on k, the lemma holds. □

Lemma 4.10. *Let $v = [a_1 \ldots a_k] \in \mathbb{N}^*$. Then for any $a, b \in \mathbb{N}$ such that $b > a$,*

$$\mu\left(\bigcup_{i=a}^{b} [v,i]\right) \leq \frac{2}{a}\prod_{i=1}^{k}\frac{2}{a_i\,(a_i+1)}$$

Proof. From Lemma 4.8, it follows that

$$\frac{\mu([v,i])}{\mu(v)} = \frac{s_k+1}{(s_k+i)(s_k+i+1)}$$

where $s_k \in [0,1]$. Therefore

$$\mu\Big(\bigcup_{i=a}^{b}[v,i]\Big) \leq \sum_{i=a}^{b}\frac{2}{i(i+1)}\mu(v) \leq \frac{2}{a}\,\mu(v).$$

From Lemma 4.9, $\mu(v) \leq \prod_{i=1}^{k}\frac{2}{a_i\,(a_i+1)}$. Therefore this lemma holds. □

4.5 Non-faithfulness of continued fraction dimension

We now show that for any $s < 1/2$, there exists an $\mathcal{F} \subseteq [0,1]$ such that the effective continued fraction dimension of $\mathcal{F}$ is greater than 0.5, but the constructive dimension of $\mathcal{F}$ is less than s.

We now proceed to show the construction of the set $\mathcal{F}$. The definition uses a parameter $\mathbf{s}$.

We first provide the stage-wise construction of a set $\mathcal{F}_k \subseteq [0,1]$, such that for each $k \in \mathbb{N}$ $\mathcal{F}_{k+1} \subseteq \mathcal{F}_k$. We then define the set $\mathcal{F}$ using an infinite intersection of the sets $\mathcal{F}_k$.

Definition 4.11. Let $0 < \mathbf{s} < 0.5$ such that $\mathbf{s} = 1/n$ for some $n \in \mathbb{N}$. Define $a_1 = 1$. For any $k \in \mathbb{N}$, such that $k > 1$, define a_k inductively as:

$$a_k = 2\Big(k\prod_{i=1}^{k-1}100a_i\Big)^{1/\mathbf{s}}.$$

For any $k \in \mathbb{N}$, define $b_k = 50.a_k$. Take $\mathcal{F}_0 = \lambda$.

We define $\mathcal{F}$ to be the set of reals $x \in [0,1]$ for which the kth number in its continued fraction expansion of x lies between a_k and b_k.

Definition 4.12. Let $\mathcal{F}_k = \{x \in [0,1] : x \in C_{[v_1\ldots v_k]}$ such that $v_i \in [a_i, b_i]$ for $i \in 1$ to $k\}$. Define

$$\mathcal{F} = \bigcap_{k=1}^{\infty}\mathcal{F}_k.$$

We use the bounds obtained from Lemma 4.8, along with basic properties of harmonic numbers to prove the following property of measures of continued fraction sub cylinders.

Lemma 4.13. *For any $x \in \mathbb{N}^*$, $s \leq 0.5$ and $a_k, b_k \in N$ such that $b_k = 50.a_k$, $\sum_{i=a_k}^{b_k} \gamma^s([x,i]) > c\gamma^s([x])$ for some $c > 1$.*

Proof. From Lemma 4.8, it follows that

$$\begin{aligned}\sum_{a_n}^{b_k} \frac{\mu^s([x,i])}{\mu^s([x])} &= \sum_{a_n}^{b_k} \left(\frac{s_k+1}{(s_k+i)(s_k+i+1)}\right)^s \\ &\geq \sum_{a_k}^{b_k} \left(\frac{1}{(i+1)(i+2)}\right)^s\end{aligned}$$

The second inequality follows from the fact that $s_k \in [0,1]$.

Using Lemma 4.7, we get that

$$\sum_{a_n}^{b_k} \frac{\gamma^s([x,i])}{\gamma^s([x])} \geq \sum_{a_k}^{b_k} \left(\frac{1}{2(i+1)(i+2)}\right)^s$$

Putting $b_k = 50a_k$ and $s \leq 0.5$, we get

$$\begin{aligned}\sum_{a_n}^{b_k} \frac{\gamma^s([x,i])}{\gamma^s([x])} &\geq \frac{1}{2} \sum_{a_k}^{50a_k} \frac{1}{i+2} \\ &= 0.5(H(50a_k+2) - H(a_k+1)).\end{aligned}$$

where H_n is the nth Harmonic number. From the fact that $\ln n \leq H_n \leq \ln n + 1$, we have

$$\begin{aligned}H(50a_k+2) - H(a_k+1) &\geq \ln(50.a_k) - \ln(2.a_k) - 1 \\ &= \ln(25) - 1.\end{aligned}$$

The lemma holds as $0.5(\ln 25 - 1)$ is greater than 1. □

Let $\tilde{d}$ be the universal continued fraction s-gale $\tilde{d}$ from Lemma 3.40. We have that for any $Y \in \mathbb{N}^\infty$ such that $\mathrm{cdim}_{\mathrm{CF}}(Y) < s$, we have $Y \in S^\infty(\tilde{d})$.

Using the bound derived above, we show that for $s = 0.5$, $\tilde{d}$ does not succeed on some sequence in $y \in \mathcal{F}$. Using this we establish that $\mathrm{cdim}_{\mathrm{CF}}(\mathcal{F}) \geq 0.5$.

Lemma 4.14. *There exists a $y \in \mathcal{F}$ such that $\mathrm{cdim}_{\mathrm{CF}}(y) \geq 0.5$.*

Proof. Let $s = 0.5$. Consider the continued fraction s-gale $\tilde{d}$ from Lemma 3.40. It follows that for all $Y \in \mathbb{N}^*$, if $\tilde{d}$ does not succeed on any $Y \in \mathbb{N}^*$, then $\mathrm{cdim}_{\mathrm{CF}}(Y) \geq s$.

Consider any $v \in \mathbb{N}^{k-1}$. From lemma 4.13, we have that for some $c > 1$, $\sum_{i=a_k}^{b_k} \gamma^s([v,i]) > c.\gamma^s([v])$.

Now if $\forall i \in [a_k, b_k]$, $\tilde{d}([v,i]) \geq \frac{1}{c}.\tilde{d}(v)$, from the s-gale condition it follows that $\tilde{d}(v)\gamma^s(v) \geq \frac{\tilde{d}(v)}{c} \sum_{i=a_k}^{b_k} \gamma^s([v,i]) > \tilde{d}(v)\gamma^s(v)$, which is a contradiction.

Therefore, it follows that for all $v \in \mathbb{N}^*$, there exists some $n_v \in [a_k, b_k]$ such that $\tilde{d}([v,i]) < \frac{1}{c}.\tilde{d}([v])$.

Let $v_0 = \lambda$, for each $i \in \mathbb{N}$, we define $v_i = [v_{i-1}, n_{v_{i-1}}]$. Take $y = \cap_{i=1}^{\infty} C_{v_i}$ and $Y = [v_1, v_2, \dots]$. Taking $\tilde{d}(\lambda) = k$ we get $\tilde{d}(Y \upharpoonright n) < \frac{k}{c^n}$. Therefore the continued fraction s-gale $\tilde{d}$ does not succeed on Y. Hence $\mathrm{cdim}_{\mathrm{CF}}(y) \geq 0.5$. □

From this, it follows that the constructive dimension of the entire set $\mathcal{F}$ is also greater than or equal to 0.5.

Lemma 4.15. $\mathrm{cdim}_{\mathrm{CF}}(\mathcal{F}) \geq 0.5$.

Proof. From Theorem 3.29, we get that $\mathrm{cdim}_{\mathrm{CF}}(\mathcal{F}) = \sup_{y \in \mathcal{F}} \mathrm{cdim}_{\mathrm{CF}}(y)$. From Lemma 4.14, it follows that there exists a $y \in \mathcal{F}$ such that $\mathrm{cdim}_{\mathrm{CF}}(y) \geq 0.5$. Combining these two, we get that $\mathrm{cdim}_{\mathrm{CF}}(\mathcal{F}) \geq 0.5$. □

Given an oracle $A \in \Sigma^\infty$, we can relativise Lemma 4.14.

Lemma 4.16. *For any $A \in \Sigma^\infty$, there exists a $y \in \mathcal{F}$ such that* $\mathrm{cdim}^A_{\mathrm{CF}}(y) \geq 0.5$.

Proof. Let $s = 0.5$. Given any $A \in \Sigma^\infty$, consider the continued fraction s-gale $\tilde{d}^A$ from Lemma 3.41. It follows that for all $Y \in \mathbb{N}^\infty$, if $\tilde{d}^A$ does not succeed on Y, then $\mathrm{cdim}^A_{\mathrm{CF}}(Y) \geq s$. The rest of the proof follows exactly as the proof of Lemma 4.14 by replacing $\tilde{d}$ with $\tilde{d}^A$. □

Using the point-to-set principle for continued fraction dimension (Corollary 4.4), it follows that the Hausdorff dimension of $\mathcal{F}$ is greater than or equal to 0.5.

Lemma 4.17. $\dim_{\mathrm{CF}}(\mathcal{F}) \geq 0.5$.

Proof. From Lemma 4.16, we have that for all $A \in \Sigma^\infty$, there exists a $Y \in \mathcal{F}$ such that $\mathrm{cdim}^A_{\mathrm{CF}}(y) \geq 0.5$. From Corollary 4.4, we get that $\dim_{\mathrm{CF}}(\mathcal{F}) = \min\limits_{A\in\Sigma^\infty} \sup\limits_{x\in\mathcal{F}} \mathrm{cdim}^A_{\mathrm{CF}}(x)$. From this, we get that $\dim_{\mathrm{CF}}(\mathcal{F}) \geq 0.5$. □

Now we show that the effective Hausdorff dimension of all points in the set $\mathcal{F}$ is less than $\mathbf{s}$. We devise a set of effective covers S_k for $\mathcal{F}$, by combining adjacent continued fraction cylinders into a single cover.

Using the bounds derived on Lebesgue measure of continued fraction cylinders, we show that for the set of covers S_k for $\mathcal{F}$, the $\mathbf{s}$-dimensional measure shrinks arbitrarily small. From this, it follows that $\mathrm{cdim}(\mathcal{F}) \leq \mathbf{s}$.

Lemma 4.18. $\mathrm{cdim}(\mathcal{F}) \leq \mathbf{s}$.

Proof. For $k \in \mathbb{N}$, let $S_k = \{ \bigcup\limits_{i=a_k}^{b_k} [v, i] \text{ for } v \in \mathcal{F}_{k-1}\}$. The largest element in S_k is $I = [(a_1 \dots a_{k-1}, a_k), [(a_1 \dots a_{k-1}, b_k)]$. The number of elements in S_k equals $\prod_{i=1}^{k-1}(b_i - a_i)$. Additionally, we have $b_i = 50a_i$ for all $i \in \mathbb{N}$. Therefore,

$$\sum_{S\in S_k} \mu^{\mathbf{s}}(S) \leq \mu^{\mathbf{s}}(I) \prod_{i=1}^{k-1} 50a_i.$$

From Lemma 4.10, it follows that $\mu(I) \leq \frac{2}{a_k} \prod\limits_{i=1}^{k-1} \frac{2}{a_i\,(a_i+1)}$. Therefore,

$$\begin{aligned}\sum_{S\in S_k} \mu^{\mathbf{s}}(S) &\leq \Big(\frac{2}{a_k} \prod_{i=1}^{k-1} \frac{2}{a_i^2}\Big)^{\mathbf{s}} \prod_{i=1}^{k-1}(50a_i) \\ &\leq \frac{2^{\mathbf{s}}}{a_k^{\mathbf{s}}} \prod_{i=1}^{k-1} 100a_i\end{aligned}$$

Since $a_k = 2(k \prod\limits_{i=1}^{k-1} 100a_i)^{1/\mathbf{s}}$, this value is less than $1/k$.

As a_k and b_k are computable, we have that S_k is also computable. Therefore, there exists $\mathbf{s}$-dimensional constructive null covers for $\mathcal{F}$, and so we have $\mathrm{cdim}(\mathcal{F}) \leq \mathbf{s}$. □

We sum up the results from Lemma 4.14 and Lemma 4.18 into the following theorem. This shows that the set of continued fraction covers is not faithful for constructive dimension.

Theorem 4.19. *Given any* $0 < \varepsilon < 0.5$, *there exists a* $y \in [0, 1]$ *such that* $\mathrm{cdim}_{\mathrm{CF}}(y) \geq 0.5$ *and* $\mathrm{cdim}(y) < \varepsilon$.

Proof. Given $0 < \varepsilon < 0.5$, take $\mathbf{s} = 1/n$ for some large enough $n \in \mathbb{N}$ such that $\mathbf{s} < \varepsilon$. Use $\mathbf{s}$ to construct the set $\mathcal{F}$ given in Definition 4.12.

From Lemma 4.14, it follows that there exists a $Y \in \mathcal{F}$ such that $\mathrm{cdim}_{\mathrm{CF}}(Y) \geq 0.5$. Let $y \in [0,1]$ be the real corresponding to Y. Therefore, $\mathrm{cdim}_{\mathrm{CF}}(y) \geq 0.5$.

From Lemma 4.18, it follows that for all $y \in \mathcal{F}$, $\mathrm{cdim}(y) \leq \mathbf{s}$. Hence $\mathrm{cdim}(y) < \varepsilon$. □

We sum up the results from Lemma 4.17 and Lemma 4.18 to show that the set of continued fraction covers is not faithful for Hausdorff dimension. This gives an alternate proof of the result by Peres and Torbin [48].

Theorem 4.20 (Peres and Torbin [48])**.** *Given any* $0 < \varepsilon < 0.5$*, there exists a* $\mathcal{F} \subseteq [0,1]$ *such that* $\dim_{\mathrm{CF}}(\mathcal{F}) \geq 0.5$ *and* $\dim(\mathcal{F}) < \varepsilon$*.*

Proof. Given $0 < \varepsilon < 0.5$, take $\mathbf{s} = 1/n$ for some large enough $n \in \mathbb{N}$ such that $\mathbf{s} < \varepsilon$. Use $\mathbf{s}$ to construct the set $\mathcal{F}$ given in Definition 4.12.

From Lemma 4.17,we have that $\dim_{\mathrm{CF}}(\mathcal{F}) \geq 0.5$.

From Lemma 4.18, we have $\mathrm{cdim}(\mathcal{F}) \leq \mathbf{s}$. Since $\dim(\mathcal{F}) \leq \mathrm{cdim}(\mathcal{F})$, we have that $\dim(\mathcal{F}) \leq \varepsilon$. □

Chapter 5

Faithfulness of Cantor Coverings

5.1 Introduction

The Cantor series expansion, introduced by Georg Cantor [6], uses a sequence of natural numbers $Q = \{n_k\}_{k\in\mathbb{N}}$ as the terms of representation. It generalises the base-b representation: whereas base-b representation use exponentials with respect to a fixed b, $\{b^n\}_{n\in\mathbb{N}}$, the Cantor series representation $Q = \{n_k\}_{k\in\mathbb{N}}$ uses factorials $\{n_1 \dots n_k\}_{k\in\mathbb{N}}$ as the basis for representation.

In this chapter, we introduce constructive Cantor covering dimension, generated by computable Cantor series expansions. Using Kolmogorov complexity techniques, we derive a log-limit condition on the terms of the Cantor series expansion that characterizes the constructive dimension faithfulness of the associated Cantor coverings (Theorem 5.9).

We then develop a combinatorial construction of sequences having Kolmogorov complexities that grow at the same rate as a given sequence relative to any given oracle (Theorem 5.11). Using this combinatorial construction and the point-to-set principle for Φ-dimension, we show that under the setting of covers generated by Cantor series expansions, the notions of constructive faithfulness and Hausdorff dimension faithfulness are equivalent (Theorem 5.16). This gives an information theoretic proof of the result by Albeverio, Ivanenko, Lebid, and Torbin [1] on the Hausdorff dimension faithfulness of the Cantor series expansion.

5.2 Constructive Cantor Covering Dimension

5.2.1 Cantor coverings over unit interval

We consider the faithfulness of a family of coverings generated by the computable Cantor series expansion [6]. We call such class of coverings *Cantor coverings*.

Given a sequence $Q = \{n_k\}_{k\in\mathbb{N}}$ with $n_k \in \mathbb{N} \setminus \{1\}$, the expression

$$x = \sum_{k=1}^{\infty} \frac{\alpha_k}{n_1.n_2 \dots n_k}$$

where $\alpha_k \in [n_k]$ is called the Cantor series expansion of the real number $x \in [0, 1]$.

Definition 5.1 (Cantor Coverings Φ_Q)**.** The class of Cantor coverings Φ_Q over the space $\mathbb{X} = [0, 1]$ generated by the Cantor series expansion $Q = \{n_k\}_{k\in\mathbb{N}}$ is the set of intervals

$$\bigcup_{k\in\mathbb{N}} \{[\frac{m}{n_0.n_1.n_2 \dots n_k}, \frac{m+1}{n_0.n_1.n_2 \dots n_k}]\}_{m\in[n_0.n_1.n_2 \dots n_k]}$$

with n_0 taken as 1.

Definition 5.2 (Computable Cantor Coverings)**.** The Cantor series expansion $Q = \{n_k\}_{k\in\mathbb{N}}$ is said to be computable if there exists a machine that generates n_k given k. We call the class of Cantor coverings Φ_Q generated by a computable Cantor series expansion Q a class of *computable Cantor coverings* over $\mathbb{X} = [0, 1]$.

Definition 5.3. For any $\mathcal{F} \subseteq [0, 1]$, the effective Cantor covering dimension of the Cantor series expansion $Q = \{n_k\}_{k\in\mathbb{N}}$ is

$$\mathrm{cdim}_{\Phi_Q}(\mathcal{F}) = \mathrm{cdim}_{\Phi}(\mathcal{F})$$

with $\Phi = \Phi_Q$. See Definition 3.20 for the Definition of cdim_{Φ}.

From Theorem 3.22, we have the following point-to-set principle for Cantor covering dimension.

Corollary 5.4. *For all $\mathcal{F} \subseteq [0, 1]$ and for all computable Cantor coverings Φ_Q,*

$$\dim_{\Phi_Q}(\mathcal{F}) = \min_{A\subseteq\mathbb{N}} \sup_{x\in\mathcal{F}} \mathrm{cdim}^{A}_{\Phi_Q}(x).$$

5.2.2 Kolmogorov complexity characterization of Cantor series dimension

We use Theorem 3.31 to show a Kolmogorov complexity characterization of constructive Φ-dimension for computable Cantor coverings, based on the terms appearing in the Cantor series expansion.

Theorem 5.5. *For any $x \in \mathbb{X}$, and any computable Cantor coverings Φ_Q generated by $Q = \{n_k\}_{k\in\mathbb{N}}$,*

$$\mathrm{cdim}_{\Phi_Q}(x) = \liminf_{k\to\infty} \frac{K(X \upharpoonright m_k)}{m_k}$$

where X is a binary expansion of x and $m_k = \lfloor \log_2(n_1.n_2 \dots n_k) \rfloor$.

Proof. From Theorem 3.31, for any $s > \mathrm{cdim}_{\Phi_Q}(x)$, we have that for infinitely many $r \in \mathbb{N}$, there exists a $U \in \Phi_Q$ such that $|U| < 2^{-r}$ and $x \in U$ and $K(U) \leq sr$. Let $|U| = n_0 \dots n_k$ for some $k \in \mathbb{N}$ and let $m_k = \lfloor \log_2(n_1.n_2 \dots n_k) \rfloor$.

We now goon to show that $K(X \upharpoonright m_k) \leq s \cdot m_k + o(m_k)$. Given an cover $U \in \Phi_Q$, there are two dyadic intervals of size less than $|U|$ which can cover U. We use the short description of U to produce a short description of the binary cover that contains x.

Since $|U| \leq \frac{1}{2^{m_k}}$, we have that $x \in [\frac{j+e}{2^{m_k}}, \frac{j+e+1}{2^{m_k}})$ for some $e \in \{-1, 0, 1\}$, and therefore $X \upharpoonright m_k = \frac{j+e}{2^{m_k}}$. Since $|U| > \frac{1}{2.2^{m_k}}$ and $|U| < 2^{-r}$, we have $m_k + 1 > r$. It can be seen that

$$K(X \upharpoonright m_k) \leq K(U) + o(m_k) \leq rs + o(m_k) \leq s \cdot m_k + o(m_k).$$

Towards the other direction, assume that for infinitely many $k \in \mathbb{N}$, $K(X \upharpoonright m_k) \leq s \cdot m_k + o(m_k)$. We use the short description of $X \upharpoonright m_k$ to produce a short description of the Cantor covering $U \in \Phi_Q$ such that $x \in U$ and $|U| < 2^{-m_k}$. We produce $\alpha_1 \dots \alpha_k$, where $\alpha_i \in [n_i]$ and $U = [\frac{a_1 \dots \alpha_k}{n_1 \dots n_k}, \frac{a_1 \dots \alpha_k + 1}{n_1 \dots 2^{n_k}}]$. Therefore $K(U) \leq s \cdot m_k + o(m_k)$. As $|U| < 2^{-m_k}$, we have $\mathrm{cdim}_{\Phi_Q}(x) \leq s$.

□

From Theorem 5.5, when the Kolmogorov complexities of any two $X, Y \in \Sigma^\infty$ align over all finite prefixes, the constructive Φ-dimensions also become equal.

Lemma 5.6. *For any $x, y \in \mathbb{X}$, $A, B \subseteq \mathbb{N}$ and any class of computable Cantor coverings Φ, if for all n, $|K^A(X \upharpoonright n) - K^B(Y \upharpoonright n)| = o(n)$, then $\mathrm{cdim}^A(x) = \mathrm{cdim}^B(y)$ and $\mathrm{cdim}^A_\Phi(x) = \mathrm{cdim}^B_\Phi(y)$. Here X and Y are the binary expansions of x and y respectively.*

5.3 Constructive Faithfulness of Cantor Coverings

In this section, we define the notion of faithfulness of a class of coverings towards constructive and Hausdorff Dimension. For Cantor coverings, we characterise the constructive dimension faithfulness using a log-limit condition of the terms appearing in the Cantor series expansion.

Then, using the point-to-set principle and properties of Kolmogorov complexity, we show that when the class of covers Φ is generated by computable Cantor series expansions, the faithfulness at the Hausdorff and constructive levels are equivalent notions.

Lemma 5.7. *A family of Cantor coverings Φ_Q generated by $Q = \{n_k\}_{k\in\mathbb{N}}$ is faithful with respect to constructive dimension if*

$$\lim_{k\to\infty} \frac{\log n_k}{\log n_1.n_2 \dots n_{k-1}} = 0.$$

Proof. Given any $x \in [0,1]$, we have that $\mathrm{cdim}_{\Phi_Q}(x) = \liminf_{k\to\infty} \frac{K(X\upharpoonright m_k)}{m_k}$ where X is a binary expansion of x and $m_k = \lfloor \log_2(n_1.n_2 \dots n_k) \rfloor$. Given $X \upharpoonright m_k$, and for any $\ell \in \mathbb{N}$, $X \upharpoonright m_k + \ell$ can be produced by supplying the additional $(m_k + \ell - m_k)$ bits of $X \upharpoonright m_k + \ell$. Therefore,

$$K(X \upharpoonright m_k + \ell) \leq K(X \upharpoonright m_k) + \ell + o(m_k + \ell).$$

Also $X \upharpoonright m_k$ can be computed given $X \upharpoonright m_k + \ell$ and m_k by trimming to m_k length. Therefore,

$$K(X \upharpoonright m_k) \leq K(X \upharpoonright m_k + \ell) + O(\log m_k).$$

If $\lim_{k\to\infty} \frac{\log n_k}{\log n_1.n_2 \dots n_{k-1}} = 0$, we have that $\lim_{k\to\infty} \frac{m_{k+1} - m_k}{m_k} = 0$.

Therefore, for all $m_k < \ell < m_{k+1}$, we have $\lim_{k\to\infty} \frac{\ell}{m_k} = 0$

Now,

$$\begin{aligned} \frac{K(X \upharpoonright m_k + \ell)}{m_k + \ell} - \frac{K(X \upharpoonright m_k)}{m_k} &\leq \frac{K(X \upharpoonright m_k) + \ell + o(m_k + \ell)}{m_k + \ell} - \frac{K(X \upharpoonright m_k)}{m_k} \\ &\leq \frac{\ell}{m_k} + \frac{o(m_k + \ell)}{m_k + \ell}. \end{aligned}$$

And,

$$\begin{aligned}\frac{K(X \upharpoonright m_k)}{m_k} - \frac{K(X \upharpoonright m_k+\ell)}{m_k+\ell} &\leq \frac{K(X \upharpoonright m_k+\ell)+o(m_k)}{m_k} - \frac{K(X \upharpoonright m_k+\ell)}{m_k+\ell} \\ &\leq \left(\frac{K(X \upharpoonright m_k+\ell)}{m_k+\ell} \cdot \frac{\ell}{m_k}\right) + \frac{o(m_k)}{m_k} \\ &\leq \frac{\ell}{m_k} + \frac{o(m_k)}{m_k}.\end{aligned}$$

From this, it follows that $\lim_{k\to\infty} \frac{K(X\upharpoonright m_k+\ell)}{m_k+\ell} - \frac{K(X\upharpoonright m_k)}{m_k} = 0$ where $m_k < \ell < m_{k+1}$.

Therefore, we have that for all $x \in [0,1]$, $\mathrm{cdim}_{\Phi_Q}(x) = \mathrm{cdim}(x)$.

□

Lemma 5.8. *A family of Cantor coverings Φ_Q generated by $Q = \{n_k\}_{k\in\mathbb{N}}$ is faithful with respect to constructive dimension only if*

$$\lim_{k\to\infty} \frac{\log n_k}{\log n_1.n_2 \ldots n_{k-1}} = 0.$$

Proof. Fix a rational $s < 1$. For a Cantor Covering Φ_Q which does not satisfy the condition, we construct an $X \in \Sigma^\infty$ such that $\mathrm{cdim}(X) < s$ and $\mathrm{cdim}_{\Phi_Q}(X) = s$.

For all $k \in \mathbb{N}$, let $m_k = \lfloor \log_2(n_1.n_2 \ldots n_k) \rfloor$. Let $s = p/q$ for some $p, q \in \mathbb{N}$. Let $R_s \in \Sigma^\infty$ be the sequence obtained by diluting p bits of a Martin-Löf random with $q-p$ bits of 0's. We see that $\lim_{n\to\infty} K(R_s \upharpoonright n)/n = s$.

Let $\limsup_{k\to\infty} \frac{m_{k+1}-m_k}{m_k} > \epsilon$ for some rational $\epsilon > 0$. Let $\{k_i\}_{i\in\mathbb{N}}$ be the corresponding indices.

We construct X stage wise ensuring that the following requirements are satisfied.

1. For all $k \in \mathbb{N}$, $K[X : m_k] = s \cdot m_k \pm o(m_k)$.
2. For some $\epsilon' > 0$, for all $i \in \mathbb{N}$, some $m_{k_i} < m''_{k_i} < m_{k_i+1}$,

$$K(X \upharpoonright m''_{k_i}) \leq (s-\epsilon') \cdot m''_{k_i}.$$

Condition 1 guarantees that $\mathrm{cdim}_\Phi(X) = s$. Condition 2 guarantees that $\mathrm{cdim}(X) \leq s-\epsilon'$, thereby ensuring that Φ is not faithful for constructive dimension.

We provide the stage wise construction of X. We start with $X_0 = \lambda$. At the ith stage, we have fixed $X[: m_{k_{i-1}+1}]$. Fill $X[m_{k_{i-1}+1} : m_{k_i}]$ from the next bits of R_s. This ensures Condition 1 is met at start of stage i. For the sake of notational convenience, we use m_k to denote m_{k_i}.

We are given that $m_{k+1} > (1+\epsilon)m_k$. Let $m'_k = (1+\epsilon)m_k$.

We choose a value m''_k such that $m_k < m''_k < m'_k$. At the ith stage, fill $X[m_k : m''_k]$ with 0 and $X[m''_k : m'_k]$ from a Martin-Löf random source R, independent of all other sources used so far.

Using the symmetry of information of Kolmogorov complexity (see [28]), it follows that upto $o(m_k)$ terms,

$$K(X \upharpoonright m'_k) = m'_k - m''_k + s \cdot m_k.$$

We choose m''_k such that $m'_k - m''_k = s(m'_k - m_k)$. This ensures that $K(X \upharpoonright m'_k) = s \cdot m'_k$.

Finally, we set $X[m'_k : m_{k+1}]$ using the subsequent bits of R_s, ensuring that Condition 1 is satisfied at the end of stage i.

Analysing the Kolmogorov complexity at m''_k, we have

$$K(X \upharpoonright m''_k) \leq s \cdot m_k + o(m''_k).$$

Substituting the value of m''_k and ignoring $o()$ terms, we get

$$\begin{aligned}\frac{K(X \upharpoonright m''_k)}{m''_k} &\leq \frac{s \cdot m_k}{m'_k(1-s) + s \cdot m_k} \\ &= \frac{s \cdot m_k}{m_k(s + (1+\epsilon)(1-s))} \\ &= \frac{s}{1+\epsilon(1-s)}.\end{aligned}$$

Since $\epsilon(1-s) > 0$, we have that Condition 2 is satisfied.

Therefore, we have $\mathrm{cdim}(X) < s$ and $\mathrm{cdim}_{\Phi_Q}(X) = s$. □

Theorem 5.9. *A family of Cantor coverings Φ_Q generated by $Q = \{n_k\}_{k\in\mathbb{N}}$ is faithful with respect to constructive dimension if and only if*

$$\lim_{k\to\infty} \frac{\log n_k}{\log n_1.n_2 \ldots n_{k-1}} = 0. \tag{5.1}$$

The Cantor series expansion is a generalization of the base-b representation, which is the special case when $n_k = b$ for all $k \in \mathbb{N}$. That is $Q_b = \{b\}_{n\in\mathbb{N}}$. Since the condition in Theorem 5.9 is satisfied by Q_b for any $b \in \mathbb{N}$, we have the following result by Hitchcock and Mayordomo about the base invariance of constructive dimension.

Corollary 5.10 (Hitchcock and Mayordomo [22])**.** *For any $x \in [0,1]$ and $k, l \in \mathbb{N} \setminus \{1\}$, $\mathrm{cdim}_{(k)}(x) = \mathrm{cdim}_{(l)}(x)$. where $\mathrm{cdim}_{(k)}(x)$ represents the constructive dimension of x with respect to its base-k representation.*

Note that condition (5.1) classifies the Cantor series expansions on the basis of constructive dimension faithfulness. As an example, when $n_k = 2^k$, condition (5.1) holds, and therefore $Q = \{2^k\}_{k\in\mathbb{N}}$ is faithful for constructive dimension. However, when $n_k = 2^{2^k}$, condition (5.1) does not hold, and therefore $Q = \{2^{2^k}\}_{k\in\mathbb{N}}$ is not faithful for constructive dimension.

5.4 Kolmogorov Complexity Construction

In this section we give a technical construction which is crucial in proving the results in section 5.5. Theorem 5.11 says that given an infinite sequence X and an oracle A, for any oracle B, there exists a sequence Y whose relativised Kolmogorov complexity (of prefixes) with respect to B is similar to the relativised Kolmogorov complexity (of prefixes) of X with respect to A.

Theorem 5.11. *For all $X \in \Sigma^\infty$ and $A \in \Sigma^\infty$, given a $B \in \Sigma^\infty$, there exists a $Y \in \Sigma^\infty$ such that for all $n \in \mathbb{N}$, $|K^A(X \upharpoonright n) - K^B(Y \upharpoonright n)| = o(n)$ and $\mathrm{cdim}^B(Y) = \mathrm{cdim}(Y)$.*

Proof. We construct Y in a stage-wise manner. We start with the empty string $\sigma_0 = \lambda$ and define finite strings σ_1, σ_2, $\sigma_3 \ldots$ inductively. Finally we set $Y = \sigma_0\sigma_1\sigma_2\sigma_3\ldots$.

There exists a constant c_K such that for sufficiently large n, there exists some $w \in \Sigma^n$ with, $K^B(w) \geq n - c_K$. Let M be any sufficiently large natural number such that $\log M \geq c_K$. For any $m \geq 1$, we do the following in stage m. If $m = 1$ let $s_1 = 0$ and $e_1 = M^2$. Otherwise we set $s_m = (M + m - 1)^2$ and $e_m = (M + m)^2$.

If $K^A(X[s_m : e_m] \mid X[0 : s_m - 1]) \geq \log(M + m)$, then let $\sigma_m = w_m z_m$ where w_m is any $K^A(X[s_m : e_m] \mid X[0 : s_m - 1])$-length string such that, $K^B(w_m \mid \sigma_0\sigma_1 \ldots \sigma_{m-1}) \geq K^A(X[s_m : e_m] \mid X[0 : s_m - 1]) - c_K$ and z_m is the string consisting of $e_m - s_m - K^A(X[s_m : e_m] \mid X[0 : s_m - 1])$ number of zeroes. If $K^A(X[s_m : e_m] \mid X[0 : s_m - 1]) < \log(M + m)$

then we let σ_m equal to the string consisting of $e_m - s_m$ number of zeroes. We define $Y = \sigma_0\sigma_1\sigma_2\sigma_3\ldots$. We show that Y satisfies the required properties.

To prove the first property, it is enough to show that

$$\left|K^A(X \upharpoonright (M+m-1)^2) - K^B(Y \upharpoonright (M+m-1)^2)\right| \leq c(M+m-1)\log(M+m-1)^2.$$

In the above $c = c_K + 2 + c^A_{sym} + c^B_{sym}$ where c^A_{sym} and c^B_{sym} are the constants from the asymptotic error term in the symmetry of information of Kolmogorov complexity with oracles A and B respectively (see Theorem 3.10.2 in [12]). The above statement implies that

$$\left|K^A(X \upharpoonright n) - K^B(Y \upharpoonright n)\right| \leq 4c \cdot \sqrt{n}\log n$$

which proves the first part. The proof follows by induction on n. Consider the case when $m = 1$. By the choice of σ_1 and the symmetry of information,

$$K^B(Y \upharpoonright M^2) = K^B(w_1) + K^B(z_1 \mid w_1) + O(\log M^2)$$

Now,

$$\left|K^A(X \upharpoonright M^2) - K^B(Y \upharpoonright M^2)\right| \leq c_K + \log M^2 + c^B_{sym}\log M^2 \leq c\log M^2$$

which proves the base case. Now, assume the statement for $n \geq 1$. Using the symmetry of information,

$$\begin{aligned} K^B(Y \upharpoonright (M+m)^2) = {} & K^B(Y \upharpoonright (M+m-1)^2) \\ & + K^B(w_m \mid Y \upharpoonright (M+m-1)^2) \\ & + K^B(z_m \mid Y \upharpoonright (M+m-1)^2, w_m) + O(\log(M+m)^2) \end{aligned}$$

Similarly,

$$\begin{aligned} K^A(X \upharpoonright (M+m)^2) = {} & K^A(X \upharpoonright (M+m-1)^2) \\ & + K^A(X[(M+m-1^2) : (M+m)^2] \mid X \upharpoonright (M+m-1)^2) \\ & + O(\log(M+m)^2). \end{aligned}$$

Hence, using the induction hypothesis,

$$\begin{aligned}
&\left|K^A(X \upharpoonright (M+m-1)^2) - K^B(Y \upharpoonright (M+m-1)^2)\right| \\
&\leq c(M+m-1)\log(M+m-1)^2 + c_K \\
&+ 2\log(M+m)^2 + c^A_{sym}\log(M+m)^2 + c^B_{sym}\log(M+m)^2 \\
&\leq c(M+m-1)\log(M+m)^2 + (c_K + 2 + c^A_{sym} + c^B_{sym})\log(M+m)^2 \\
&= c(M+m)\log(M+m)^2.
\end{aligned}$$

Now, we prove the second part. It is enough to show that,

$$\left|K(Y \upharpoonright (M+m-1)^2) - K^B(Y \upharpoonright (M+m-1)^2)\right| \leq c'(M+m-1)\log(M+m-1)^2.$$

for every $m \geq 1$. Here $c' = c_K + 2 + c_{sym} + c^B_{sym}$ where c_{sym} is the constant in symmetry of information for non-relativised Kolmogorov complexity. This implies that

$$\left|K(Y \upharpoonright n) - K^B(Y \upharpoonright n)\right| \leq 4c' \cdot \sqrt{n}\log n$$

which gives us the required conclusion. Consider the case when $m = 1$. Now,

$$K(Y \upharpoonright M^2) = K(w_1) + K(z_1 \mid w_1) + O(\log M^2).$$

By the choice of w_1, it is easy to verify that

$$\left|K(w_1) - K^B(w_1)\right| \leq c_K.$$

Therefore,

$$\left|K(Y \upharpoonright M^2) - K^B(Y \upharpoonright M^2)\right| \leq c_K + 2\log M^2 + c^B_{sym}\log M^2 + c_{sym}\log M^2$$

which proves the base case. Now, assume the statement for $m \geq 1$. We have,

$$\begin{aligned}
K^B(Y \upharpoonright (M+m)^2) = {} & K^B(Y \upharpoonright (M+m-1)^2) \\
& + K^B(w_m \mid Y \upharpoonright (M+m-1)^2) \\
& + K^B(z_m \mid Y \upharpoonright (M+m-1)^2, w_m) + O(\log(M+m)^2).
\end{aligned}$$

Similarly,

$$\begin{aligned} K(Y \upharpoonright (M+m)^2) &= K(Y \upharpoonright (M+m-1)^2) \\ &\quad + K(w_m \mid Y \upharpoonright (M+m-1)^2) \\ &\quad + K(z_m \mid Y \upharpoonright (M+m-1)^2, w_m) + O(\log(M+m)^2) \end{aligned}$$

By the choice of w_m,

$$\left|K(w_m \mid Y \upharpoonright (M+m-1)^2) - K^B(w_m \mid Y \upharpoonright (M+m-1)^2)\right| \leq c_K$$

Therefore using the induction hypothesis,

$$\begin{aligned} &\left|K(Y \upharpoonright (M+m)^2) - K^B(Y \upharpoonright (M+m)^2)\right| \\ &\leq c'(M+m-1)\log(M+m-1)^2 + c_K \\ &+ 2\log(M+m)^2 + c_{sym}\log(M+m)^2 + c^B_{sym}\log(M+m)^2 \\ &\leq c'(M+m-1)\log(M+m)^2 + (c_K + 2 + c_{sym} + c^B_{sym})\log(M+m)^2 \\ &= c'(M+m)\log(M+m)^2. \end{aligned}$$

The proof of the second part is thus complete. □

5.5 Equivalence of Faithfulness of Cantor Coverings at Constructive and Hausdorff Levels

We first show that if a class of computable Cantor coverings Φ is faithful with respect to constructive dimension, then Φ is also faithful with respect to Hausdorff dimension.

Lemma 5.12. *For any class of computable Cantor coverings Φ, if for all $\mathcal{F} \subseteq \mathbb{X}$, $\mathrm{cdim}(\mathcal{F}) = \mathrm{cdim}_\Phi(\mathcal{F})$, then for all $\mathcal{F} \subseteq \mathbb{X}$, $\dim(\mathcal{F}) = \dim_\Phi(\mathcal{F})$.*

Proof. We first show that if $\mathrm{cdim}(\mathcal{F}) = \mathrm{cdim}_\Phi(\mathcal{F})$ for all $\mathcal{F} \subseteq \mathbb{X}$, then for all $A \subseteq \mathbb{N}$ and $x \in [0,1] \setminus \mathbb{Q}$, $\mathrm{cdim}^A(x) = \mathrm{cdim}^A_\Phi(x)$.

Let $B = \emptyset$. From Theorem 5.11, we have that for all $x \in [0,1]$, and $A \subseteq \mathbb{N}$, there exists a $y \in [0,1]$, such that for all $n \in \mathbb{N}$, $|K^A(X \upharpoonright n) - K(Y \upharpoonright n)| = o(n)$. Here X and Y are binary expansions of x and y respectively. Therefore from Lemma 5.6, we have that $\mathrm{cdim}^A(x) = \mathrm{cdim}(y)$ and $\mathrm{cdim}^A_\Phi(x) = \mathrm{cdim}_\Phi(y)$.

Since Φ is faithful with respect to constructive dimension, we have that $\mathrm{cdim}(y) = \mathrm{cdim}_\Phi(y)$. Therefore we have that $\mathrm{cdim}^A(x) = \mathrm{cdim}^A_\Phi(x)$.

Let $\mathcal{F} \subset \mathbb{X}$ be arbitrary. From the point-to-set principle for dimension of Cantor Coverings (Corollary 5.4), we have that

$$\dim_\Phi(\mathcal{F}) = \min_{A\subseteq\mathbb{N}} \sup_{x\in\mathcal{F}} \mathrm{cdim}^A_\Phi(x) = \min_{A\subseteq\mathbb{N}} \sup_{x\in\mathcal{F}} \mathrm{cdim}^A(x) = \dim(\mathcal{F}).$$

The last equality follows from the point-to-set principle for Hausdorff dimension (Corollary 3.26). □

To prove the converse, we require the construction of set $\mathcal{I}_s$ that contains all points in $\mathbb{X}$ having constructive dimension equal to s.

Definition 5.13. Given $s \in [0, \infty)$, define $\mathcal{I}_s = \{x \in \mathbb{X} \mid \mathrm{cdim}(x) = s\}$.

Lutz [31] showed that the Hausdorff dimension of $\mathcal{I}_s$ is equal to s. We provide a simple alternate proof of this using the point-to-set principle.

Lemma 5.14 (Lutz [31])**.** *For all $s \in [0, 1]$, $\dim(\mathcal{I}_s) = s$.*

Proof. Since for all $x \in \mathcal{I}_s$, by definition $\mathrm{cdim}(x) = s$, from [31], we have that $\mathrm{cdim}(I_s) = \sup_{x\in I_s} \mathrm{cdim}(x)$, and so we have $\mathrm{cdim}(\mathcal{I}_s) = s$. From [31], we have that $\dim(\mathcal{I}_s) \leq \mathrm{cdim}(\mathcal{I}_s)$, from which it follows that $\dim(\mathcal{I}_s) \leq s$.

Consider any $x \in \mathcal{I}_s$. We have that $\mathrm{cdim}(x) = s$. Let $A = \emptyset$. From Theorem 5.11, we have that for all $B \subseteq \mathbb{N}$, there exists a $y \in [0, 1]$ such that for all $n \in \mathbb{N}$, $|K^A(X \upharpoonright n) - K^B(Y \upharpoonright n)| = o(n)$ having $\mathrm{cdim}^B(Y) = \mathrm{cdim}(Y)$. Here X and Y are binary expansions of x and y respectively. From Lemma 5.6, we have that $\mathrm{cdim}^B(y) = \mathrm{cdim}^A(x) = \mathrm{cdim}(x) = s$.

Therefore, we have that for all $B \subseteq \mathbb{N}$, there exits a $y \in \mathbb{X}$ such that $\mathrm{cdim}(y) = \mathrm{cdim}^B(y) = s$. Hence $Y \in \mathcal{I}_s$ and therefore from the point-to-set principle for Hausdorff dimension (Corollary 3.26), we have that $\dim(\mathcal{I}_s) = \min_{B\subseteq\mathbb{N}} \sup_{y\in\mathcal{I}_s} \mathrm{cdim}^B(y) \geq s$. □

We now show that if a class of computable Cantor coverings Φ is faithful with respect to Hausdorff dimension, then Φ is also faithful with respect to constructive dimension.

Lemma 5.15. *For any class of computable Cantor coverings Φ, if for all $\mathcal{F} \subseteq \mathbb{X}$, $\dim(\mathcal{F}) = \dim_\Phi(\mathcal{F})$, then for all $\mathcal{F} \subseteq \mathbb{X}$, $\mathrm{cdim}(\mathcal{F}) = \mathrm{cdim}_\Phi(\mathcal{F})$.*

Proof. Given an $x \in \mathbb{X}$ having $\mathrm{cdim}(x) = s$, consider the set $\mathcal{I}_s$ from Definition 5.13. From Lemma 5.14, we have $\dim(\mathcal{I}_s) = s$. Since Φ is faithful with respect to Hausdorff dimension, we have that $\dim_\Phi(\mathcal{I}_s) = s$.

We now go on to show that for any $x \in \mathbb{X}$ having cdim $= s$ (or equivalently, for any $x \in \mathcal{I}_s$), $\mathrm{cdim}_\Phi(x) \leq s$. This along with Lemma 3.43 shows that $\mathrm{cdim}(x) = \mathrm{cdim}_\Phi(x)$. From Lemma 3.46, we have that for all $\mathcal{F} \subseteq \mathbb{X}$, $\mathrm{cdim}(\mathcal{F}) = \mathrm{cdim}_\Phi(\mathcal{F})$.

We first show that for all $B \subseteq \mathbb{N}$, there exists a $y \in \mathcal{I}_s$ having $\mathrm{cdim}_\Phi^B(y) = \mathrm{cdim}_\Phi(x)$. Let $A = \emptyset$. From Theorem 5.11, we have that for all $x \in [0,1]$, and $B \subseteq \mathbb{N}$, there exists a $y \in [0,1]$, such that for all $n \in \mathbb{N}$, $|K^A(X \upharpoonright n) - K^B(Y \upharpoonright n)| = o(n)$. Here X and Y are binary expansions of x and y respectively. Setting $A = \emptyset$ and using Lemma 5.6, we have that $\mathrm{cdim}_\Phi(x) = \mathrm{cdim}_\Phi^B(y)$.

We now show that $\dim_\Phi(\mathcal{I}_s) \geq \mathrm{cdim}_\Phi(X)$. From the point-to-set principle for dimension of Cantor Coverings (Corollary 5.4), we have that $\dim_\Phi(\mathcal{I}_s) = \min\limits_{B \subseteq \mathbb{N}} \sup\limits_{y \in \mathcal{I}_s} \mathrm{cdim}_\Phi^B(y)$. From the argument given above we have that for all $B \subseteq \mathbb{N}$, there exists a $y \in \mathcal{I}_s$ having $\mathrm{cdim}_\Phi^B(y) = \mathrm{cdim}_\Phi(x)$. So it follows that $\dim_\Phi(\mathcal{I}_s) \geq \mathrm{cdim}_\Phi(x)$. Since we have already shown that $\dim_\Phi(\mathcal{I}_s) = s$, it follows that $\mathrm{cdim}_\Phi(x) \leq s$. □

Therefore, we have the following theorem which states that for the classes of Cantor coverings Φ, faithfulness with respect to Hausdorff and constructive dimensions are equivalent.

Theorem 5.16. *For any class of computable Cantor coverings Φ,*

$$\forall \mathcal{F} \subseteq \mathbb{X}\ ;\ \dim(\mathcal{F}) = \dim_\Phi(\mathcal{F}) \iff \forall \mathcal{F} \subseteq \mathbb{X}\ ;\ \mathrm{cdim}(\mathcal{F}) = \mathrm{cdim}_\Phi(\mathcal{F}).$$

We have the log-limit condition for constructive dimension faithfulness of Cantor coverings (Theorem 5.9). We also have the equivalence of faithfulness of Cantor coverings at the classical and constructive levels (Theorem 5.16). Using these two, we give an alternate, information theoretic proof of the result by Albeverio, Ivanenko, Lebid and Torbin [1] that the Hausdorff dimension faithfulness of Cantor coverings can be determined using same log-limit condition over the terms n_k in Q.

Theorem 5.17 (Albeverio, Ivanenko, Lebid and Torbin [1])**.** *A family of computable Cantor coverings Φ_Q generated by $Q = \{n_k\}_{k \in \mathbb{N}}$ is faithful with respect to Hausdorff dimension if and only if*

$$\lim_{k \to \infty} \frac{\log n_k}{\log n_1 \cdot n_2 \ldots n_{k-1}} = 0.$$

Chapter 6

One-way functions and Polynomial-time Dimension

6.1 Introduction

In this chapter, we study the notion of constructive dimension of infinite binary sequences, with feasible resource bounds placed on the computation.

Polynomial-time dimension (denoted $\mathrm{cdim_P}$) quantifies the density of information of infinite sequences measured using polynomial time betting algorithms called s-gales. An alternate quantification of the notion of polynomial time density of information is using polynomial-time Kolmogorov complexity rate (denoted $\mathcal{K}_{\mathrm{poly}}$). Hitchcock and Vinodchandran in 2004 [20] show that the polynomial-time dimension ($\mathrm{cdim_P}$), is greater than or equal to the asymptotic lower rate of polynomial-time Kolmogorov complexity ($\mathcal{K}_{\mathrm{poly}}$). They asked ([20], see also Stull [53]) whether the converse relationship, $\mathrm{cdim_P} \leq \mathcal{K}_{\mathrm{poly}}$ also holds. This question has thus far resisted resolution.

In this chapter, we address an this open problem. Using the assumption that one-way functions exist, we show that the existence of one-way functions implies existence of sequences for which $\mathcal{K}_{\mathrm{poly}}(X) < \mathrm{cdim_P}(X)$. Thereby we give a negative answer to the open question posed by Hitchcock and Vinodchandran [20] and Stull [53].

6.2 Notation

Σ denotes the binary alphabet $\{0,1\}$. For any function $f : \Sigma^\infty \to \Sigma^\infty$ and $\mathcal{S} \subseteq \Sigma^\infty$ let $f(\mathcal{S})$ denote $\{f(X) : X \in \mathcal{S}\}$. For $w \in \Sigma^*$, C_w denotes the *cylinder set* of w, that is the set of all infinite sequences in Σ^∞ that begin with w. Let $\mathcal{B}(\Sigma^\infty)$ denote the Borel σ-algebra over Σ^∞ generated by the set of all cylinder sets $\{C_w : w \in \Sigma^*\}$.

We use poly(n) to denote the set of polynomial (time) functions $\bigcup_{c\in\mathbb{N}}\{n^c\}$. When the context is clear, we also use poly to denote poly(n). We use U_n to denote the uniform probability distribution over Σ^n.

6.3 Background

Time Bounded Kolmogorov complexity

We define time bounded Kolmogorov complexity.

Definition 6.1 (t-time bounded Kolmogorov complexity [25, 28])**.** Let t be a time-constructible function, and $x \in \Sigma^*$ be a finite binary string. The t-time bounded Kolmogorov complexity of x is

$$K_t(x) = \min\{|\Pi| \text{ such that } \Pi \in \Sigma^* \text{ and } \mathcal{U}(\Pi) = x \text{ in } t(|x|) \text{ steps}\}.$$

where $\mathcal{U}$ is any fixed universal Turing machine.

In the study of effective versions of dimension, we divide $K_t(x)$ by $|x|$. So our results are unaffected by the choice between prefix-free Kolmogorov complexity and plain Kolmogorov complexity. We use plain Kolmogorov complexity in the rest of the paper.

Polynomial-time dimension

We first give the formal definition of polynomial time dimension using polynomial time s-gales. Towards this end, we first define the concepts of s-gales, supergales, martingales and supermartingales.

Definition 6.2 ([30, 32])**.** For $s \in [0,\infty)$,

1. An s-supergale is a function $d : \Sigma^* \to [0, \infty)$ such that $d(\lambda) < \infty$ and,

$$d(w) \geq \frac{d(w0) + d(w1)}{2^s}$$

   for every finite string $w \in \Sigma^*$.

2. A supermartingale is a 1-supergale.

3. An s-gale is a function $d : \Sigma^* \to [0, \infty)$ with $d(\lambda) < \infty$ that satisfies the inequality in 1 with an equality for every $w \in \Sigma^*$.

4. A martingale is a 1-gale.

Unless specified otherwise, for every s-supergale d, we assume that the initial capital $d(\lambda) = 1$.

Definition 6.3 ([30, 32])**.** The *success set* of an s-supergale d is

$$S^\infty(d) = \left\{ X \in \mathbb{N}^\infty \mid \limsup_{n \to \infty} d(X \upharpoonright n) = \infty \right\}.$$

Now, we define time bounded supergales. We first define $t(n)$-time computable s-gales.

Definition 6.4 ($t(n)$-time computability [30])**.** An s-supergale $d : \Sigma^* \to [0, \infty) \cap \mathbb{Q}$ is $t(n)$-time computable if there exist a $t(n)$-time computable function $f : \Sigma^* \times 1^* \to \mathbb{Q}$ such that $|d(w) - f(w, 1^n)| \leq 2^{-n}$ for every $w \in \Sigma^*$ and $n \in \mathbb{N}$.

Another notion of time bounded computability is that of exact $t(n)$-time computability.

Definition 6.5 (Exact $t(n)$-time computability [30])**.** An s-supergale $d : \Sigma^* \to [0, \infty) \cap \mathbb{Q}$ is exact $t(n)$-time computable if there exist a $t(n)$-time computable function $f : \Sigma^* \to \mathbb{Q}$ such that $d(w) = f(w)$ for every $w \in \Sigma^*$.

Now, we define polynomial-time dimension and strong dimension.

Definition 6.6 (Polynomial-time dimension [30])**.** The polynomial-time dimension of $\mathcal{F} \subseteq \Sigma^\infty$ is defined as

$$\mathrm{cdim_P}(\mathcal{F}) = \inf\{s \mid (\exists k) \text{ there is an exact } n^k\text{-time computable } s\text{-gale } d \text{ such that } \mathcal{F} \subseteq S^\infty(d)\}.$$

For a sequence $X \in \Sigma^\infty$, define $\mathrm{cdim_P}(X) = \mathrm{cdim_P}(\{X\})$.

We now make an important remark. Polynomial time dimension is defined in terms of n^k-time computable s-gales that need not be exact computable. But these definitions are in fact equivalent. This is a direct consequence of the following lemma from [30] which is also useful in proving the main results in this paper.

Lemma 6.7 (Exact Computation Lemma [30]). *If there is a polynomial time computable s-gale d such that $2^s \in \mathbb{Q}$, then there exists an exact polynomial time computable s-gale $\tilde{d}$ such that $S^\infty(d) \subseteq S^\infty(\tilde{d})$.*

We will often use the terminology polynomial-time computable gales to refer to exact $t(n)$-time computable s-gales where $t(n) \in \text{poly}(n)$. Similar comment applies in the case of exact polynomial-time computability and other notions related to gales such as martingales and supergales.

Now, we define the asymptotic polynomial-time density of information of a subset $\mathcal{F} \subseteq \Sigma^\infty$.

Definition 6.8 (Hitchcock and Vinodchandran [23]). For any $\mathcal{F} \subseteq \Sigma^\infty$, define

$$\mathcal{K}_{\text{poly}}(\mathcal{F}) = \inf_{t \in poly} \sup_{X \in \mathcal{F}} \liminf_{n \to \infty} \frac{K_t(X \upharpoonright n)}{n}.$$

For individual sequences, we define $\mathcal{K}_{\text{poly}}(X) = \mathcal{K}_{\text{poly}}(\{X\})$.

Definition 6.9 (Hitchcock and Vinodchandran [23]). For any $X \in \Sigma^\infty$, define

$$\mathcal{K}_{\text{poly}}(X) = \inf_{t \in poly} \liminf_{n \to \infty} \frac{K_t(X \upharpoonright n)}{n}.$$

Measure over the Cantor Space

Definition 6.10 (Measure over the Cantor Space [12, 47]). Let Σ^∞ be the Cantor space. The measure of a cylinder set C_w determined by a finite string $w \in \Sigma^n$ is defined as:

$$\mu(C_w) = 2^{-n}.$$

Recall that C_w denotes the set of infinite sequences in Σ^∞ that begin with w. Using routine measure theoretic arguments, μ uniquely extends to a probability measure over $(\Sigma^\infty, \mathcal{B}(\Sigma^\infty))$. In the rest of the paper we work in the measure space $(\Sigma^\infty, \mathcal{B}(\Sigma^\infty), \mu)$.

One-way functions

The following is the definition of one-way functions secure against uniform probabilistic polynomial time adversaries.

Definition 6.11 (One-way functions [16, 54, 15])**.** A function $f_n : \Sigma^* \to \Sigma^*$ is a one-way function (OWF) if:

1. There is a constant b such that f_n is computable in time n^b for sufficiently large n.
2. For every probabilistic polynomial time algorithm A an d every constant $c > 0$:

$$\Pr[A(f_n(U_n)) \in f_n^{-1}(f_n(U_n))] \leq \frac{1}{n^c}$$

for all sufficiently large n, where the probability is taken over the choice of U_n and the internal randomness of algorithm A.

Pseudorandom generators

In order to define pseudorandom generators, we first define computational indisitinguishability.

Definition 6.12 (Computational Indistinguishability [16, 54])**.** The ensembles of random variables $\{X_n\}_{n\in\mathbb{N}}$ and $\{Y_n\}_{n\in\mathbb{N}}$ are computationally indistinguishable if for every probabilistic polynomial time algorithm T and every constant $c > 0$,

$$|\Pr[T(X_n) = 1] - \Pr[T(Y_n) = 1]| \leq \frac{1}{n^c}$$

where the probabilities are taken over the distributions of either X_n or Y_n and the internal randomness of the algorithm T.

Definition 6.13 (Pseudorandom generators [16, 54, 15])**.** A sequence of functions $G_m : \Sigma^{d(m)} \to \Sigma^m$ is a pseudorandom generator (PRG) if:

1. $d(m) < m$
2. There exist an uniform and deterministic polynomial time algorithm M such that $M(m, x) = G_m(x)$.
3. $\{G(U_{d(n)})\}_{n\in\mathbb{N}}$ and $\{U_n\}_{n\in\mathbb{N}}$ are computationally indistinguishable.

One-way functions and Pseudorandom generators

The equivalence theorem stated below demonstrates that one-way functions and pseudorandom generators are fundamentally equivalent.

Theorem 6.14 (Equivalence of One-way Functions and PRGs [18, 16, 15])**.** *The following are equivalent:*

1. *One-way functions exist.*
2. *There exist a pseudorandom generator with seed length* $d(m) = m - 1$.
3. *For every polynomial-time computable constant* $\epsilon \in (0, 1)$*, there exists a pseudorandom generator with seed length* $d(m) = \epsilon.m$.

6.4 Proof Outline

We prove that if one-way functions exist, then there exist sets $\mathcal{F} \subseteq \Sigma^\infty$ such that $\mathrm{cdim}_{\mathrm{P}}(\mathcal{F}) > \mathcal{K}_{\mathrm{poly}}(\mathcal{F})$. We show that if $\mathrm{cdim}_{\mathrm{P}}(\mathcal{F}) = \mathcal{K}_{\mathrm{poly}}(\mathcal{F})$ for all $\mathcal{F} \subseteq \Sigma^\infty$, then we can break the pseudorandom generators $G_n : \Sigma^{sn} \to \Sigma^n$ for $s = 2^{-m}$ and $m > 1$. From Theorem 6.14, it follows that one-way functions do not exist.

We now sketch how the proof is organised. We assume the existence of pseudorandom generators $G_n : \Sigma^{sn} \to \Sigma^n$ where $s = 2^{-m}$ for some $m \in \mathbb{N}$. In Section 6.5, we construct a mapping g from $\Sigma^\infty \to \Sigma^\infty$ and show (Lemma (6.18)) that $\mathcal{K}_{\mathrm{poly}}(g(\Sigma^\infty)) \leq s$. From the assumption that $\mathrm{cdim}_{\mathrm{P}}(\mathcal{F}) \leq \mathcal{K}_{\mathrm{poly}}(\mathcal{F})$, we get that $\mathrm{cdim}_{\mathrm{P}}(g(\Sigma^\infty)) \leq s$. From this, we get an exact $t(n)$-time bounded s'-gale d' for any $s' > s$ that succeeds on all $Y \in g(\Sigma^\infty)$.

In Section 6.6.1 we convert d' to a martingale $\tilde{d}$ such that in all $Y \in g(\Sigma^\infty)$ there is a guarantee multiplicative growth (Condition 6.1) on the capital between infinitely many blocks (according to the design of g) of Y.

In Section 6.6.2, we use the Borel-Cantelli Lemma to make the points at which the multiplicative growth uniform, over a significant (inverse polynomial) fraction of $Y \in g(\Sigma^\infty)$.

In Section 6.6.3, we design a distinguisher Algorithm A for the PRG $\{G_n\}_{n\in\mathbb{N}}$ using Condition 6.1 over the martingale $\tilde{d}$. In Section 6.6.4, we use the uniform bound on fraction of Y satisfying multiplicative growth to show that A succeeds on atleast an inverse polynomial

fraction of PRG output. We use the the Kolmogorov inequality on Martingales to show that A succeeds on less than an inverse exponential fraction of a truly random input.

In Section 6.6.5, we tie everything together to show that A breaks the PRG $\{G_n\}_{n\in\mathbb{N}}$. Thus, PRG's and therefore, equivalently one-way functions cannot exist if $\mathrm{cdim}_{\mathrm{P}}(\mathcal{F}) = \mathcal{K}_{\mathrm{poly}}(\mathcal{F})$ for all $\mathcal{F} \subseteq \Sigma^\infty$.

6.5 A mapping between infinite sequences using PRGs

A constant stretch pseudorandom generator maps an input of length sn (with $s < 1$) to a finite string of length n. However, the notions of dimension operate on infinite strings. Therefore, one of the main constructions we require in proving our results is an extension of pseudorandom generators to a mapping between infinite strings.

6.5.1 Definition of g

Given a PRG $G_n : \Sigma^{sn} \to \Sigma^n$ where $s = 2^{-m}$ for some $m \in \mathbb{N}$, we construct a mapping g from $\Sigma^\infty \to \Sigma^\infty$.

To define $Y = g(X)$, for some $X \in \Sigma^\infty$, we first partition X to successive finite blocks of size 2^n for all $n \in \mathbb{N}$. We then use the family $\{G_n\}_{n\in\mathbb{N}}$ to map blocks of size 2^n within X to successive blocks of size 2^{m+n} within Y.

Definition 6.15. Let $G_n : \Sigma^{sn} \to \Sigma^n$ where $s = 2^{-m}$ for some $m \in \mathbb{N}$. For $w \in \Sigma^{2^n-1}$, for some $n \in \mathbb{N}$, define $g_n(w)$ as follows. For $1 \leq i \leq 2^m$, we set $g_n(w)[i] = 0$.

For all $k \leq n$ let

$$g_n(w)[2^{m+k} : 2^{m+k} - 1] = G_{2^{m+k}}(w[2^k : 2^{k+1} - 1]).$$

We define g as the limit of the mappings g_n as $n \to \infty$.

Definition 6.16. For every $X \in \Sigma^*$ let $g(X) = \cap_{n=1}^{\infty} C_{g(X\restriction 2^n-1)}$.

6.5.2 Analysis of $K_t(g(X) \restriction 2^n)$

Now, we prove the following lemma that the polynomial time bounded Kolmogorov complexity of prefixes of sequences in the image set of g is low.

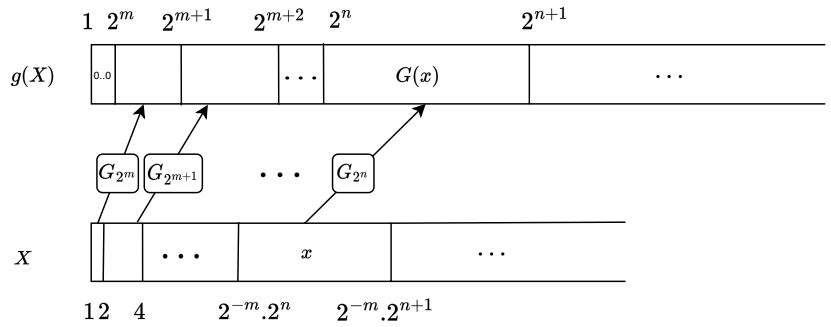


FIGURE 6.1: Illustration of the construction of $g(X)$. The first 2^m bits of the $g(X)$ are 0. Thereafter, for every for every $n > m$, the block $X[s.2^{n-1}, s.2^n - 1]$ is mapped to $G_{2^{n-1}}(X[s.2^{n-1}, s.2^n - 1])$ in the output string $g(X)$.

Lemma 6.17. *For $s = 2^{-m}$ for some $m > 1$, let $g : \Sigma^\infty \to \Sigma^\infty$ be constructed from a PRG $G_n : \Sigma^{sn} \to \Sigma^n$ that runs in time $t_G(n)$ as in Definition 6.15. There exists a constant $c \in \mathbb{N}$ such that for all $X \in \Sigma^*$ and for all $n \in \mathbb{N}$,*

$$K_{t_g}(g(X) \upharpoonright 2^n) \leq s.2^n + c$$

where $t_g(n) = t_G(n)n$.

Proof. Consider the algorithm M which implements the mapping in Definition 6.15 as follows.

Algorithm 1 Algorithm M

Input: A string w.
if $w = \lambda$ **then**
    Output λ
else
    Output 0^{2^m}
end if
for $i = m$ to $\lfloor \log(|w|2^m) \rfloor - 1$ **do**
    Output $G_{2^i}(w[2^{-m} \cdot 2^i, 2^{-m} \cdot 2^{i+1} - 1])$
end for

Since G_n runs in $t_G(n)$ time, it follows that M runs in $t_g(n) = t_G(n) \cdot n$ time. Also, for any $X \in \Sigma^\infty$ and $n \geq m$, M outputs the string $g(X) \upharpoonright 2^n$ on input $X \upharpoonright 2^{-m} \cdot 2^n = X \upharpoonright c \cdot 2^n$. Therefore, it follows that $K_{t_g}(g(X) \upharpoonright 2^n) \leq s \cdot 2^n + c$ where the constant c depends only on

the length of any representation of M in the prefix language of the fixed universal Turing machine. □

From this, it immediately follows that the $\mathcal{K}_{\text{poly}}$ dimension of the set of sequences in the range of g is less than or equal to s.

Lemma 6.18. *For $s = 2^{-m}$ for some $m > 1$, let $g : \Sigma^\infty \to \Sigma^\infty$ be constructed from a PRG $G_n : \Sigma^{sn} \to \Sigma^n$ that runs in polynomial-time. Then, we have*

$$\mathcal{K}_{\text{poly}}(g(\Sigma^\infty)) \leq s.$$

6.6 One-way functions and polynomial-time dimension of sets

In this section, we prove that if one-way functions exist, then there exist sets $\mathcal{F} \subseteq \Sigma^\infty$ such that $\text{cdim}_{\text{P}}(\mathcal{F}) > \mathcal{K}_{\text{poly}}(\mathcal{F})$.

We show that if $\text{cdim}_{\text{P}}(\mathcal{F}) = \mathcal{K}_{\text{poly}}(\mathcal{F})$ for all $\mathcal{F} \subseteq \Sigma^\infty$, then we can break the pseudorandom generators $G_n : \Sigma^{sn} \to \Sigma^n$ for $s = 2^{-m}$ and $m > 1$. From Theorem 6.14, it follows that one-way functions do not exist.

6.6.1 Martingales and growth rates

We use Lemma 6.19 and Lemma 6.20 stated below to convert an exact polynomial time s'-gale d' to an exact-polynomial time martingale $\tilde{d}$ with a guarantee that its capital grows *significantly* between blocks of length 2^n for infinitely many n.

Lemma 6.19. *For $s' \in [0, \infty)$ if $d' : \Sigma^\infty \to [0, \infty)$ is an exact $t(n)$-time computable s'-gale, then for any s'' satisfying $s' \leq s'' \leq 1$ and $2^{s''} \in \mathbb{Q}$, there exists an exact $t(n)poly(n)$-time computable martingale $\tilde{d} : \Sigma^\infty \to [0, \infty)$ such that for any $X \in S^\infty(d')$,*

$$\limsup_{n\to\infty} \frac{\tilde{d}(X \upharpoonright n)}{2^{(1-s'')n}} = \infty.$$

Proof. Let $s'' \geq s'$ such that $2^{s''} \in \mathbb{Q}$. We know that d' is an exact $t(n)$-time s''-supergale that succeeds on X (see Observation 4.4 from [30]). Let $\bar{d}$ be defined as

$\bar{d}(w) = d'(w)2^{(1-s'')|w|}$. From Observation 3.2 in [32], it follows that $\bar{d}$ is a supermartingale such that,

$$\limsup_{n\to\infty} \frac{\bar{d}(X \upharpoonright n)}{2^{(1-s'')n}} = \infty.$$

Since $2^{s''}$ is rational, so is $2^{(1-s'')}$. Hence, $\bar{d}$ is an exact $t(n) \cdot poly(n)$-time computable supermartingale. Now , using exactly the same argument in the proof of Lemma 4.3 from [30], we construct an exact $t(n)poly(n)$-computable martingale $\tilde{d}$ such that $\tilde{d}(w) \geq \bar{d}(w)$ for all $w \in \Sigma^*$. Using this inequality we obtain that,

$$\limsup_{n\to\infty} \frac{\tilde{d}(X \upharpoonright n)}{2^{(1-s'')n}} \geq \limsup_{n\to\infty} \frac{\bar{d}(X \upharpoonright n)}{2^{(1-s'')n}} = \infty.$$

□

Lemma 6.20. *For $s' < 1/2$, if $d' : \Sigma^\infty \to [0,\infty)$ is an exact $t(n)$-time computable s'-gale, then for any $\tilde{s}$ such that $\tilde{s} > 2s'$ and $2^{\tilde{s}} \in \mathbb{Q}$, there exists an exact $t(n) \cdot poly(n)$-time computable martingale $\tilde{d} : \Sigma^\infty \to [0,\infty)$ such that for any $X \in S^\infty(d')$, there exists infinitely many $n \in \mathbb{N}$ such that*

$$\tilde{d}(X \upharpoonright 2^n) > 2^{(1-\tilde{s})2^{n-1}} \tilde{d}(X \upharpoonright 2^{n-1}) \tag{6.1}$$

Proof. Choose $s'' \in (s', 1/2)$ such that $s' < 2s' < 2s'' < \tilde{s}$ and $2^{s''} \in \mathbb{Q}$. Let $\tilde{d}$ be the exact $t(n)poly(n)$-time computable martingale from Lemma 6.19 such that,

$$\limsup_{n\to\infty} \frac{\tilde{d}(X \upharpoonright n)}{2^{(1-s'')n}} = \infty.$$

We show that $\tilde{d}$ satisfies the required property. On the contrary, assume that there exists $m > 0$ such that for every $n > m$,

$$\tilde{d}(X \upharpoonright 2^n) \leq 2^{(1-\tilde{s})2^{n-1}} \tilde{d}(X \upharpoonright 2^{n-1}).$$

Consider any $n > m$. Let $\ell \leq 2^n$. We get that,

$$\begin{aligned}\tilde{d}(X \upharpoonright 2^n + \ell) &\leq 2^\ell 2^{(1-\tilde{s})\sum_{i=m}^{n-1} 2^i} \tilde{d}(X \upharpoonright 2^m) \\ &= 2^\ell 2^{(1-\tilde{s})(2^n - 2^m)} \tilde{d}(X \upharpoonright 2^m).\end{aligned}$$

If $\tilde{d}(X \upharpoonright 2^n + \ell) \geq 2^{(2^n+\ell)(1-s'')}$, then

$$2^{\ell s''} \geq \frac{2^{(1-\tilde{s})2^m}}{\tilde{d}(X \upharpoonright 2^m)} 2^{2^n(\tilde{s}-s'')}.$$

Since $\ell \leq 2^n$, we obtain

$$2^{2^n(\tilde{s}-2s'')} \leq \frac{\tilde{d}(X \upharpoonright 2^m)}{2^{(1-\tilde{s})2^m}}. \tag{6.2}$$

If there exist infinitely many k such that $\tilde{d}(X \upharpoonright k) \geq 2^{(1-s'')k}$ then there exist infinitely many n satisfying 6.2. But, since $\tilde{s} > 2s''$, 6.2 cannot be true for large enough n and therefore we obtain a contradiction. Hence, it must be the case that for infinitely many n,

$$\tilde{d}(X \upharpoonright 2^n) > 2^{(1-\tilde{s})2^{n-1}} \tilde{d}(X \upharpoonright 2^{n-1})$$

which is the required conclusion. □

6.6.2 Borel-Cantelli on fast growing Martingales

Now, we define a $\{0,1\}$-valued function f which checks the fast growing condition for prefixes of infinite strings.

Definition 6.21. Let $f : \Sigma^\infty \times \mathbb{N} \to \{0,1\}$ be defined as follows,

$$f(X,n) = \begin{cases} 1 & \text{if } \tilde{d}(g(X \upharpoonright s.2^n)) > 2^{(1-\tilde{s})2^n} \tilde{d}(g(X \upharpoonright s.2^{n-1})) \\ 0 & \text{otherwise.} \end{cases}$$

Lemma 6.22. *Let $f : \Sigma^\infty \times \mathbb{N} \to \{0,1\}$ and $\mathcal{S} \subseteq \Sigma^\infty$ such that $\mu(\mathcal{S}) > 0$. If for every $X \in \mathcal{S}$, there exist infinitely many $n \in \mathbb{N}$ such that $f(X,n) = 1$. Then for any $c > 1$, there exist infinitely many $n \in \mathbb{N}$, such that $\mu(\{X : f(X,n) = 1\}) > n^{-c}$.*

Proof. Let $A_n = \{X \in \Sigma^\infty : f(X,n) = 1\}$. We know that,

$$\mathcal{S} \subseteq \limsup_{n\to\infty} A_n = \bigcap_{i=0}^{\infty} \bigcup_{n=i}^{\infty} A_n.$$

Therefore, $\mu(\limsup_{n\to\infty} A_n) \geq \mu(\mathcal{S}) > 0$. Since for $c > 1$, the series $\sum n^{-c}$ is convergent, using the Borel-Cantelli Lemma (see [**?**]), we obtain that $\mu(A_n) > n^{-c}$ for infinitely many n, which is the required conclusion. □

Lemma 6.23. *Given $\tilde{d} : \Sigma^\infty \to [0, \infty)$, if for all sequences $Y \in g(\mathcal{S})$ where $\mu(\mathcal{S}) > 0$, and some $\tilde{s} > 0$, there exists infinitely many $n \in \mathbb{N}$ such that*

$$\tilde{d}(Y \upharpoonright 2^n) > 2^{(1-\tilde{s})2^{n-1}} \tilde{d}(Y \upharpoonright 2^{n-1}).$$

Then it holds that for infinitely many n,

$$\Pr_{x \in \Sigma^{s.2^n}}[\tilde{d}(g(x)) > 2^{(1-\tilde{s})2^n} \tilde{d}(g(x \upharpoonright s.2^{n-1}))] \geq \frac{1}{n^2}. \tag{6.3}$$

Proof. We have that for all $Y \in g(\mathcal{S})$ where $\mu(\mathcal{S}) > 0$, there exists infinitely many $n \in \mathbb{N}$ such that $\tilde{d}(Y \upharpoonright 2^n) > 2^{(1-\tilde{s})2^{n-1}} \tilde{d}(Y \upharpoonright 2^{n-1})$. From the definition of the function f (Definition 6.21), we have that for all $X \in \mathcal{S}$, there exist infinitely many $n \in \mathbb{N}$ such that $f(X, n) = 1$.

Since $\mu(\mathcal{S}) > 0$, using Lemma 6.22, we obtain that for any $c > 1$ there exist infinitely many $n \in \mathbb{N}$ such that $\mu(\{X : f(X, n) = 1\}) > n^{-c}$. Then, from the definition of f, we have

$$\mu(\{X : \tilde{d}(g(X \upharpoonright s.2^n)) > 2^{(1-\tilde{s})2^n} \tilde{d}(g(X \upharpoonright s.2^{n-1}))\} > n^{-c}.$$

From this, it follows that

$$\Pr_{x \in \Sigma^{s.2^n}}[\tilde{d}(g(x)) > 2^{(1-\tilde{s})2^n} \tilde{d}(g(x \upharpoonright s.2^{n-1}))] > n^{-c}.$$

Putting $c = 2$, we have

$$\Pr_{x \in \Sigma^{s.2^n}}[\tilde{d}(g(x)) > 2^{(1-\tilde{s})2^n} \tilde{d}(g(x \upharpoonright s.2^{n-1}))] > n^{-2}.$$

□

6.6.3 A Distinguisher Algorithm for PRGs

Let $G_n : \Sigma^{sn} \to \Sigma^n$, where $s = 2^{-m}$ for some $m > 1$ be a PRG. Let $t_G(n)$ be the running time of G_n and $t_g(n) = n.t_G(n)$ be the time needed to construct g.

Given a martingale $\tilde{d} : \Sigma^\infty \to [0, \infty)$, we describe the probabilistic distinguisher algorithm A that attempts to breaks G_n, using $\tilde{d}$.

We assume that A takes as an input a string w of length 2^n for $n \geq m$. A uses $s.2^n$ random bits and uses the mapping g (Definition 6.15) to generate a string $w' = g(r)$.

A checks if $\tilde{d}(w'w) \geq 2^{|w|(1-\tilde{s})}d(w')$ (condition 6.1) and returns 1 (flags w as output of PRG) if and only if the condition is true. The algorithm for A is given below.

Algorithm 2 Algorithm A

Input: A string w where $|w| = 2^n$
$r \leftarrow \text{random}(s \cdot 2^n)$ ▷ Generate a random seed of length $s \cdot 2^n$
for $i = m$ to $n-1$ **do**
    $v_i \leftarrow G_{2^i}(r[s \cdot 2^i, s \cdot 2^{i+1} - 1])$ ▷ Use r and G_{2^i} to get a PRG output of length 2^i
end for
$w' \leftarrow 0^{2^m} v_1 \dots v_n$ ▷ Concatenate all v_i
return $\tilde{d}(w'w) \geq 2^{(1-\tilde{s})|w|} \cdot \tilde{d}(w')$

6.6.4 Analysis of the Distinguisher Algorithm

We first analyze the running time of A. Let $t_G(n)$ be the running time of G and $t_g(n) = n \cdot t_G(n)$ be the time needed to construct g. Let $\tilde{s} < 1$ such that $2^{\tilde{s}} \in \mathbb{Q}$ and $\tilde{d} : \Sigma^\infty \to [0, \infty)$ be the $t_{\tilde{d}}(n)$-time computable martingale.

Since $2^{\tilde{s}} \in \mathbb{Q}$, we obtain that the running time of A is $t_A(n) \leq n + t_G(n) \log n + t_{\tilde{d}}(n) + c$, where c is a constant. Hence if $t_{\tilde{d}}(n) \in \text{poly}(n)$, then $t_A(n) \in \text{poly}(n)$.

6.6.4.1 Performance of the algorithm on outputs of PRGs

We now analyze $\Pr_{x \in \Sigma^{s \cdot 2^n}}[A(G_{2^n}(x)) = 1]$. We show that for infinitely many n, this probability is greater than $1/p(n)$ for some polynomial p, if the martingale $\tilde{d}$ used in A satisfies the fast growth condition (Condition 6.1). We apply the Borel Cantelli Lemma ()

Lemma 6.24. *Let $\tilde{d} : \Sigma^* \to [0, \infty)$ be a martingale such that for some rational $\tilde{s} < 1$ and for any $Y \in g(\mathcal{S})$, with $\mu(\mathcal{S}) > 0$, there exists infinitely many $n \in \mathbb{N}$ such that*

$$\tilde{d}(Y \upharpoonright 2^n) > 2^{(1-\tilde{s})2^{n-1}} \tilde{d}(Y \upharpoonright 2^{n-1}).$$

Then for the distinguisher algorithm A given in Algorithm 2 that uses $\tilde{d}$, we have that for infinitely many $n \in \mathbb{N}$, and $N = 2^n$,

$$\Pr_{(x) \in \Sigma^{s \cdot N}}[A(G_N(x)) = 1] \geq \frac{1}{N^2}. \tag{6.4}$$

Proof. Since $\mu(\mathcal{S}) > 0$, applying the Borel-Cantelli Lemma (Lemma 6.23), it holds that for infinitely many n,

$$\Pr_{x\in\Sigma^{s\cdot 2^n}}[\tilde{d}(g(x)) > 2^{(1-\tilde{s})2^n}\tilde{d}(g(x \upharpoonright s.2^{n-1}))] \geq \frac{1}{n^2}.$$

Let $n+1$ be one of the lengths at which the condition 6.3 holds. Observe that,

$$\begin{aligned}
\Pr_{(x,r)\in\Sigma^{s\cdot 2^n}\times\Sigma^{s\cdot 2^n}}[A(G_{2^n}(x),r)) = 1] &= \Pr_{(x,r)\in\Sigma^{s\cdot 2^n}\times\Sigma^{s\cdot 2^n}}[\tilde{d}(g(r)G_{2^n}(x)) \geq 2^{(1-\tilde{s})2^n}\tilde{d}(g(r))] \\
&= \Pr_{(x,r)\in\Sigma^{s\cdot 2^n}\times\Sigma^{s\cdot 2^n}}[\tilde{d}(g(rx)) \geq 2^{(1-\tilde{s})2^n}\tilde{d}(g(r))] \\
&= \Pr_{x'\in\Sigma^{s\cdot 2^{n+1}}}[\tilde{d}(g(x')) > 2^{(1-\tilde{s})2^n}\tilde{d}(g(x' \upharpoonright s.2^{n-1}))] \\
&\geq \frac{1}{(n+1)^2}.
\end{aligned}$$

Hence, for large enough n, taking $N = 2^n$,

$$\Pr_{(x)\in\Sigma^{s\cdot N}}[A(G_N(x)) = 1] \geq \frac{1}{N^2}.$$

□

6.6.4.2 Performance of the algorithm on uniformly random inputs

We now analyze $\Pr_{x\in\Sigma^{2^n}}[A(x) = 1]$. We use the Kolmogorov inequality for s-supergales to show that this is inverse exponentially small in input size.

Lemma 6.25 (Kolmogorov Inequality [30])**.** *Let $d : \Sigma^\infty \to [0,\infty)$ be a s-supergale. For any $w' \in \Sigma^*$ and $n, c \in \mathbb{N}$, the number of strings $w \in \Sigma^n$ such that $d(w'w) \geq c.d(w')$ is less than or equal to $2^{sn}/c$.*

Proof. Since d is a s-supergale, using a simple induction argument, it follows that

$$d(w') \geq \frac{1}{2^{sn}}\sum_{w\in\Sigma^n} d(w'w).$$

If $d(w') = 0$, then the lemma follows trivially. Assume that $d(w') > 0$. Let M be the number of strings $w \in \Sigma^n$ such that $d(w'w) \geq c.d(w')$. It follows that,

$$d(w') \geq \frac{1}{2^{sn}} \cdot M \cdot c \cdot d(w').$$

Therefore, we obtain that $M \leq 2^{sn}/c$, which is the required conclusion. □

Setting $s = 1$, we obtain the upper bound $2^n/c$ when d is a martingale in the above lemma.

Corollary 6.26 (Kolmogorov Inequality for Martingales [56, 12, 30])**.** *Let $d : \Sigma^\infty \to [0, \infty)$ be a martingale. For any $w' \in \Sigma^*$ and $n, c \in \mathbb{N}$, the number of strings $w \in \Sigma^n$ such that $d(w'w) \geq c.d(w')$ is less than or equal to $2^n/c$.*

Lemma 6.27. *Let $\tilde{d} : \Sigma^* \to [0, \infty)$ be any martingale. Then for the distinguisher algorithm A given in Algorithm 2 that uses $\tilde{d}$, we have that for any $n \in \mathbb{N}$, and $N = 2^n$,*

$$\Pr_{x \in \Sigma^N}[A(x) = 1] < \frac{1}{2^{N(1-\tilde{s})}}. \tag{6.5}$$

Proof. For any $n \in \mathbb{N}$, take any $r \in \Sigma^{s.2^n}$, and let $w' = g(r)$. Using the Kolmogorov Inequality for Martingales (Corollary 6.26), the number of strings $w \in \Sigma^{2^n}$ such that $\tilde{d}(w'w) \geq 2^{|w|.(1-\tilde{s})}\tilde{d}(w')$ is less than $2^{2^n}/2^{2^n(1-\tilde{s})}$.

Let $N = 2^n$. For any fixed $r \in \Sigma^{s.N}$, we have that

$$\Pr_{x \in \Sigma^N}[A(x, r) = 1] < 2^{-N(1-\tilde{s})}.$$

Therefore, it follows that for any $n \in \mathbb{N}$,

$$\Pr_{x \in \Sigma^N}[A(x) = 1] < \frac{1}{2^{N(1-\tilde{s})}}.$$

□

6.6.5 One way Functions and Polynomial-time dimension of sets

We now tie together the results obtained till now to show that if $\mathrm{cdim}_{\mathrm{P}}(\mathcal{F}) = \mathcal{K}_{\mathrm{poly}}(\mathcal{F})$ for all sets $\mathcal{F} \subseteq \Sigma^\infty$, then it is not possible that one-way functions exist.

Theorem 6.28. *If for all $\mathcal{F} \subseteq \Sigma^\infty$,*

$$\mathrm{cdim}_{\mathrm{P}}(\mathcal{F}) = \mathcal{K}_{\mathrm{poly}}(\mathcal{F}),$$

then one-way functions do not exist.

Proof. Assume that for all $\mathcal{F} \subseteq \Sigma^\infty$, we have that

$$\mathrm{cdim}_{\mathrm{P}}(\mathcal{F}) \leq \mathcal{K}_{\mathrm{poly}}(\mathcal{F}).$$

We show that using this assumption, we can break pseudorandom generators $G_n : \Sigma^{sn} \to \Sigma^n$ for $s = 2^{-m}$ and $m > 1$. From Theorem 6.14, it follows that one-way functions do not exist.

Consider mapping g from $\Sigma^\infty \to \Sigma^\infty$ in Section 6.5. Let $\mathcal{F}_s \subseteq \Sigma^\infty$ be the set of sequences in the range of g, that is $\mathcal{F}_s = g(\Sigma^\infty)$. From Lemma 6.18, we have that $\mathcal{K}_{\text{poly}}(\mathcal{F}_s) \leq s$. From our assumption, it follows that $\text{cdim}_{\text{P}}(\mathcal{F}_s) \leq s$.

From this, we get that for all $s' > s$, there exists an exact $t(n)$-time s'-gale d' that runs in time for some polynomial $t(n)$ and succeeds on all $Y \in \mathcal{F}_s$.

From Lemma 6.20, there exists an exact $t(n) \cdot \text{poly}(n)$-time computable martingale $\tilde{d} : \Sigma^\infty \to [0, \infty)$ such that for all $Y \in \mathcal{F}_s$, there exists infinitely many $n \in \mathbb{N}$ such that

$$\tilde{d}(Y \upharpoonright 2^n) > 2^{(1-\tilde{s})2^{n-1}} \tilde{d}(Y \upharpoonright 2^{n-1}).$$

From Lemma 6.24, for the distinguisher algorithm A (Algorithm 2) for infinitely many $N \in \mathbb{N}$,

$$\Pr_{x \in \Sigma^{s \cdot N}}[A(G_N(x)) = 1] - \Pr_{x \in \Sigma^N}[A(x) = 1] > \frac{1}{N^2} - \frac{1}{2^{N(1-\tilde{s})}}.$$

From Lemma 6.27, for any $n \in \mathbb{N}$, and $N = 2^n$,

$$\Pr_{x \in \Sigma^N}[A(x) = 1] < \frac{1}{2^{N(1-\tilde{s})}}.$$

As $\tilde{d}$ runs in polynomial-time, the distinguisher algorithm A runs also runs in polynomial-time. Let c be any constant such that $t_A(n) \leq n^c$ and for all large enough N,

$$\frac{1}{N^2} - \frac{1}{2^{N(1-\tilde{s})}} \geq \frac{1}{N^c}$$

Therefore, we have that for some $c \in \mathbb{N}$ and for infinitely many $N \in \mathbb{N}$,

$$\Pr_{x \in \Sigma^{s \cdot N}}[A(G_N(x)) = 1] - \Pr_{x \in \Sigma^N}[A(x) = 1] \geq \frac{1}{N^c}. \tag{6.6}$$

It follows that G_N is not a $(N^c, 1/N^c)$ PRG for infinitely many N. Therefore the family $\{G_n\}_{n \in \mathbb{N}}$ is not a PRG. Hence we have shown that PRG's $G_n : \Sigma^{sn} \to \Sigma^n$ do not exist

for $s = 2^{-m}$. Therefore from Theorem 6.14, we obtain that one-way functions do not exist. □

6.7 One-way functions and polynomial-time dimension of sequences

In this section, we now show the stronger result (compared to Theorem 6.28) that if one-way functions exist, there are *sequences* for which $\mathrm{cdim}_{\mathrm{P}}$ is strictly greater than $\mathcal{K}_{\mathrm{poly}}$.

Now we take the weaker assumption that for all $X \in \Sigma^\infty$, $\mathrm{cdim}_{\mathrm{P}}(X) \leq \mathcal{K}_{\mathrm{poly}}(X)$.

We assume the existence of pseudorandom generators $G_n : \Sigma^{sn} \to \Sigma^n$ where $s = 2^{-m}$ for some $m \in \mathbb{N}$. We use the same mapping g from $\Sigma^\infty \to \Sigma^\infty$ in Section 6.5. We observe that for all $X \in \Sigma^\infty$, $\mathcal{K}_{\mathrm{poly}}(g(X)) \leq s$. From the assumption that $\mathrm{cdim}_{\mathrm{P}}(g(X)) \leq \mathcal{K}_{\mathrm{poly}}(g(X))$, we get that for all $X \in \Sigma^\infty$, $\mathrm{cdim}_{\mathrm{P}}(g(X)) \leq s$.

From this, we get that for all rational $s' > s$ and for all $X \in \Sigma^\infty$ there exists an exact computable s'-gale d'_X that runs in time $t_{d'_X}(n) \in \mathrm{poly}(n)$, such that d'_X succeeds on $g(X)$. The major technical obstacle in proceeding along the lines of the proof of Theorem 6.28 is that we do not have a single s'-gale d' that succeeds on $g(X)$ for all $X \in \Sigma^\infty$.

To overcome this, using a gale combination technique, we construct a set $\mathcal{S} \subseteq \Sigma^\infty$ with $\mu(\mathcal{S}) > 0$ such that there exists a single s-gale d' that succeeds on all $Y \in g(\mathcal{S})$. This suffices to apply the Borel Cantelli Lemma (Lemma 6.23) and so the rest of the proof is similar to the proof of Theorem 6.28.

6.7.1 Martingale Combination

Lemma 6.29. *For any time $t(n) \in \mathrm{poly}(n)$, and $s' > 0$ with $2^{s'} \in \mathbb{Q}$, there exists an exact $t(n) \cdot n \log(n)$-time computable s'-gale d' such that for all exact $t(n)$-time computable s'-gales d', there exist a constant c such that for all $X \in \Sigma^\infty$ and $n \in \mathbb{N}$, $d'(X \upharpoonright n) \geq c \cdot d(X \upharpoonright n)$.*

Proof. Let $\{M_i\}$ be a standard computable enumeration of all Turing machines. The idea of the proof is that we consider the $t(n)$ time s'-gales among the first $\log n$ machines to place bets on strings of length n. The remaining martingales are assumed to bet evenly on all strings of length n.

We will define the s'-gale, conditioned on the length of the input string x. When the length of the input string x satisfies $2^n \leq |x| < 2^{n+1}$, we consider the machines $M_1 \dots M_n$ to decide $d'(x)$. For $n \in N$, let $\mathcal{D}_n$ be the set of machines in $\{M_1, \dots M_n\}$ such that for each $i \in \{0, 1, \dots 2^n\}$, $M(x \upharpoonright i)$ runs in time $t(i)$ and $M(\lambda) = 1$ and $2^s \cdot M(x \upharpoonright i) = M((x \upharpoonright i)0) + M((x \upharpoonright i)1)$.

The remaining machines M_i for $i \geq n+1$ are assumed to bet *evenly* on x. So, for $i \geq n+1$, we force the bet placed by M_i on x to be $2^{-n(1-s')}$. The actual bets placed by martingale M_i is taken into consideration when $M_i \in \mathcal{D}_n$, which happens for strings x of length more than 2^i. In that case, we need to ensure that the capital of M_i at $x \upharpoonright 2^i$ is forced to be $2^{-i(1-s)}$.

Therefore, given $x \in \Sigma^*$, such that $2^n \leq |x| < 2^{n+1}$, we define

$$d'(x) = \sum_{M_i \in \{M_1, \dots M_n\}} 2^{-i} \cdot M'_i(x) + 2^{-(n-1)}.2^{-n(1-s')}$$

where $M'_i(x)$ is defined as follows.

Case 1: $M_i \in \mathcal{D}_n$. In this case, when $M_i(x \upharpoonright 2^i) > 0$, we define $M'_i(x) = 2^{-i(1-s')}.(M_i(x)/M_i(x \upharpoonright 2^i))$. If $M_i(x \upharpoonright 2^i) = 0$, then it holds that $M_i(x) = 0$, so we just take $M'_i(x) = 0$.

Case 2: $M_i \notin \mathcal{D}_n$. In this case, let j be the least number such that $M_i(x \upharpoonright j)$ takes more than $t(j)$ time to run, or the s-gale condition gets violated, that is $2^s \cdot M_i(x \upharpoonright j) \neq M_i((x \upharpoonright j)0) + M_i((x \upharpoonright j)1)$. In this case, we freeze $M'_i s$ capital at index j, and M_i is forced to bet evenly after that. Note that as in the previous case, we have to force M_i to bet evenly up to $x \upharpoonright 2^i$. Therefore if $j < 2^i$, we take $M'_i(x) = 2^{-i(1-s)}$. Otherwise $M'_i(x) = M'_i(x \upharpoonright j)$, where $M'_i(x \upharpoonright j) = 2^{-i(1-s')} \cdot (M_i(x \upharpoonright j)/M_i(x \upharpoonright 2^i))$.

Consider any exact t-time computable s'-gale d. Let M_k be any $t(n)$-time machine such that $d(w) = M_k(w)$ for every $w \in \Sigma^*$. Then, it follows from the construction of d that

$$d'(w) \geq \frac{2^{-k} \cdot 2^{-k(1-s')}}{\min\limits_{\{w' \in \Sigma^{2^k} : d'(w') > 0\}} d'(w')} \cdot d(w)$$

for all w with $|w| > 2^k$. Let c be the constant on the right hand side. Note that the membership of any machine from $\{M_1, M_2, \dots M_n\}$ in $\mathcal{D}_n$ is decidable in time $t(2^n) \cdot n \cdot 2^n$. Therefore, d is an exact $t(n) \cdot n \cdot \log(n)$-time computable s'-gale. □

Therefore, there exists an exact n^{k+c}-time computable s'-gale d' that succeeds on all sequences in $g(\mathcal{S})$. Using Lemma 6.20, we convert the s'-gale d' to a martingale $\tilde{d}$.

Lemma 6.30. *If for all $X \in \Sigma^\infty$, we have that $\mathrm{cdim_P}(X) \leq \mathcal{K}_{\mathrm{poly}}(X)$, then for any $0 < s < 1/2$, there exists an $\tilde{s} < 1$ such that $2^{\tilde{s}} \in \mathbb{Q}$, a set $\mathcal{S}$ such that $\mu(S) > 0$ and an exact poly(n)-time computable martingale $\tilde{d} : \Sigma^\infty \to [0, \infty)$ such that for any $X \in g(\mathcal{S})$, there exists infinitely many $n \in \mathbb{N}$ such that*

$$\tilde{d}(X \upharpoonright 2^n) > 2^{(1-\tilde{s})2^{n-1}} \tilde{d}(X \upharpoonright 2^{n-1}) \tag{6.7}$$

6.7.2 One way Functions and Polynomial-time dimension of sequences

We now show that if $\mathrm{cdim_P}(X) = \mathcal{K}_{\mathrm{poly}}(X)$ for all sequences $X \in \Sigma^\infty$, then it is not possible that one-way functions exist.

Theorem 6.31. *If for all $X \in \Sigma^\infty$,*

$$\mathrm{cdim_P}(X) = \mathcal{K}_{\mathrm{poly}}(X),$$

then one-way functions do not exist.

Proof. Assume that for all $\mathcal{F} \subseteq \Sigma^\infty$, we have that

$$\mathrm{cdim_P}(\mathcal{F}) \leq \mathcal{K}_{\mathrm{poly}}(\mathcal{F}).$$

We show that using this assumption, we can break pseudorandom generators $G_n : \Sigma^{sn} \to \Sigma^n$ for $s = 2^{-m}$ and $m > 1$. From Theorem 6.14, it follows that one-way functions do not exist.

Consider mapping g from $\Sigma^\infty \to \Sigma^\infty$ in Section 6.5. From Lemma 6.17, we have that for all $X \in \Sigma^\infty$, $\mathcal{K}_{\mathrm{poly}}(g(X)) \leq s$. From our assumption, it follows that $\mathrm{cdim_P}(g(X)) \leq s$.

From this, we get that for all $s' > s$ such that $2^{s'} \in \mathbb{Q}$ and for all $X \in \Sigma^\infty$ there exists an exact computable s'-gale d'_X that runs in time $t_{d'_X}(n) \in \mathrm{poly}(n)$, such that d'_X succeeds on $g(X)$.

We first partition sequences in the input space of g into sets, based on the running time of the s'-gales that succeeds on it. Define

$$R_i = \{X \in \Sigma^\infty : \text{ there exists an } n^i\text{-time } s'\text{-gale } d \text{ such that } g(X) \in S^\infty(d)\}$$

and let $S_i = R_i \setminus R_{i-1}$. Note that $\{S_i\}_{i\in\mathbb{N}}$ is a partition of Σ^∞ into countably many disjoint subsets. As $\sum_{i=1}^{\infty} \mu(S_i) = 1$, there exists an $k \in \mathbb{N}$ such that $\sum_{i=1}^{k} \mu(S_i) > 2/3$. Define $\mathcal{S} = \bigcup_{i=1}^{k} S_i$.

Using Lemma 6.29, we get that there exists an n^{k+2}-time computable s'-gale $\hat{d}$ that succeeds on all $Y \in g(\mathcal{S})$.

From Lemma 6.20, there exists an exact $t(n) \cdot \text{poly}(n)$-time computable martingale $\tilde{d} : \Sigma^\infty \to [0, \infty)$ such that for all $Y \in g(\mathcal{S})$, there exists infinitely many $n \in \mathbb{N}$ such that

$$\tilde{d}(Y \upharpoonright 2^n) > 2^{(1-\tilde{s})2^{n-1}} \tilde{d}(Y \upharpoonright 2^{n-1}).$$

From Lemma 6.24, for the distinguisher algorithm A (Algorithm 2) for infinitely many $N \in \mathbb{N}$,

$$\Pr_{x\in\Sigma^{s.N}}[A(G_N(x)) = 1] - \Pr_{x\in\Sigma^N}[A(x) = 1] > \frac{1}{N^2} - \frac{1}{2^{N(1-\tilde{s})}}.$$

From Lemma 6.27, for any $n \in \mathbb{N}$, and $N = 2^n$,

$$\Pr_{x\in\Sigma^N}[A(x) = 1] < \frac{1}{2^{N(1-\tilde{s})}}.$$

As $\tilde{d}$ runs in polynomial-time, the distinguisher algorithm A runs also runs in polynomial-time. Let c be any constant such that $t_A(n) \leq n^c$ and for all large enough N,

$$\frac{1}{N^2} - \frac{1}{2^{N(1-\tilde{s})}} \geq \frac{1}{N^c}$$

Therefore, we have that for some $c \in \mathbb{N}$ and for infinitely many $N \in \mathbb{N}$,

$$\Pr_{x\in\Sigma^{s.N}}[A(G_N(x)) = 1] - \Pr_{x\in\Sigma^N}[A(x) = 1] \geq \frac{1}{N^c}. \tag{6.8}$$

It follows that G_N is not a $(N^c, 1/N^c)$ PRG for infinitely many N. Therefore the family $\{G_n\}_{n\in\mathbb{N}}$ is not a PRG. Hence we have shown that PRG's $G_n : \Sigma^{sn} \to \Sigma^n$ do not exist for $s = 2^{-m}$. Therefore from Theorem 6.14, we obtain that one-way functions do not exist. □

Chapter 7

7. Finite-state Relative Dimension and Relative Block entropy rates

7.1 Introduction

Dai, Lathrop, Lutz, and Mayordomo [9] extended the notion of constructive dimension to the setting where finite-state resource bounds are imposed on the computation. The corresponding notion, called finite-state dimension measures the density of information in a sequence using betting algorithms called finite-state s-gales. A result due to Schnorr and Stimm [49] implies that a sequence is Borel normal if and only if it has finite-state dimension 1, establishing connections to metric number theory and probability.

Bourke, Hitchcock and Vinodchandran [5] gave an equivalent characterisation of finite-state dimension using block entropy rates. This notion has several other equivalent characterizations using finite-state compressibility ratios [10], and automatic Kolmogorov complexity [26].

In this chapter, we formulate the notion of finite-state relative dimension using finite-state gamblers with finite oracle access. We then show that the finite-state relative dimension of a sequence is equivalent to its disjoint relative block entropy rate. We also establish similar results for finite-state strong dimension. We also compare finite-state relative dimension with other related notions, such as finite-state independence [2] and finite-state mutual dimension [7].

7.2 Finite-State Relative Dimension

The finite-state relative dimension of an infinite sequence $X \in \Sigma_1^\infty$ with respect to another an infinite sequence $Y \in \Sigma_2^\infty$ represents the finite-state asymptotic information density of X relative to Y. We formulate finite-state relative dimension using an extension of the finite-state gambler (see [9]) which we call as finite-state relative gambler. In contrast to the classic finite-state gambler, the finite-state relative gambler betting on an infinite sequence X is given access to the characters of another infinite sequence Y. The gambler may utilize the information obtained from the characters of Y while betting on X. To preserve the finite-state nature of the model, the gambler is allowed access to only a finite window of characters of Y. This finite window shifts forward in Y, in a sliding manner as the gambler processes X.

Definition 7.1. A k-account *finite-state relative gambler* (FSRG) with window length ℓ is a 5-tuple $G = (Q, \delta, \vec{\beta}, q_0, \vec{c})$ where:

- Q is a nonempty, finite set of states,
- Σ_1 and Σ_2 are finite alphabets,
- $\delta : Q \times \Sigma_2^\ell \times \Sigma_1 \to Q$ is the transition function,
- $\vec{\beta} : Q \times \Sigma_2^\ell \times \Sigma_1 \to (\mathbb{Q} \cap [0,1])^k$ is the account-wise betting function, such that $\forall q \in Q, y \in \Sigma_2^\ell$, $i \in \{1 \dots k\}$, we have $\sum_{a \in \Sigma_1} \beta_i(q, y, a) = 1$,
- $q_0 \in Q$ is the initial state, and
- $\vec{c} = (c_1, \dots c_k)$ is the initial capital per account, where each $c_i = 1/k$.

Let a finite-state relative gambler (FSRG) G be in state $q \in Q$. After reading the oracle word $y \in \Sigma_2^\ell$, the gambler places the bet $\vec{\beta}(q, y, x)$ on the next character $x \in \Sigma_1$ of X. After processing x, the input pointer to X as well as Y move forward by a single position. Further, G enters the state $\delta(q, y, x)$.

Now, we define the extended transition function $\delta^* : Q \times \Sigma_2^* \times \Sigma_1^* \to Q$. Given $q \in Q$, $y \in \Sigma_2^*$ and $x \in \Sigma_1^*$, $\delta^*(q, y, x)$ specifies the state the gambler reaches from state q on input string x and oracle string y. Since the oracle input requires a *ℓ-window lookahead*,

$\delta^*(q, y, x)$ is undefined unless $|y| = |x| + \ell - 1$. For q, y and x satisfying this condition

$$\delta^*(q, y, x) = \begin{cases} \delta(q',\ y[|y| - \ell : |y|],\ x[|x| - 1]) & \text{if } |x| > 1, \\ \delta(q, y, x) & \text{if } |x| = 1. \end{cases}$$

where $q' = \delta^*(q, y[: |y| - 1], x[: |x| - 1])$

Analogous to [9], we define the concept of an s-gale induced by a FSRG and the corresponding success sets.

Definition 7.2. Let $G = (Q, \delta, \vec{\beta}, q_0, \vec{c})$ be an k-account FSRG with a window length ℓ, and with oracle access to Y. The $s - gale$ induced by the ith account of G is the function $d^{(s)}_{G_i,Y} : \Sigma_1^* \to [0, \infty)$ defined by $d^{(s)}_{G_i,Y}(\lambda) = c_i$ and for every $w \in \Sigma_1^*$ and $x \in \Sigma_1$, by

$$d^{(s)}_{G_i,Y}(wx) = |\Sigma_1|^s \times \beta_i(q, y, x) \times d^{(s)}_{G_i,Y}(w)$$

where $y = Y[|w| - 1 : |w| + \ell - 1]$ and $q = \delta^*(q_0, Y[: |w| + \ell - 1], X[: |w|])$.

The total $s - gale$ (or simply the $s - gale$) of G is the function

$$d^{(s)}_{G,Y} = \sum_{i=1}^{k} d^{(s)}_{G_i,Y}.$$

We call an s-gale induced by a finite-state relative gambler as a *finite-state relative s-gale.*

Intuitively, when the automata is in state q, after reading the oracle word $y \in \Sigma_2^\ell$ and the next character $x \in \Sigma_1$, the s-gale function of the ith account gets multiplied by a factor of $|\Sigma_1|^s \times \beta_i(q, y, x)$. As remarked in [9], this corresponds to "betting under inflation", since the expected value after any bet is less than that before the bet.

Definition 7.3. We say that the finite-state relative s-gale $d^{(s)}_{G,Y}$ *succeeds* on a sequence $X \in \Sigma_1^\infty$ if

$$\limsup_{n \to \infty} d^{(s)}_{G,Y}(X \upharpoonright n) = \infty.$$

The *success set* of $d^{(s)}_{G,Y}$ is

$$S^\infty[d^{(s)}_{G,Y}] = \{X \in \Sigma_1^\infty \mid d^{(s)}_{G,Y} \text{ succeeds on X } \}.$$

Definition 7.4. For any $X \in \Sigma_1^\infty$ and $Y \in \Sigma_2^\infty$, $\mathcal{G}^Y(X)$ is the set of all $s \in [0, \infty)$ such that there is a finite-state relative s-gale $d^{(s)}_{G,Y}$ for which $X \in S^\infty[d^{(s)}_{G,Y}]$.

Definition 7.5. The *finite-state relative dimension* of a sequence $X \in \Sigma_1^\infty$ with respect to a sequence $Y \in \Sigma_2^\infty$ is

$$\dim_{FS}^Y(X) = \inf \mathcal{G}^Y(X).$$

Definition 7.6. We say that the finite-state relative s-gale $d_{G,Y}^{(s)}$ *strongly succeeds* on a sequence $X \in \Sigma_1^\infty$ if

$$\liminf_{n\to\infty} d_{G,Y}^{(s)}(X \upharpoonright n) = \infty.$$

The *strong success set* of $d_{G,Y}^{(s)}$ is

$$S_{str}^\infty[d_{G,Y}^{(s)}] = \{X \in \Sigma_1^\infty \mid d_{G,Y}^{(s)} \text{ strongly succeeds on X } \}.$$

For any $X \in \Sigma_1^\infty$ and $Y \in \Sigma_2^\infty$, $\mathcal{G}_{str}^Y(X)$ is the set of all $s \in [0, \infty)$ such that there is a finite-state relative s-gale $d_{G,Y}^{(s)}$ for which $X \in S_{str}^\infty[d_{G,Y}^{(s)}]$.

Definition 7.7. The *finite-state relative strong dimension* of a sequence $X \in \Sigma_1^\infty$ with respect to a sequence $Y \in \Sigma_2^\infty$ is

$$\mathrm{Dim}_{FS}^Y(X) = \inf \mathcal{G}_{str}^Y(X).$$

Dai et al. showed the equivalence between multi-account finite-state gamblers and single account finite-state gamblers while defining finite-state dimension (see Theorem 4.5 in [9]). It is straightforward to verify that the proof from [9] relativizes and extends to the setting of a finite-state relative gamblers as well. Hence we use the model of single account FSRG in the rest of the paper, unless specified otherwise.

7.3 Relative block entropy rates and Finite-state relative dimension

We formulate the notion of relative disjoint block entropy rates, extending the definitions given by Bourke, Hitchcock and Vinodchandran [5]. We characterize finite-state relative dimension using relative block entropy rates, establishing that the notion of relative dimension is mathematically robust.

7.3.1 Relative Block entropy rates

Let $n, k, \ell \in \mathbb{N}$, where $n = k\ell$. Given $x \in \Sigma_1^n$ and $y \in \Sigma_2^n$ and $u \in \Sigma_1^\ell$ and $v \in \Sigma_2^\ell$, let

$$N(v, y) = |\{0 \leq i < k \mid y[i\ell : (i+1)\ell] = v\}|$$

be the number of times v occurs in ℓ length blocks of y and let

$$N(u, x \ ; \ v, y) \ = \ \left|\left\{0 \ \leq \ i \ < \ k \ \mid \ x[i\ell \ : \ (i+1)\ell] \ = \ u \ \ \wedge \ \ y[i\ell \ : \ (i+1)\ell] \ = \ v\right\}\right|.$$

be the number of times u and v occur concurrently in the disjoint ℓ length blocks of x and y respectively.

The *block frequency* of v in y is defined as

$$P(v, y) = \frac{N(v, y)}{k}.$$

The *joint block frequency* of u in x with v in y is defined as

$$P(u, x \ ; \ v, y) = \frac{N(u, x \ ; \ v, y)}{k}.$$

When $N(v, y) \neq 0$, the *conditional block frequency* of u in x relative to v in y is defined as

$$P(u, x \mid v, y) = \frac{N(u, x \ ; \ v, y)}{N(v, y)}.$$

For any $v \in \Sigma_2^\ell$ such that $N(v, y) \neq 0$, it follows that $\sum_{u \in \Sigma_1^\ell} P(u, x \mid v, y) = 1$.

Definition 7.8. For two finite sequences $x \in \Sigma_1^n$ and $y \in \Sigma_2^n$, the *ℓ-length block entropy rate* of x relative to y is defined as:

$$H_\ell(x|y) = -\frac{1}{\ell \ log(|\Sigma_1|)} \sum_{v \in \Sigma_2^\ell} \sum_{u \in \Sigma_1^\ell} P(u, x \ ; \ v, y) \ \log(P(u, x \mid v, y)).$$

with the summation taken only over u, v such that $N(u, x \ ; \ v, y) \neq 0$.

Definition 7.9. For $X \in \Sigma_1^\infty$ and $Y \in \Sigma_2^\infty$, the ℓ-length *block entropy rate* of X relative to Y is defined as:

$$H_\ell(X|Y) = \liminf_{k\to\infty} H_\ell(X[:k\ell] \mid Y[:k\ell]).$$

The *block entropy rate* of X relative to Y is defined as[1] :

$$H(X|Y) = \inf_{\ell\in\mathbf{N}} H_\ell(X|Y).$$

Definition 7.10. For $X \in \Sigma_1^\infty$ and $Y \in \Sigma_2^\infty$, the ℓ-length *upper block entropy rate* of X relative to Y is defined as:

$$\overline{H}_\ell(X|Y) = \limsup_{k\to\infty} H_\ell(X[:k\ell] \mid Y[:k\ell]).$$

The *upper block entropy rate* of X relative to Y is defined as:

$$\overline{H}(X|Y) = \inf_{\ell\in\mathbf{N}} \overline{H}_\ell(X|Y).$$

7.3.2 Upper bounding dimension using entropy

For two sequences $X \in \Sigma_1^\infty$ and $Y \in \Sigma_2^\infty$, we prove that $\dim_{FS}^Y(X) \leq H(X \mid Y)$ by constructing a finite-state relative s-gale for any $s > H(X|Y)$, that succeeds on X. Similarly, we prove that $\mathrm{Dim}_{FS}^Y(X) \leq \overline{H}(X|Y)$by constructing an s'- gale for any $s' > \overline{H}(X|Y)$ that strongly succeeds on X.

The following lemma gives the logarithm of the capital attained by a finite-state relative gambler G betting on ℓ length blocks of X. Given a positive joint probability distribution $Q(x, y)$, the conditional distribution $Q(x|y)$ can be obtained as $Q(x|y) = \frac{Q(x,y)}{Q(y)}$ for all $Q(y) \neq 0$ where $Q(y) = \sum_{u\in\Sigma_1^\ell} Q(u, y)$.

Lemma 7.11. *Given a block length ℓ and a probability distribution $Q(x, y) : \Sigma_1^\ell \times \Sigma_2^\ell \to \mathbb{Q}\cap(0, 1)$, there exists a finite-state relative gambler G such that for any $X \in \Sigma_1^\infty, Y \in \Sigma_2^\infty$, and for any $s > 0$, for the s-gale induced by G,*

$$\log(d_{G,Y}^{(s)}(X[:k\ell]) = k\ell \log|\Sigma_1|\Big(s + \frac{1}{\ell\log|\Sigma_1|}\sum_{y\in\Sigma_2^\ell}\sum_{x\in\Sigma_1^\ell} P_{k\ell}(x, y)\log(Q(x|y))\Big).$$

[1] We later show that the $\inf_\ell$ can be replaced with a stronger $\lim_{\ell\to\infty}$ for both block entropy rate and upper block entropy rate.

where $P_{k\ell}(x, y) = P(x, X[:k\ell]\ ;\ y, Y[:k\ell])$.

Proof. The idea is to design a relative gambler having a window length ℓ that makes a bet corresponding to $Q(x|y)$ on the string $x \in \Sigma_1^\ell$ when it receives the sequence $y \in \Sigma_2^\ell$ from the oracle.

Consider the following finite-state relative gambler $G = (Q, \delta, \beta, q_0)$ with window length ℓ, where

- $Q = (\Sigma_2^\ell \times \Sigma_1^{<\ell}) \cup \{q_\lambda\}$
- For all $(y, w) \in Q$, $y' \in \Sigma_2^\ell$ and $a \in \Sigma_1$

$$\delta(q_\lambda, y, a) = (y, a)$$
$$\delta((y, w), y', a) = \begin{cases} (y, wa), & \text{if } |w| < \ell - 1 \\ q_\lambda, & \text{if } |w| = \ell - 1 \end{cases}$$

- For all $(y, w) \in Q$, $y' \in \Sigma_2^\ell$ and $a \in \Sigma_1$

$$\beta(q_\lambda, y, a) = Q(a|y)$$
$$\beta((y, w), y', a) = \frac{Q(wa|y)}{Q(w|y)}$$

  where for $|w| < \ell$, $Q(w|y) = \sum_{w' \in \Sigma_1^{\ell - |w|}} Q(ww'|y)$.

- $q_0 = q_\lambda$

After processing each ℓ length block of X, the automata resets to state q_λ, and the bets placed on the subsequent ℓ characters of X depend only on the characters from Y received at this state. It can be seen that if $y \in \Sigma_2^\ell$ is the oracle word received at state q_λ, for each $x \in \Sigma_1^\ell$, a cumulative bet corresponding to $Q(x|y)$ is placed on the next ℓ characters of input.

Therefore, from the definition of s-gale, it follows that for any $k \in \mathbb{N}$:

$$d_{G,Y}^{(s)}(X[:k\ell]) = \prod_{y \in \Sigma_2^\ell} \prod_{x \in \Sigma_1^\ell} \{|\Sigma_1|^{s.\ell} \times \beta(x|y)\}^{\hat{N}(x,y)}$$

where $\hat{N}(x,y) = N(x, X[:k\ell]\ ;\ y, Y[:k\ell])$, the number of times x and y occur concurrently in the disjoint ℓ length blocks of $X[:k\ell]$ and $Y[:k\ell]$ respectively. Taking logarithms,

$$\begin{aligned}
\log(d^{(s)}_{G,Y}(X[:k\ell]) &= \sum_{y\in\Sigma_2^\ell}\sum_{x\in\Sigma_1^\ell} \hat{N}(x,y)\big(\log|\Sigma_1|\ s\ell + \log(Q(x|y))\big) \\
&= ks\ell\log|\Sigma_1| + k\sum_{y\in\Sigma_2^\ell}\sum_{x\in\Sigma_1^\ell}\frac{\hat{N}(x,y)}{k}\log(Q(x|y)) \\
&= k\ell\log|\Sigma_1|\Big(s + \frac{1}{\ell\log|\Sigma_1|}\sum_{y\in\Sigma_2^\ell}\sum_{x\in\Sigma_1^\ell} P_{k\ell}(x,y)\log(Q(x|y))\Big).
\end{aligned}$$

□

Lemma 7.12 ensures that all elements in a probability space are close to a strictly positive probability distribution.

Lemma 7.12. *For every probability distribution P over a finite space Ω and any $\epsilon < \frac{1}{|\Omega|}$, there exists a probability distribution Q on Ω satisfying the following:*

1. *For every $\omega \in \Omega$, $Q(\omega) \geq \epsilon$.*
2. $\max_{\omega\in\Omega} |P(w) - Q(w)| \leq |\Omega|\epsilon$.

Proof. We construct the required Q by increasing $P(\omega)$ at any ω with $P(\omega) < \epsilon$ to ϵ by *shifting* sufficient amount of weight from those ω with $P(\omega) \geq \epsilon$. These modifications are done so that Q is a probability distribution on Ω satisfying the required conditions.

We define Q as follows. For any ω with $P(\omega) < \epsilon$, let $Q(\omega) = \epsilon$. For ω satisfying $P(\omega) \geq \epsilon$ let

$$Q(\omega) = \epsilon + (P(\omega) - \epsilon)\left(1 - \frac{\sum_{\omega':P(\omega')<\epsilon}(\epsilon - P(\omega'))}{\sum_{\omega':P(\omega')\geq\epsilon}(P(\omega') - \epsilon)}\right)$$

The term $\sum_{\omega':P(\omega')\geq\epsilon}(P(\omega')-\epsilon)$ is positive because $\epsilon < \frac{1}{|\Omega|}$. $Q(\omega)$ defined above is a positive quantity greater than or equal to ϵ because

$$\begin{aligned}\sum_{\omega':P(\omega')<\epsilon}(\epsilon - P(\omega')) &= \sum_{\omega':P(\omega')<\epsilon}\epsilon - 1 + \sum_{\omega':P(\omega')\geq\epsilon}P(\omega') \\ &= \epsilon|\Omega| - \sum_{\omega':P(\omega')\geq\epsilon}\epsilon - 1 + \sum_{\omega':P(\omega')\geq\epsilon}P(\omega') \\ &= (\epsilon|\Omega|-1) + \sum_{\omega':P(\omega')\geq\epsilon}(P(\omega')-\epsilon) \\ &\leq \sum_{\omega':P(\omega')\geq\epsilon}(P(\omega')-\epsilon).\end{aligned}$$

The last inequality follows since $\epsilon < \frac{1}{|\Omega|}$. It is straightforward to verify that Q is a probability distribution on Ω and it satisfies

$$\max_{\omega:P(\omega)<\epsilon}|P(\omega)-Q(\omega)| \leq \epsilon.$$

Consider any ω such that $P(\omega)\geq\epsilon$. We have

$$\begin{aligned}|P(\omega)-Q(\omega)| &= \frac{P(\omega)-\epsilon}{\sum_{\omega':P(\omega')\geq\epsilon}(P(\omega')-\epsilon)}\sum_{\omega':P(\omega')<\epsilon}(\epsilon-P(\omega')) \\ &\leq \sum_{\omega':P(\omega')<\epsilon}(\epsilon-P(\omega')) \leq \epsilon|\Omega|.\end{aligned}$$

The proof of the lemma is thus complete. □

In Lemma 7.13, we show that when all the bets placed are strictly positive, a particular Kullback-Leibler Divergence score between the empirical probability distribution and a close enough bet is small. In Lemma 7.14 we use this condition to show that when the relative entropy falls, the s-gale corresponding to a gambler that places a close enough bet to the empirical distribution succeeds.

Lemma 7.13. *Given $\epsilon > 0$, and a probability distribution $Q(x,y)$ defined over $\Omega = \Sigma_1^\ell \times \Sigma_2^\ell$. If for all $x,y \in \Sigma_1^\ell \times \Sigma_2^\ell$ $Q(x,y)\geq\epsilon$, then for all $P(x,y)$ such that $\max_{x,y}|P(x,y)-Q(x,y)| < 2|\Omega|\epsilon$, we have*

$$\frac{1}{\ell\log|\Sigma_1|}\sum_{y\in\Sigma_2^\ell}\sum_{x\in\Sigma_1^\ell}P(x,y)\,[\,\log P(x|y)-\log Q(x|y)\,] < 2|\Omega|^4\epsilon.$$

Proof. It is easy to see that the term on the left expands to

$$\frac{1}{\ell \log |\Sigma_1|} \sum_{y \in \Sigma_2^\ell} \sum_{x \in \Sigma_1^\ell} P(x,y) \log \left(\frac{P(x,y)}{Q(x,y)} \right) - \frac{1}{\ell \log |\Sigma_1|} \sum_{y \in \Sigma_2^\ell} P(y) \log \left(\frac{P(y)}{Q(y)} \right)$$

As the second term is the KL divergence $D_{KL}(P(y)||Q(y))$, which is non-negative [8], we have that the above expression is less than or equal to the first term, $D_{KL}(P(x,y)||Q(x,y))$.

Define $\alpha_Q = \min_{x,y} Q(x,y)$. Using the inverse of Pinsker's Inequality [17], we see that

$$D_{KL}(P(x,y)||Q(x,y)) \leq \frac{2}{\alpha_Q} \delta(P(x,y), Q(x,y))^2$$

where $\delta(P(x,y), Q(x,y))$ is the total variation distance [27] and it can be expressed as $\frac{1}{2} \sum_{x,y} |P(x,y) - Q(x,y)|$.

Using the conditions given, it follows that $\delta(P(x,y), Q(x,y)) < |\Omega|^2 \epsilon$ and $\alpha_Q \geq \epsilon$. Therefore,

$$D_{KL}(P(x,y)||Q(x,y)) < 2|\Omega|^4 \epsilon.$$

□

Lemma 7.14. *For every $X \in \Sigma_1^\infty$ and $Y \in \Sigma_2^\infty$, $\dim_{FS}^Y(X) \leq H(X|Y)$.*

Proof. It suffices to show that for any $s > H(X|Y)$, there exists an s-gale corresponding to a gambler G that succeeds on X. By definition, there exists a block length $\ell \in \mathbb{N}$ such that $\liminf_{k \to \infty} H_\ell(X[:k\ell] \mid Y[:k\ell]) < s'$ for some $s' < s$.

Let $S = \{P(x,y) : \Sigma_1^\ell \times \Sigma_2^\ell \to [0,1] : \sum_{x,y} P(x,y) = 1\}$ be the set of joint probability distributions defined over $x, y \in \Omega$, where $\Omega = \Sigma_1^\ell \times \Sigma_2^\ell$.

Given $0 < \epsilon < \frac{1}{|\Omega|}$, consider the set $T = \{P(x,y) \in S : \forall x, y \; P(x,y) \geq \epsilon\}$. From Lemma 7.12 it follows that the union of $2\epsilon|\Omega|$ open boxes around elements of T cover the compact set S, that is

$$\bigcup_{Q(x,y) \in T} \{P(x,y) : \max_{x,y} |P(x,y) - Q(x,y)| < 2\epsilon|\Omega|\} = S.$$

Choose $\{Q_i(x,y) : 1 \leq i \leq \mathbf{n}\}$ to be the set of representative elements in the finite subcover of this open cover of S. Consider the $\mathbf{n}$-account finite state relative gambler G,

with each account placing bets corresponding to a particular $Q_i(x, y)$. Proceeding with the construction in Lemma 7.11, in the account corresponding to $Q(x, y)$, we have that

$$\log(d^{(s)}_{G,Y}(X[:k\ell]) = k\ell \log|\Sigma_1|\Big(s + \frac{1}{\ell \log|\Sigma_1|} \sum_{y\in\Sigma_2^\ell} \sum_{x\in\Sigma_1^\ell} P_{k\ell}(x, y) \log(Q(x|y))\Big)$$

where $P_{kl}(x, y) = P(x, X[:k\ell]\ ;\ y, Y[:k\ell])$.

Since $\liminf_{k\to\infty} H_\ell(X[:k\ell] \mid Y[:k\ell]) < s'$, it follows that for infinitely many $k \in \mathbb{N}$,

$$-\frac{1}{\ell \log|\Sigma_1|} \sum_{y\in\Sigma_2^\ell} \sum_{x\in\Sigma_1^\ell} P(x, X[:k\ell]\ ;\ y, Y[:k\ell]) \log(P(x, X[:k\ell] \mid y, Y[:k\ell])) < s'.$$

Applying Lemma 7.13, we see that in the account corresponding to $Q_i(x, y)$ such that $\max_{x,y} |P(x, y) - Q(x, y)| < 2|\Omega|\epsilon$, we have

$$\log(d^{(s)}_{G,Y}(X[:k\ell]) > k\ell \log|\Sigma_1| \ \big(s - s' - 2|\Omega|^4\epsilon\big).$$

Choose an $\epsilon < \frac{1}{|\Omega|}$ such that $s > s' + 2|\Omega|^4\epsilon$. Since k is unbounded, the **n**-account $s - gale$ succeeds on X relative to Y. □

Lemma 7.15. *For every $X \in \Sigma_1^\infty$ and $Y \in \Sigma_2^\infty$, $\mathrm{Dim}^Y_{FS}(X) \leq \overline{H}(X|Y)$.*

Proof. For any $s > \overline{H}(X|Y)$, there exists a block length $\ell \in \mathbb{N}$ such that $\limsup_{k\to\infty} H_\ell(X[:k\ell] \mid Y[:k\ell]) < s'$ for some $s' < s$.

Proceeding in the same lines as Lemma 7.14, we get an **n**-account finite-state gambler [9], such that for sufficiently large $k \in \mathbb{N}$, in one among the accounts,

$$\log(d^{(s)}_{G,Y}(X[:k\ell]) > k\ell \log|\Sigma_1|\big(s - s' - 2|\Omega|^4\epsilon\big).$$

Choose an $\epsilon < \frac{1}{|\Omega|}$ such that $s > s' + 2|\Omega|^4\epsilon$. Hence the **n** account finite-state gambler strongly succeeds on X relative to Y. □

7.3.3 Upper bounding entropy using dimension

Let an s-gale $d^{(s)}_{G,Y}$ corresponding to a finite-state gambler G succeed on $X \in \Sigma_1^\infty$. For a block length $L \in \mathbb{N}$, we construct the *stretched finite-state relative gambler* G_L, to keep track of the state encountered after every run of input of length L on G.

Definition 7.16. Let $G = (Q, \delta, \beta, q)$ be a finite-state relative gambler with window length ℓ. For any $L \in \mathbb{N}$ such that $L > \ell$, we define the *L-length stretched FSRG* of G to be $G_L = (Q', \delta', \beta', q'_0)$, where:

- $Q' = Q \times [L]$.
- For all $(q, n) \in Q'$, $y \in \Sigma_2^\ell$ and $x \in \Sigma_1$,

$$\delta'((q,n),y,x) = \begin{cases} (\delta(q,y,x), n+1) & \text{if } n < L-1 \\ (\delta(q,y,x), 0) & \text{if } n = L-1 \end{cases}$$

- For all $(q, n) \in Q'$, $y \in \Sigma_2^\ell$ and $x \in \Sigma_1$,

$$\beta'((q,n),y,x) = \begin{cases} \beta(q,y,x), & \text{if } n \le L-\ell \\ 1/|\Sigma_1|, & \text{if } n > L-\ell \end{cases}$$

- $q'_0 = (q_0, 0)$.

Lemma 7.17. *Let the s-gale $d^{(s)}_{G,Y}$ corresponding to gambler G succeeds on X, and let ℓ be the oracle window length of G. Then for any $L > \ell$ and $s' = s + \frac{\ell-1}{L}$, the s'-gale corresponding to the stretched FSRG, $d^{(s')}_{G_L,Y}$ succeeds on X.*

Proof. $d^{(s')}_{G_L,Y}$ succeeds on X, because after processing k blocks of length L,

$$\begin{aligned} d^{(s')}_{G_L,Y}(X[:kL])) &= |\Sigma_1|^{kL(s+\frac{\ell-1}{L})} \prod_{i=1}^{k} \left\{ \prod_{j=0}^{L-l} \beta_{(i,j)} \prod_{j=L-\ell+1}^{L-1} (1/|\Sigma_1|) \right\} \\ &= \frac{1}{|\Sigma_1|^{k(\ell-1)}} |\Sigma_1|^{k(\ell-1)} |\Sigma_1|^s \prod_{i=1}^{k} \left\{ \prod_{j=0}^{L-l} \beta_{(i,j)} \prod_{j=L-\ell+1}^{L-1} 1 \right\} \\ &\ge d^{(s)}_{G_L,Y}(X[:kL]). \end{aligned}$$

□

We now show that if s-gale $d^{(s)}_{G,Y}$ succeeds on X, then for any sufficiently large block length L, $H_L(X|Y) \leq s + c/L$, where c is a constant which does not depend on L.

Lemma 7.18. *For every $X \in \Sigma_1^\infty$ and $Y \in \Sigma_2^\infty$, we have $H(X|Y) \leq \dim^Y_{FS}(X)$.*

Proof. Consider any s such that $s > \dim^Y_{FS}(X)$. It suffices to show that $H(X|Y) \leq s$. By definition, there exists a finite-state relative s-gale $d^{(s)}_{G,Y}$ that succeeds on X. Let ℓ be the oracle window length of the corresponding finite-state relative gambler G. For any $L \in \mathbb{N}$ such that $L > \ell$, consider the *L-length stretched FSRG* G_L given in Definition 7.16. From Lemma 7.17, it follows that taking $s' = s + \frac{\ell-1}{L}$, the s'-gale corresponding to G_L, $d^{(s')}_{G_L,Y}$ succeeds on X.

After processing each L length block of characters from input X, G_L reaches the state $(q, 0)$ for some $q \in Q$. Let $x = X[iL : (i+1)L]$, the ith disjoint block of length L in X for some $i \in \mathbb{N}$. The bets placed by G_L on the last $L - \ell$ characters of x are constant. Therefore, the cumulative bet $\beta(q, y, x)$ placed by G_L on x is dependent only on the state $(q, 0)$ encountered while processing the block and the corresponding L characters $y = Y[iL : (i+1)L]$ of the oracle input Y. For any $q \in Q, x \in \Sigma_1^L$ and $y \in \Sigma_2^L$, let $N(q, y, x)$ be the number of times an L length block in X is x, in Y it is y and the starting state of the block was $(q, 0)$. Therefore,

$$d^{(s')}_{G_L,Y}(X[: kL]) = \prod_{q\in Q} \prod_{y\in\Sigma_2^L} \prod_{x\in\Sigma_1^L} \{|\Sigma_1|^{s'.L} \times \beta(q, y, x)\}^{N(q,y,x)}.$$

Take $N(q, y) = \sum_{x\in\Sigma_1^L} N(q, y, x)$ and $N(q) = \sum_{y\in\Sigma_2^L} N(q, y)$. Hence,

$$\begin{aligned}\log(d^{(s')}_{G_L,Y}(X[: kL])) &= s'kL \log|\Sigma_1| + \sum_{q\in Q} \sum_{y\in\Sigma_2^L} \sum_{x\in\Sigma_1^L} N(q, y, x)\ \log(\beta(q, y, x)) \\ &= s'kL \log|\Sigma_1| + \ k \sum_{q\in Q} \frac{N(q)}{k} \sum_{y\in\Sigma_2^L} \frac{N(q, y)}{N(q)} \sum_{x\in\Sigma_1^L} \frac{N(q, y, x)}{N(q, y)} \log(\beta(q, y, x)).\end{aligned}$$

Taking $P(q) = \frac{N(q)}{k}$, $P(y|q) = \frac{N(q,y)}{N(q)}$ and $P(x|y, q) = \frac{N(q,y,x)}{N(q,y)}$, stipulating that these take values 0 if the denominators are zero,

$$\begin{aligned}\log(d^{(s')}_{G_L,Y}(X[: kL])) = k\Big(&s'L \log|\Sigma_1| \\ &+ \sum_{q\in Q} P(q) \sum_{y\in\Sigma_2^L} P(y|q) \sum_{x\in\Sigma_1^L} P(x|y, q)\ \log(\beta(q, y, x))\Big).\end{aligned}$$

$d^{(s')}_{G_L,Y}$ succeeds on X only if for infinitely many $k \in \mathbb{N}$,

$$-\sum_{q\in Q} P(q) \sum_{y\in\Sigma_2^L} P(y|q) \sum_{x\in\Sigma_1^L} P(x|y,q)\ \log(\beta(q,y,x)) < s'L\log|\Sigma_1|.$$

For all $q \in Q$ and $y \in \Sigma_2^L$, $\sum_{x\in\Sigma_1^L} \beta(q,y,x) = 1$ and $\sum_{x\in\Sigma_1^L} P(x|y,q) = 1$.

Therefore, the KL divergence of these distributions [8] $D_{KL}(P(x|y,q)||\beta(q,y,x))$ over values in x is well defined and is non-negative. So it follows that

$$-\sum_{q\in Q} P(q) \sum_{y\in\Sigma_2^L} P(y|q) \sum_{x\in\Sigma_1^L} P(x|y,q)\ \log(P(x|y,q)) < s'L\log|\Sigma_1|.$$

Using result from information theory that $H(X|Y) \leq H(X|Q,Y) + H(Q|Y)$ [8], it follows that,

$$-\sum_{y\in\Sigma_2^L} P(y) \sum_{x\in\Sigma_1^L} P(x|y)\ \log(P(x|y)) < s'L\log|\Sigma_1| + H(Q|Y).$$

$H(Q|Y)$ is at most $\log(n)$, where $n = |Q|$, the number of states in the original gambler G.

Dividing by $L\log|\Sigma_1|$ and substituting the value of s', we have that for infinitely many $k \in \mathbb{N}$,

$$H_L(X[:kL] \mid Y[:kL]) \ < \ s + \frac{l-1}{L} + \frac{\log(n)}{L\log|\Sigma_1|}.$$

Taking lim inf over k,

$$H_L(X|Y) \leq \ s + \frac{l-1}{L} + \frac{\log(n)}{L\log|\Sigma_1|}. \tag{7.1}$$

Taking infimum over all L,

$$H(X|Y) \leq s.$$

□

Lemma 7.19. *For every $X \in \Sigma_1^\infty$ and $Y \in \Sigma_2^\infty$, $\overline{H}(X|Y) \leq \mathrm{Dim}^Y_{FS}(X)$.*

Proof. The proof follows along the lines of the proof of Lemma 7.18. Using the condition that an s-gale strongly succeeding on X, we have that for sufficiently large $k \in \mathbb{N}$,

$$H_L(X[:kL] \mid Y[:kL]) \;<\; s + \frac{l-1}{L} + \frac{\log(n)}{L \log|\Sigma_1|}. \tag{7.2}$$

From which it follows that, $\overline{H}(X|Y) \leq s$. □

We originally defined block entropy rates by taking infimum of $H_\ell(X|Y)$ over all block lengths ℓ. But as a consequence of the proof, we show that the infimum is in fact a limit as $\ell \to \infty$.

Theorem 7.20. *For any two sequences $X \in \Sigma_1^\infty$ and $Y \in \Sigma_2^\infty$,*

$$\dim_{FS}^{Y}(X) = \lim_{\ell\to\infty} H_\ell(X|Y).$$

Proof. Lemma 7.14 established that $\inf_\ell H_\ell(X|Y) \geq \dim_{FS}^{Y}(X)$, therefore

$$\liminf_{\ell\to\infty} H_\ell(X|Y) \geq \dim_{FS}^{Y}(X). \tag{7.3}$$

From (7.1) in Lemma 7.18, it follows that

$$\limsup_{\ell\to\infty} H_\ell(X|Y) \leq \dim_{FS}^{Y}(X). \tag{7.4}$$

The theorem follows immediately from (7.3) and (7.4). □

Theorem 7.21. *For any two sequences $X \in \Sigma_1^\infty$ and $Y \in \Sigma_2^\infty$,*

$$\mathrm{Dim}_{FS}^{Y}(X) = \lim_{\ell\to\infty} \overline{H_\ell}(X|Y).$$

Proof. It follows from Lemma 7.15 that

$$\liminf_{\ell\to\infty} \overline{H}_\ell(X|Y) \geq \mathrm{Dim}_{FS}^{Y}(X). \tag{7.5}$$

From (7.2) in Lemma 7.19, it follows that

$$\limsup_{\ell\to\infty} \overline{H}_\ell(X|Y) \leq \mathrm{Dim}_{FS}^{Y}(X). \tag{7.6}$$

The theorem follows immediately from (7.5) and (7.6). □

Note: Shen [50] shows how this equivalence can be obtained easily by extending existing characterizations of normality and dimension. He also shows characterizations in terms of gambling, superadditive complexity measures and compressibility.

7.3.4 Multiple Oracles

Consider the case when the automaton has access to multiple oracle sequences, say $Y_1 \cdots Y_n$, where each $Y_i \in \Sigma_2^\infty$. The finite-state relative dimension and relative block entropy rates in this scenario are defined by considering a single oracle sequence $Y \in (\Sigma_2^n)^\infty$, such that $Y = Y_1 \times Y_2 \cdots \times Y_n$.

Definition 7.22. The finite-state relative dimension of a sequence X with respect to a sequences $Y_1 \cdots Y_n$ is defined as $\dim_{FS}^{Y_1 \cdots Y_n}(X) = \dim_{FS}^{Y}(X)$.

Definition 7.23. The *relative block entropy rate* of X with respect to $Y_1, \ldots, Y_n$ is defined as

$$H(X|Y_1 \cdots Y_n) = H(X|Y).$$

Therefore from Theorem 7.20 it follows that,

Theorem 7.24. *For sequences $X \in \Sigma_1^\infty$ and $Y_1 \ldots Y_n \in \Sigma_2^\infty$*

$$\dim_{FS}^{Y_1 \cdots Y_n}(X) = H(X|Y_1 \cdots Y_n)$$
$$\mathrm{Dim}_{FS}^{Y_1 \cdots Y_n}(X) = \overline{H}(X|Y_1 \cdots Y_n)$$

7.4 Properties of of finite-state relative dimension

The following are some basic properties of finite-state relative dimension.

Lemma 7.25. *For any $X \in \Sigma_1^\infty$, $Y \in \Sigma_2^\infty$ and $a \in \Sigma_2$,*

1. $\dim_{FS}^{Y}(X) \leq \dim_{FS}(X)$.
2. $\dim_{FS}^{X}(X) = 0$.
3. $\dim_{FS}^{a^\infty}(X) = \dim_{FS}(X)$.
4. $\dim_{FS}^{Y}(X) \leq \mathrm{Dim}_{FS}^{Y}(X)$.

Proof. Statement 1 follows from the fact that a finite-state relative s-gale can ignore the oracle input Y and place the same bets on a sequence X that a normal s-gale places on it.

For Statement 2, we have the complete information about bits of X given by the oracle. Therefore for any $s > 0$, we can design a finite-state s-gale of window length 1 that places all its capital on the input given by the oracle.

For the proof of Statement 3, consider a relative s-gale corresponding to G' having window length ℓ that wins on a sequence X given a^∞ as the oracle. We can construct a normal s-gale G that places the same bets that G' places on a character of X when the input from oracle is a^ℓ . It is easy to see then that the s-gale corresponding to G succeeds on X.

Statement 4 follows from the fact that $\liminf a_n \leq \limsup a_n$ for any sequence $\langle a_n \rangle$. □

7.5 Comparison with other related notions

In this section, we look at relationship between finite-state relative dimension and the notions of finite-state mutual dimension [7] and finite-state independence [2].

7.5.1 Relation to Mutual Dimension

Case and Lutz [7] introduce finite-state mutual dimension as an analogue of mutual dimension in the a finite state setting. Intuitively, finite-state mutual dimension between two sequences measures the finite-state density of information shared between the sequences.

We show that for the case of regular sequences X and Y, adding finite-state relative dimension of X given $Y$and finite-state mutual dimension between X and Y, we get the finite state dimension of X. For non-regular sequences, we provide upper and lower bounds on finite-state mutual dimension using finite state dimension of X and finite-state relative dimension of X given Y.

Definition 7.26 (Case and Lutz [7])**.** For two sequences $X, Y \in \Sigma^\infty$, the *finite-state mutual dimension* between X and Y,

$$\mathrm{mdim}_{FS}(X;Y) = \lim_{\ell\to\infty} \liminf_{k\to\infty} I_\ell(X[:k\ell]\ ;\ Y[:k\ell])$$

where for two finite sequences $X, Y \in \Sigma^{k\ell}$,

$$I_\ell(X\ ;\ Y) = H_\ell(X) - H_\ell(X \mid Y).$$

The *finite-state strong mutual dimension* $\mathrm{Mdim}_{FS}(X;Y)$ between X and Y is defined similarly as $\mathrm{mdim}_{FS}(X;Y)$ by replacing the lim inf in the definition with a lim sup. Using (7.26) and routine real analytic arguments it follows that the following relationships between finite-state relative dimension and finite-state mutual dimension hold:

Lemma 7.27. *For all* $X \in \Sigma^\infty$,

1. $\dim_{FS}(X) - \mathrm{Dim}^Y_{FS}(X) \leq \mathrm{mdim}_{FS}(X:Y) \leq \dim_{FS}(X) - \dim^Y_{FS}(X)$.
2. $\mathrm{Dim}_{FS}(X) - \mathrm{Dim}^Y_{FS}(X) \leq \mathrm{Mdim}_{FS}(X:Y) \leq \mathrm{Dim}_{FS}(X) - \dim^Y_{FS}(X)$.

If the sequences X and Y are regular, then by definition, $\dim_{FS}(X) = \mathrm{Dim}_{FS}(X)$ and $\dim^Y_{FS}(X) = \mathrm{Dim}^Y_{FS}(X)$ and so,

$$\dim^Y_{FS}(X) = \dim_{FS}(X) - \mathrm{mdim}_{FS}(X:Y).$$

7.5.2 Relation to finite-state independence

Becher, Carton and Heiber [2] define finite-state independence between two strings. They use a joint compression model comprising of a one-to-one finite-state transducer with auxiliary input. Two strings X and Y are said to be finite-state independent when one does not help to compress the other in this model. However, due to the presence of ε-transitions, the input tape of X and Y need not be read in tandem. Our model is more restrictive, hence more pairs of sequences are relatively finite-state random in our model.

Definition 7.28. We call a sequence $A \in \Sigma^\infty$ *relatively normal* to another sequence $B \in \Sigma^\infty$ if and only if $\dim^B_{FS}(A) = 1$.

We require the finite-state analogue of the van Lambalgen's Theorem [43].

Theorem 7.29 (Akhil, Nandakumar, Pulari [43])**.** *Let* $A, B \in \Sigma^\infty$. *A is normal and B is normal relative to A if and only if $A \oplus B$ is normal.*

We now show that for two normal sequences $X, Y \in \Sigma^\infty$, finite state independence implies relative normality, but the converse does not hold.

Lemma 7.30. *Given any two normal sequences* $X \in \Sigma^\infty$ *and* $Y \in \Sigma^\infty$, *if X and Y are finite-state independent, then X and Y are relatively normal. But, there are normal sequences X and Y that are not finite-state independent but X and Y are relatively normal.*

Proof. By Theorem 4.1 in [2], if two normal sequences X and Y are finite-state independent, then $X \oplus Y$ is normal. By Theorem 7.29, we have that X and Y are relatively normal.

By Theorem 4.3 in [2], there exists two normal strings X and Y such that $X \oplus Y$ is normal but X and Y are not finite-state independent. Since $X \oplus Y$ is normal, from Theorem 7.29, we have that X and Y are relatively normal. □

Conclusion

The first part of this thesis investigated the notions of constructive Φ-dimension and faithfulness of constructive dimension in the setting of continued fractions and Cantor coverings.

In chapter 3, we formulate constructive Φ-dimension using covers and showed equivalent characterisations using s-gales and Kolmogorov complexity. We also show a point-to-set principle for Φ-dimension.

In Chapter 4, we show that the set of continued fraction coverings is not faithful for constructive dimension. Thus we show that constructive continued fraction dimension is not equivalent to constructive dimension. We also show that this proof relativises, and using the point-to-set principle, show that the set of continued fraction coverings is not faithful for Hausdorff dimension.

In Chapter 5, we explored constructive dimension faithfulness of covers generated by Cantor series expansions. We charaterise the constructive dimension faithfulness of Cantor coverings, using a log-limit condition on the terms appearing in the expansion. We also show that for computable Cantor coverings, faithfulness at the constructive and Hausdorff dimension levels are equivalent.

The second part of this thesis investigated the notion of polynomial-time dimension. In Chapter 6, we show that two notions of polynomial-time density of information, via s-gales (dimension) and via Kolmogorov complexity rates (decompression) are not equivalent if One-way functions exist. Hence we gave a conditional resolution to the open question of robustness of polynomial-time dimension [52].

The third part of this thesis looked at the notion of dimension using finite-state resource bounds. In Chapter 7, we formulate the notion of finite-state relative dimension using s-gales with a finite-window oracle access. We also show an equivalent characterisation using relative block entropy rates.

The following are some problems that remain open.

1. Are the faithfulness at constructive and Hausdorff levels equivalent for all computable family of covering sets Φ?

2. What is the packing dimension analogue of faithfulness, is there any relationship between faithfulness of Hausdorff dimension and packing dimension?

3. Can weaker hardness assumptions, such as P $\neq$ NP, DistNP $\not\subseteq$ AvgP or DistNP $\not\subseteq$ AvgBPP can be characterized in terms of the separation of polynomial-time dimension, as defined using time-bounded s-gales and time-bounded Kolmogorov complexity ?

4. Do the notions of polynomial-time randomness-defined via betting algorithms and Kolmogorov incompressibility of prefixes [52] capture equivalent concepts of randomness ?

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